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Semiconductor device having a groove portion in an oxide layer of a transistor channel

Abstract

A semiconductor device that can be miniaturized or highly integrated is provided. The semiconductor device includes a transistor. The transistor includes a first conductor, a first insulator over the first conductor, an oxide provided with a groove portion over the first insulator, a second conductor and a third conductor disposed in a region that does not overlap with the groove portion in the oxide, a second insulator disposed between the second conductor and the third conductor and in the groove portion in the oxide, and a fourth conductor over the second insulator. A bottom surface of the fourth conductor is lower than a bottom surface of the second conductor and a bottom surface of the third conductor. In a cross-sectional view of the transistor in the channel length direction, an end portion of a bottom surface of the groove portion has a curvature.

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Background/Summary

TECHNICAL FIELD

(1) One embodiment of the present invention relates to a transistor, a semiconductor device, and an electronic device. One embodiment of the present invention relates to a method of manufacturing a semiconductor device. One embodiment of the present invention relates to a semiconductor wafer and a module.

(2) Note that in this specification and the like, a semiconductor device generally means a device that can function by utilizing semiconductor characteristics. A semiconductor element such as a transistor, a semiconductor circuit, an arithmetic device, and a storage device are each one embodiment of a semiconductor device. It can be sometimes said that a display device (a liquid

crystal display device, a light-emitting display device, or the like), a projection device, a lighting device, an electro-optical device, a power storage device, a storage device, a semiconductor circuit, an imaging device, an electronic device, and the like include a semiconductor device.
(3) Note that one embodiment of the present invention is not limited to the above technical field. One embodiment of the invention disclosed in this specification and the like relates to an object, a method, or a manufacturing method. Another embodiment of the present invention relates to a process, a machine, manufacture, or a composition of matter.

BACKGROUND ART

(4) In recent years, demand for an integrated circuit in which transistors and the like are integrated with high density has risen with reductions in the size and weight of an electronic device. As a means of integrating transistors with high density, miniaturization of a transistor has been developed. However, miniaturization of a transistor might cause a short-channel effect.
(5) To prevent the occurrence of a short-channel effect by miniaturization of a transistor, a semiconductor device whose channel length is effectively lengthened by forming a channel in a U-shaped groove provided on a surface of a semiconductor substrate is disclosed (see Patent Document 1).
(6) A technique by which a transistor is formed using a semiconductor thin film formed over a substrate having an insulating surface has been attracting attention. The transistor is applied to a wide range of electronic devices such as an integrated circuit (IC) or an image display device (also simply referred to as a display device). A silicon-based semiconductor material is widely known as a semiconductor thin film applicable to the transistor; in addition, an oxide semiconductor has been attracting attention as another material.

REFERENCE

Patent Document

(7) [Patent Document 1] Japanese Published Patent Application No. H9-148576

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(8) With reductions in the size and weight of an electronic device, miniaturization and high integration of a semiconductor device included in the electronic device is required. Miniaturization and high integration of a semiconductor device need to reduce the area occupied by transistors included in the semiconductor device. In order to reduce the area occupied by the transistors, it is effective to shorten a distance between a source and a drain, for example. However, when the distance between a source and a drain is shortened, a short-channel effect might occur.
(9) The short-channel effect refers to degradation of electrical characteristics which becomes obvious along with miniaturization of a transistor (a decrease in channel length). The short-channel effect results from the effect of an electric field of a drain on a source. Specific examples of the short-channel effect include a decrease in threshold voltage, an increase in subthreshold swing value (S value), an increase in leakage current, and the like. S value means the amount of change in a gate voltage in the subthreshold region when the drain voltage keeps constant and the drain current changes by one digit.
(10) An object of one embodiment of the present invention is to provide a semiconductor device that can be miniaturized or highly integrated. An object of one embodiment of the present invention is to provide a semiconductor device having favorable electrical characteristics. An object of one embodiment of the present invention is to provide a semiconductor device with less variations in transistor characteristics. An object of one embodiment of the present invention is to provide a highly reliable semiconductor device. An object of one embodiment of the present invention is to provide a semiconductor device with a high on-state current. An object of one embodiment of the present invention is to provide a semiconductor device with low power consumption.
(11) Note that the description of these objects does not preclude the existence of other objects. One embodiment of the present invention does not necessarily achieve all these objects. Objects other

than these are apparent from the description of the specification, the drawings, the claims, and the like and objects other than these can be derived from the description of the specification, the drawings, the claims, and the like.

Means for Solving the Problems

(12) One embodiment of the present invention is a semiconductor device including a transistor. The transistor includes a first conductor, a first insulator over the first conductor, an oxide provided with a groove portion over the first insulator, a second conductor and a third conductor disposed in a region that does not overlap with the groove portion in the oxide, a second insulator disposed between the second conductor and the third conductor and in the groove portion in the oxide, and a fourth conductor over the second insulator. A bottom surface of the fourth conductor is lower than a bottom surface of the second conductor and a bottom surface of the third conductor. In a cross-sectional view of the transistor in the channel length direction, an end portion of a bottom surface of the groove portion has a curvature.

(13) In the above semiconductor device, the depth of the groove portion is preferably greater than or equal to 5 nm and less than or equal to 30 nm.

(14) Another embodiment of the present invention is a semiconductor device including a transistor. The transistor includes a first conductor, a first insulator over the first conductor, a first oxide over the first insulator, a second oxide over the first oxide, a second conductor and a third conductor over the second oxide, a third oxide disposed between the second conductor and the third conductor, a second insulator over the third oxide, a fourth conductor over the second insulator, and a third insulator over the second conductor and the third conductor. A top surface of the fourth conductor is substantially aligned with a top surface of the second insulator and a top surface of the third oxide. The second oxide includes a first groove portion. The third insulator includes a second groove portion. A sidewall of the first groove portion is substantially aligned with a sidewall of the second groove portion. A bottom surface of the fourth conductor is lower than a bottom surface of the second conductor and a bottom surface of the third conductor. In a cross-sectional view of the transistor in the channel length direction, an end portion of a bottom surface of the first groove portion has a curvature.

(15) In the above semiconductor device, the depth of the first groove portion is preferably greater than or equal to 5 nm and less than or equal to 30 nm.

(16) In the above semiconductor device, the second oxide preferably contains indium, and the third oxide preferably contains indium, an element M (M is gallium, aluminum, yttrium, or tin), and zinc. An atomic ratio of the indium to a metal element which is a main component in the second oxide is preferably larger than an atomic ratio of the indium to a metal element which is a main component in the third oxide.

Effect of the Invention

(17) According to one embodiment of the present invention, a semiconductor device that can be miniaturized or highly integrated can be provided. According to one embodiment of the present invention, a semiconductor device having favorable electrical characteristics can be provided. According to one embodiment of the present invention, a semiconductor device with less variations in transistor characteristics can be provided. According to one embodiment of the present invention, a highly reliable semiconductor device can be provided. According to one embodiment of the present invention, a semiconductor device with a high on-state current can be provided. According to one embodiment of the present invention, a semiconductor device with low power consumption can be provided.

(18) Note that the description of the effects does not preclude the existence of other effects. One embodiment of the present invention does not necessarily achieve all the effects. Effects other than these are apparent from the description of the specification, the drawings, the claims, and the like and effects other than these can be derived from the description of the specification, the drawings, the claims, and the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A to FIG. **1**C are cross-sectional views of transistors of embodiments of the present invention.
(2) FIG. **2**A is a graph showing the relation between an effective channel length and a Vsh of a transistor. FIG. **2**B is a graph showing the relation between an effective channel length and the maximum value of gm of a transistor. FIG. **2**C is a graph showing the relation between an effective channel length and an S value of a transistor.
(3) FIG. **3** is a graph showing the relation between an effective channel length and DIBL of a transistor.
(4) FIG. **4** is a graph showing the relation between an effective channel length and ∂ Vsh/∂ Vbg of a transistor.
(5) FIG. **5**A is a graph showing the relation between the depth of a groove portion and a Vsh of a transistor. FIG. **5**B is a graph showing the relation between the depth of a groove portion and the maximum value of gm of a transistor. FIG. **5**C is a graph showing the relation between the depth of a groove portion and an S value of a transistor.
(6) FIG. **6** is a graph showing the relation between the depth of a groove portion and DIBL of a transistor.
(7) FIG. **7**A is a top view of a semiconductor device of one embodiment of the present invention.
(8) FIG. **7**B to FIG. **7**D are cross-sectional views of a semiconductor device of one embodiment of the present invention.
(9) FIG. **8**A is a view illustrating classification of crystal structures of IGZO. FIG. **8**B is a graph showing an XRD spectrum of a quartz glass. FIG. **8**C is a graph showing an XRD spectrum of crystalline IGZO.
(10) FIG. **9**A is a top view illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention. FIG. **9**B to FIG. **9**D are cross-sectional views illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention.
(11) FIG. **10**A is a top view illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention. FIG. **10**B to FIG. **10**D are cross-sectional views illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention.
(12) FIG. **11**A is a top view illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention. FIG. **11**B to FIG. **11**D are cross-sectional views illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention.
(13) FIG. **12**A is a top view illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention. FIG. **12**B to FIG. **12**D are cross-sectional views illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention.
(14) FIG. **13**A is a top view illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention. FIG. **13**B to FIG. **13**D are cross-sectional views illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention.
(15) FIG. **14**A is a top view illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention. FIG. **14**B to FIG. **14**D are cross-sectional views illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention.
(16) FIG. **15**A is a top view illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention. FIG. **15**B to FIG. **15**D are cross-sectional views illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention.
(17) FIG. **16**A is a top view illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention. FIG. **16**B to FIG. **16**D are cross-sectional views illustrating a method of manufacturing a semiconductor device of one embodiment of the present invention.

(18) FIG. **17**A is a top view of a semiconductor device of one embodiment of the present invention. FIG. **17**B to FIG. **17**D are cross-sectional views of a semiconductor device of one embodiment of the present invention.
(19) FIG. **18**A and FIG. **18**B are cross-sectional views of semiconductor devices of embodiments of the present invention.
(20) FIG. **19** is a cross-sectional view illustrating a structure of a storage device of one embodiment of the present invention.
(21) FIG. **20** is a cross-sectional view illustrating a structure of a storage device of one embodiment of the present invention.
(22) FIG. **21** is a cross-sectional view illustrating a structure of a storage device of one embodiment of the present invention.
(23) FIG. **22** is a cross-sectional view of a semiconductor device of one embodiment of the present invention.
(24) FIG. **23**A and FIG. **23**B are cross-sectional views of a semiconductor device of one embodiment of the present invention.
(25) FIG. **24** is a cross-sectional view of a semiconductor device of one embodiment of the present invention.
(26) FIG. **25** is a cross-sectional view of a semiconductor device of one embodiment of the present invention.
(27) FIG. **26**A is a block diagram illustrating a structure example of a storage device of one embodiment of the present invention. FIG. **26**B is a perspective view illustrating a structure example of a storage device of one embodiment of the present invention.
(28) FIG. **27**A to FIG. **27**H are circuit diagrams illustrating configuration examples of storage devices of embodiments of the present invention.
(29) FIG. **28** is a view illustrating a hierarchy of various kinds of storage devices.
(30) FIG. **29**A and FIG. **29**B are schematic views of a semiconductor device of one embodiment of the present invention.
(31) FIG. **30**A and FIG. **30**B are views illustrating examples of electronic components.
(32) FIG. **31**A to FIG. **31**E are schematic views of storage devices of embodiments of the present invention.
(33) FIG. **32**A to FIG. **32**H are view illustrating electronic devices of embodiments of the present invention.
(34) FIG. **33**A is a view illustrating a structure of a sample for cross-sectional observation. FIG. **33**B and FIG. **33**C are views each showing a selected area diffraction pattern of a sample for cross-sectional observation.
(35) FIG. **34** is a view showing a bright field image of a sample for cross-sectional observation.
(36) FIG. **35**A is a view showing a selected area diffraction pattern of a sample for cross-sectional observation. FIG. **35**B to FIG. **35**D are views showing a dark field image near the surface of the metal oxide film utilizing the spot of a diffraction wave (009).
(37) FIG. **36**A is a view showing a selected area diffraction pattern of a sample for cross-sectional observation. FIG. **36**B to FIG. **36**D are views each showing a dark field image near the silicon oxide film utilizing the spot of a diffraction wave (009).
(38) FIG. **37**A is a view showing a selected area diffraction pattern of a sample for plan-view observation. FIG. **37**B to FIG. **37**D are views each showing a dark field image of the metal oxide film utilizing the spot of a diffraction wave (100).

MODE FOR CARRYING OUT THE INVENTION

(39) Hereinafter, embodiments will be described with reference to the drawings. However, the embodiments can be implemented with many different modes, and it will be readily appreciated by those skilled in the art that modes and details thereof can be changed in various ways without departing from the spirit and scope thereof. Thus, the present invention should not be interpreted as

being limited to the following description of the embodiments.
(40) In the drawings, the size, the layer thickness, or the region is exaggerated for clarity in some cases. Therefore, the size, the layer thickness, or the region is not necessarily limited to the illustrated scale. Note that the drawings are schematic views showing ideal examples, and embodiments of the present invention are not limited to shapes or values shown in the drawings. For example, in the actual manufacturing process, a layer, a resist mask, or the like might be unintentionally reduced in size by treatment such as etching, which might not be reflected in the drawings for easy understanding of the invention. Furthermore, in the drawings, the same reference numerals are used in common for the same portions or portions having similar functions in different drawings, and repeated description thereof is omitted in some cases. Furthermore, the same hatch pattern is used for the portions having similar functions, and the portions are not especially denoted by reference numerals in some cases.
(41) Furthermore, especially in a top view (also referred to as a “plan view”), a perspective view, or the like, the description of some components might be omitted for easy understanding of the invention. The description of some hidden lines and the like might be omitted.
(42) In this specification and the like, the ordinal numbers such as first and second are used for convenience and do not denote the order of steps or the stacking order of layers. Therefore, for example, description can be made even when “first” is replaced with “second”, “third”, or the like, as appropriate. In addition, the ordinal numbers in this specification and the like do not correspond to the ordinal numbers which are used to specify one embodiment of the present invention in some cases.
(43) In this specification and the like, terms for describing arrangement, such as “over” and “under”, are used for convenience in describing a positional relation between components with reference to drawings. The positional relation between components is changed as appropriate in accordance with a direction in which the components are described. Thus, without limitation to terms described in this specification, the description can be changed appropriately depending on the situation.
(44) When this specification and the like explicitly state that X and Y are connected, for example, the case where X and Y are electrically connected, the case where X and Y are functionally connected, and the case where X and Y are directly connected are regarded as being disclosed in this specification and the like. Accordingly, without being limited to a predetermined connection relation, for example, a connection relation shown in drawings or text, a connection relation other than one shown in drawings or text is regarded as being disclosed in the drawings or the text. Here, X and Y each denote an object (e.g., a device, an element, a circuit, a wiring, an electrode, a terminal, a conductive film, or a layer).
(45) In this specification and the like, a transistor is an element having at least three terminals of a gate, a drain, and a source. In addition, the transistor includes a region where a channel is formed (hereinafter also referred to as a channel formation region) between the drain (a drain terminal, a drain region, or a drain electrode) and the source (a source terminal, a source region, or a source electrode), and current can flow between the source and the drain through the channel formation region. Note that in this specification and the like, a channel formation region refers to a region through which a current mainly flows.
(46) Furthermore, functions of a source and a drain might be switched when a transistor of opposite polarity is employed or a direction of current flow is changed in circuit operation, for example. Therefore, the terms “source” and “drain” can sometimes be interchanged with each other in this specification and the like.
(47) Note that a channel length refers to, for example, a distance between a source (a source region or a source electrode) and a drain (a drain region or a drain electrode) in a region where a semiconductor (or a portion where current flows in a semiconductor when a transistor is in an on state) and a gate electrode overlap with each other or a channel formation region in a top view of

the transistor. Note that in one transistor, channel lengths in all regions do not necessarily have the same value. In other words, the channel length of one transistor is not fixed to one value in some cases. Thus, in this specification, the channel length is any one of the values, the maximum value, the minimum value, or the average value in a channel formation region.

(48) A channel width refers to, for example, the length of a region where a semiconductor (or a portion where a current flows in a semiconductor when a transistor is on) and a gate electrode overlap with each other or a channel formation region in a direction perpendicular to a channel length direction in a top view of the transistor. Note that in one transistor, channel widths in all regions do not necessarily have the same value. In other words, the channel width of one transistor is not fixed to one value in some cases. Thus, in this specification, the channel width is any one of the values, the maximum value, the minimum value, or the average value in a channel formation region.

(49) Note that in this specification and the like, depending on the transistor structure, a channel width in a region where a channel is actually formed (hereinafter also referred to as an “effective channel width”) is sometimes different from a channel width shown in a top view of a transistor (hereinafter also referred to as an “apparent channel width”). For example, when a gate electrode covers a side surface of a semiconductor, an effective channel width is greater than an apparent channel width, and its influence cannot be ignored in some cases. For example, in a miniaturized transistor whose gate electrode covers a side surface of a semiconductor, the proportion of a channel formation region formed in the side surface of the semiconductor is increased in some cases. In that case, the effective channel width is greater than the apparent channel width.

(50) In such a case, the effective channel width is sometimes difficult to estimate by actual measurement. For example, estimation of an effective channel width from a design value requires assumption that the shape of a semiconductor is known. Accordingly, in the case where the shape of a semiconductor is not known accurately, it is difficult to measure the effective channel width accurately.

(51) In this specification, the simple term “channel width” refers to apparent channel width in some cases. Alternatively, in this specification, the simple term “channel width” refers to effective channel width in some cases. Note that values of channel length, channel width, effective channel width, apparent channel width, and the like can be determined, for example, by analyzing a cross-sectional TEM image and the like.

(52) Note that impurities in a semiconductor refer to, for example, elements other than the main components of the semiconductor. For example, an element with a concentration lower than 0.1 atomic % can be regarded as an impurity. When an impurity is contained, for example, the density of defect states in a semiconductor increases and the crystallinity decreases in some cases. In the case where the semiconductor is an oxide semiconductor, examples of an impurity which changes the characteristics of the semiconductor include Group 1 elements, Group 2 elements, Group 13 elements, Group 14 elements, Group 15 elements, transition metals other than the main components of the oxide semiconductor, and the like; hydrogen, lithium, sodium, silicon, boron, phosphorus, carbon, nitrogen, and the like are given as examples. Note that water also serves as an impurity in some cases. In addition, in the case of an oxide semiconductor, oxygen vacancies (referred to as V.sub.O in some cases) are formed by entry of impurities in some cases, for example.

(53) Note that in this specification and the like, silicon oxynitride is a material that contains more oxygen than nitrogen in its composition. Moreover, silicon nitride oxide is a material that contains more nitrogen than oxygen in its composition.

(54) In this specification and the like, the term “insulator” can be replaced with an insulating film or an insulating layer. Furthermore, the term “conductor” can be replaced with a conductive film or a conductive layer. Moreover, the term “semiconductor” can be replaced with a semiconductor film or a semiconductor layer.

(55) In this specification and the like, "parallel" indicates a state where two straight lines are placed at an angle greater than or equal to −10° and less than or equal to 10°. Accordingly, the case where the angle is greater than or equal to −5° and less than or equal to 5° is also included. Furthermore, "substantially parallel" indicates a state where two straight lines are placed at an angle greater than or equal to −30° and less than or equal to 30°. Moreover, "perpendicular" indicates a state where two straight lines are placed at an angle greater than or equal to 80° and less than or equal to 100°. Accordingly, the case where the angle is greater than or equal to 85° and less than or equal to 95° is also included. Moreover, "substantially perpendicular" indicates a state where two straight lines are placed at an angle greater than or equal to 60° and less than or equal to 120°.
(56) In this specification and the like, a metal oxide is an oxide of metal in a broad sense. Metal oxides are classified into an oxide insulator, an oxide conductor (including a transparent oxide conductor), an oxide semiconductor (also simply referred to as an OS), and the like. For example, in the case where a metal oxide is used in a semiconductor layer of a transistor, the metal oxide is referred to as an oxide semiconductor in some cases. That is, an OS transistor can also be called a transistor including a metal oxide or an oxide semiconductor.
(57) In this specification and the like, normally off means drain current per micrometer of channel width flowing through a transistor being 1×10.sup.−20 A or less at room temperature, 1×10.sup.−18 A or less at 85° C., or 1×10.sup.−16 A or less at 125° C. when a potential is not applied to a gate or a ground potential is applied to the gate.
Embodiment 1
(58) In this embodiment, an example of a semiconductor device including a transistor of one embodiment of the present invention will be described.
(59) <Structure Example 1 of Transistor>
(60) FIG. **1**A to FIG. **1**C are cross-sectional views of transistors of embodiments of the present invention.
(61) As illustrated in FIG. **1**A, a transistor of one embodiment of the present invention includes a conductor **205** disposed over a substrate (not illustrated), an insulator **224** disposed over the conductor **205**, an oxide **230***a* disposed over the insulator **224**, an oxide **230***b* disposed over the oxide **230***a*, a conductor **242***a* and a conductor **242***b* disposed over the oxide **230***b*, an insulator **250** disposed over the oxide **230***b* and between the conductor **242***a* and the conductor **242***b*, and a conductor **260** disposed over the insulator **250**.
(62) The conductor **260** functions as a first gate (also referred to as a top gate) electrode, and the conductor **205** functions as a second gate (also referred to as a back gate) electrode. The insulator **250** functions as a first gate insulator, and the insulator **224** functions as a second gate insulator. The conductor **242***a* functions as one of a source and a drain, and the conductor **242***b* functions as the other of the source and the drain. The oxide **230***b* functions as a channel formation region. Note that a channel formation region of the transistor is formed in the oxide **230***b* in the vicinity of the interface with the insulator **250**. The channel formation region may be formed in the oxide **230***a*.
(63) The transistor described above includes a top gate and a back gate. Applying different potentials to the top gate and the back gate can adjust the threshold voltage of the transistor including the top gate and the back gate. Applying a negative potential to the back gate can increase the threshold voltage of the transistor, which can reduce the off-state current, for example. In other words, the drain current at the time when the potential applied to the top gate is 0 V can be reduced by applying a negative potential to the back gate.
(64) In the transistor, a metal oxide functioning as a semiconductor (hereinafter, also referred to as an oxide semiconductor) is preferably used in a channel formation region. The transistor including an oxide semiconductor in the channel formation region has an extremely small leakage current in a non-conduction state; hence, a low-power semiconductor device can be provided. An oxide semiconductor can be deposited by a sputtering method or the like, and thus can be used for a transistor included in a highly integrated semiconductor device.

(65) In the case where an oxide semiconductor is used in the channel formation region of a transistor, an i-type (intrinsic) or substantially i-type oxide semiconductor with a low carrier concentration is preferably used. When the oxide semiconductor with a low carrier concentration is used in the channel formation region of a transistor, the off-state current of the transistor can be kept low or the reliability of the transistor can be improved. Note that an oxide semiconductor will be described in detail in Embodiment 2.
(66) In the transistor of one embodiment of the present invention, in a cross-sectional view of the transistor in the channel length direction, it is preferable that a groove portion (also referred to as a trench, an opening, or the like in some cases) be provided in the oxide **230***b* and the insulator **250** and the conductor **260** be embedded in the groove portion. At this time, the insulator **250** is provided to cover the inner wall (a sidewall and a bottom surface) of the groove portion and the conductor **260** is provided to fill the groove portion with the insulator **250** therebetween.
(67) In the transistor of one embodiment of the present invention, a hollow-shaped portion may be provided in the oxide **230***b* and the insulator **250** and the conductor **260** may be embedded in the hollow-shaped portion. Alternatively, a top surface of a region of the oxide **230***b* which overlaps with neither the conductor **242***a* nor the conductor **242***b* may be lower than a top surface of a region of the oxide **230***b* which overlaps with the conductor **242***a* or the conductor **242***b*, and the insulator **250** and the conductor **260** may be provided over the region of the oxide **230***b* which overlaps with neither the conductor **242***a* nor the conductor **242***b.*
(68) With the above structures, the channel length can be effectively longer than the channel length in a plan view of the transistor. Accordingly, an effective channel length can be long while the distance between a source and a drain is kept short. Therefore, a semiconductor device with a reduced short-channel effect and favorable electrical characteristics can be provided. In addition, a semiconductor device that can be miniaturized or highly integrated can be provided.
(69) The transistor with any of the above structures can have high controllability by the back gate compared with a transistor with a structure in which a groove portion is not provided in the oxide **230***b* (also referred to as a planar structure in some cases). Thus, a semiconductor device with less variations in transistor characteristics can be provided. In addition, a highly reliable semiconductor device can be provided.
(70) Note that a portion between a sidewall of the groove portion and a bottom surface of the groove portion (also referred to as an end portion of the bottom surface of the groove portion or a lower end portion of the sidewall of the groove portion) may be bent or may have a curvature. The oxide **230***b* may have a hollow curved shape.
(71) When the insulator **250** is provided in the groove portion whose end portion of the bottom surface has a curvature, in a cross-sectional view of the transistor of one embodiment of the present invention in the channel length direction, at least part of the lower portion of the insulator **250** has a curvature. Alternatively, the insulator **250** has a curved shape which is convex downward. In some cases, the conductor **260** fills the groove portion with the insulator **250** therebetween, so that at least part of lower portion of the conductor **260** has a curvature. Alternatively, the conductor **260** on the oxide **230***b* side has a curved shape projecting downward in some cases. Note that when the lower portion of the insulator **250** has a small curvature, the lower portion of the conductor **260** on the oxide **230***b* side does not have a curvature in some cases. Alternatively, the conductor **260** on the oxide **230***b* side has a curved shape which is convex downward in some cases.
(72) Here, as illustrated in FIG. **1**A, in a cross-sectional view of the transistor in the channel length direction, the center of curvature at the end portion of the bottom surface of the groove portion is denoted by C and the radius of the curvature is denoted by R. At this time, the end portion of the bottom surface of the groove portion is expressed as being curved with the radius of curvature R in some cases.
(73) In the cross-sectional view of the transistor in the channel length direction, the depth of the groove portion provided in the oxide **230***b* is called D**1**, as illustrated in FIG. **1**A. Note that the

depth D**1** is also a difference between the top surface of the region of the oxide **230***b* which overlaps with the conductor **242***a* or the conductor **242***b* and the top surface of the region of the oxide **230***b* which overlaps with the conductor **260**.
(74) The depth D**1** is preferably greater than 0 nm and more preferably larger than the thickness of the insulator **250**. Specifically, the depth D**1** is preferably greater than 0 nm and less than or equal to 100 nm, more preferably greater than or equal to 2 nm and less than or equal to 50 nm, more preferably greater than or equal to 5 nm and less than or equal to 30 nm. With this structure, an electric field of the drain is blocked by the gate electrode, whereby a short-channel effect can be reduced and a semiconductor device with favorable electrical characteristics can be provided. A semiconductor device that can be miniaturized or highly integrated can be provided.
(75) As illustrated in FIG. **1**A, in the cross-sectional view of the transistor in the channel length direction, the thickness (film thickness) of the oxide **230***b* in a region overlapping with the conductor **260** is called D**2**. That is, the thickness of the oxide **230***b* in a region overlapping with the conductor **242***a* or the conductor **242***b* is the sum of the depth D**1** and the thickness D**2**. A difference between a bottom surface of the conductor **242***a* or the conductor **242***b* and a bottom surface of the conductor **260** is called D**3**.
(76) In the transistor of one embodiment of the present invention, the bottom surface of the conductor **260** may be lower than the bottom surface of the conductor **242***a* or the conductor **242***b*. That is, D**3** may be greater than 0 nm. Accordingly, the electric field of the conductor **260** functioning as the first gate electrode easily acts on the oxide **230***b* in the vicinity of a sidewall of the groove portion. Thus, the on-state current of the transistor can be increased and the frequency characteristics can be improved.
(77) As illustrated in FIG. **1**A, in the cross-sectional view of the transistor in the channel length direction, the width of the groove portion provided in the oxide **230***b* is called L**1**. Note that the width L**1** is also a distance between the conductor **242***a* and the conductor **242***b*. In the bottom surface of the groove portion provided in the oxide **230***b*, the length of a region that is not curved is called L**2**.
(78) The radius of curvature R is preferably large. For example, it is preferable that the radius of curvature R be larger than 0 nm and smaller than or equal to the width L**1** or smaller than or equal to the depth D**1**. When the radius of curvature is larger than 0 nm, that is, the end portion of the bottom surface of the groove portion has a curvature, the coverage of the groove portion with the insulator **250** and the conductor **260**, which are to be formed in a later step, can be improved.
(79) As illustrated in FIG. **1**B, in the cross-sectional view of the transistor in the channel length direction, the groove portion provided in the oxide **230***b* may have a tapered shape. Due to the tapered shape of the groove portion, even when the radius of curvature R is small, the coverage of the groove portion with the insulator **250** and the conductor **260**, which are to be formed in a later step, can be improved.
(80) Note that a side surface of the conductor **242***a* facing to the conductor **242***b* and a side surface of the conductor **242***b* facing to the conductor **242***a* may each have a tapered shape with respect to the substrate surface. At this time, the side surface of the conductor **242***a* may be substantially aligned with a sidewall of the groove portion in the oxide **230***b*. The side surface of the conductor **242***b* may be substantially aligned with the sidewall of the groove portion in the oxide **230***b*.
(81) As illustrated in FIG. **1**C, in the cross-sectional view of the transistor in the channel length direction, an end portion of the bottom surface of the groove portion provided in the oxide **230***b* may have a pointed shape. Note that the shape has a radius of curvature of 0 and the curvature cannot be defined. In the case where the transistor in which the end portion of the bottom surface of the groove portion has a curvature and the transistor in which the end portion of the bottom surface of the groove portion has a pointed shape have the same depth D**1** of the groove portion, the effective channel length in the transistor having a pointed shape can be longer than that in the transistor in which the end portion of the bottom surface of the groove portion has a curvature.

Therefore, a semiconductor device with a reduced short-channel effect and favorable electrical characteristics can be provided. In addition, a semiconductor device that can be miniaturized or highly integrated can be provided.
(82) Note that in this specification, regardless of the shape of the end portion of the bottom surface of the groove portion, a structure in which the insulator **250** functioning as the first gate insulator and the conductor **260** functioning as the first gate electrode are embedded in the groove portion in the oxide **230***b* is called a U-shaped structure in some cases. In the above structure, a channel whose effective channel length is ensured by having a broken line shape or a curved shape is called a buried channel, a recessed channel, or the like in some cases. The shape is called U-shaped in some cases. In this case, in the transistor having the groove portion in the oxide **230***b*, the channel shape is U-shaped.
(83) Electrical characteristics of a transistor having a groove portion in the oxide **230***b* (also referred to as a transistor with a U-shaped structure), which are obtained from calculation with a device simulator, are described below. Note that in the following description, calculation is also performed on a transistor in which a groove portion is not provided in the oxide **230***b* (also referred to as a transistor with a planar structure) for comparison.
(84) <Comparison of Electrical Characteristics Between Transistor with U-Shaped Structure and Transistor with Planar Structure>
(85) First, electrical characteristics of the transistor with a U-shaped structure and the transistor with a planar structure are compared by calculation with a device simulator. Specifically, the shift voltages (Vsh), the maximum values of gm, the S values, and DIBL (Drain-Induced Barrier Lowering) of the transistors are calculated as electrical characteristics of the transistors. Note that structures of the transistors simulated in the device simulator calculation are the same as the structure of the transistor illustrated in FIG. **1**A.
(86) Here, Vsh is defined as, in the drain current-gate voltage characteristics (Id-Vg characteristics) of a transistor, Vg at which the tangent line at a point where the slope of the Id-Vg curve is the steepest intersects the straight line of Id=1 pA. Furthermore, gm is the amount of change in drain current with respect to the amount of change in gate voltage Vg, and defined as ∂Id/∂Vg. Note that the unit of gm is [S].
(87) DIBL is a phenomenon in which the threshold voltage at high drain voltage decreases (shifts to the negative direction) when the channel length is shortened, and is a kind of short-channel effect. The phenomenon is thought to occur because an influence of a drain electric field on a gate electric field cannot be ignored when the channel length is shortened and a potential barrier for carriers between a source and a drain is likely to decrease due to the drain electric field.
(88) Note that in the following description, DIBL is a value obtained by subtracting the value of Vsh at a drain voltage Vd=1.2 V from the value of Vsh at a drain voltage Vd=0.1 V.
(89) In this calculation, transistors with a U-shaped structure (a transistor **1**A to a transistor **9**A) having different depths D**1** of the groove portions each provided in the oxide **230***b* and transistors with a planar structure (a transistor **1**B to a transistor **9**B) having different distances (corresponding to the widths L**1**) between the conductor **242***a* and the conductor **242***b* were prepared.
(90) Of parameter values set in the calculation with the device simulator, Table 1 shows the parameter values which are different among the transistor **1**A to the transistor **9**A.
(91) TABLE-US-00001 TABLE 1 Tran- Depth D1 Effective channel sistor [nm] length [nm] 1A 10 31.4 2A 15 41.4 3A 20 51.4 4A 25 61.4 5A 30 71.4 6A 35 81.4 7A 40 91.4 8A 45 101.4 9A 50 111.4 10A 0 20
(92) Since each of the depths D**1** of the transistor **1**A to the transistor **9**A is greater than 0 nm, the transistor **1**A to the transistor **9**A are transistors with a U-shaped structure. In the transistor **1**A to the transistor **9**A, the width L**1** of the groove portion is 20 nm and the radius of curvature R is 10 nm. At this time, the length L**2** is 0 nm. The thickness D**2** is 15 nm. When the thickness of the insulator **250** is 5 nm, the width of the conductor **260** is 10 nm.

(93) The effective channel length is the length of an interface between the insulator **250** and the oxide **230***b*. That is, the effective channel length of each of the transistor **1**A to the transistor **9**A is calculated from the depth D**1** and the radius of curvature R. Specifically, the effective channel length of each of the transistor **1**A to the transistor **9**A is 2(D**1**−R)+πR.
(94) Of parameter values set in the calculation with the device simulator, Table 2 shows the parameter values which are different among the transistor **1**B to the transistor **9**B.
(95) TABLE-US-00002 TABLE 2 Tran- Length L1 Effective channel sistor [nm] length [nm] 1B 31.4 31.4 2B 41.4 41.4 3B 51.4 51.4 4B 61.4 61.4 5B 71.4 71.4 6B 81.4 81.4 7B 91.4 91.4 8B 101.4 101.4 9B 111.4 111.4
(96) The thickness of the insulator **250** in each of the transistor **1**B to the transistor **9**B is 5 nm, and the thickness of the conductor **260** is 15 nm. The thickness D**2** is 15 nm. Note that the effective channel length is the length of the interface between the insulator **250** and the oxide **230***b*. That is, the effective channel length of each of the transistor **1**B to the transistor **9**B is the length L**1**.
(97) The calculation with a device simulator was performed on the transistor **1**A to the transistor **9**A and the transistor **1**B to the transistor **9**B, so that electrical characteristics of the transistors were obtained. A device simulator Atlas manufactured by Silvaco, Inc. was used as the device simulator. Of parameter values set in the calculation with the device simulator, Table 3 shows the parameter values which are common among the transistor **1**A to the transistor **9**A and the transistor **1**B to the transistor **9**B.
(98) TABLE-US-00003 TABLE 3 Tran- sistor Channel width 1 μm 260 Work function 5.0 eV 250 Relative permittivity 4.1 Film thickness 5 nm 242a Work function 4.8 eV 242b Length 100 nm Film thickness 20 nm 230a Relative permittivity 15 230b Effective density of states in the 5 × 10.sup.18 cm.sup.−3 conduction band Nc Effective density of states in the 5 × 10.sup.18 cm.sup.−3 valence band Nv 230b Electron affinity 4.8 eV Band gap 2.9 eV Electron mobility 15 cm.sup.2/ (V s) Hole mobility 0.01 cm.sup.2/ (V s) Thickness D2 15 nm 230a Electron affinity 4.5 eV Band gap 3.4 eV Electron mobility 5 cm.sup.2/ (V s) Hole mobility 0.01 cm.sup.2/ (V s) Film thickness 5 nm 224 Relative permittivity 4.1 Film thickness 30 nm 205 Work function 5.0 eV Film thickness 20 nm
(99) The Id-Vg characteristics of the transistor **1**A to the transistor **9**A and the transistor **1**B to the transistor **9**B when drain voltage Vd=1.2 V, back gate voltage Vbg=0.0 V, and source voltage Vs=0.0 V are calculated to obtain the Vsh, the maximum value of gm, and the S value. In order to obtain DIBL, the Id-Vg characteristics when drain voltage Vd=0.1 V, back-gate voltage Vbg=0.0 V, and source voltage Vs=0.0 V are also calculated.
(100) FIG. **2**A shows Vsh of the transistor **1**A to the transistor **9**A and the transistor **1**B to the transistor **9**B. In FIG. **2**A, the horizontal axis represents the effective channel length [nm] and the vertical axis represents the Vsh [mV]. Note that the Vsh values of the transistors with a U-shaped structure (the transistor **1**A to the transistor **9**A) are plotted with black rhombuses. The Vsh values of the transistors with a planar structure (the transistor **1**B to the transistor **9**B) are plotted with white triangles.
(101) According to FIG. **2**A, for example, the Vsh value of the transistor with a U-shaped structure (the transistor **2**A to the transistor **9**A) having an effective channel length of 41.4 nm or larger is larger than the Vsh value (=−3244 mV) of the transistor with a planar structure (the transistor **1**B) having a length L**1** of 31.4 nm. That is, in the transistor with a U-shaped structure whose length L**1** is 20 nm, by setting the depth D**1** to be greater than or equal to 15 nm, a decrease in Vsh can be suppressed as compared with the transistor **1**B. Accordingly, the transistor with a U-shaped structure can be miniaturized and can suppress a decrease in Vsh than the transistor with a planar structure. Note that the same applies to the other effective channel lengths.
(102) FIG. **2**B shows the maximum values of gm of the transistor **1**A to the transistor **9**A and the transistor **1**B to the transistor **9**B. In FIG. **2**B, the horizontal axis represents the effective channel length [nm] and the vertical axis represents the maximum value of gm [S]. Note that the maximum

values of gm of the transistors with a U-shaped structure (the transistor **1**A to the transistor **9**A) are plotted with black rhombuses. The maximum values of gm of the transistors with a planar structure (the transistor **1**B to the transistor **9**B) are plotted with white triangles.

(103) It is found from FIG. **2**B that the maximum value of gm of the transistor with a U-shaped structure is larger than that of the transistor with a planar structure when the transistors have the same effective channel length. Thus, when the transistor has a U-shaped structure, the value of gm can be improved and the on-state current can be improved.

(104) FIG. **2**C shows calculated S values of the transistor **1**A to the transistor **9**A and the transistor **1**B to the transistor **9**B. In FIG. **2**C, the horizontal axis represents the effective channel length [nm] and the vertical axis represents the S value [mV/dec.]. Note that the S values of the transistors with a U-shaped structure (the transistor **1**A to the transistor **9**A) are plotted with black rhombuses. The S values of the transistors with a planar structure (the transistor **1**B to the transistor **9**B) are plotted with white triangles. On the basis of discussion similar to the Vsh results shown in FIG. **2**A, by employing the U-shaped structure, a transistor can be miniaturized and an increase in an S value can be suppressed as compared with the planar structure.

(105) FIG. **3** shows calculated DIBL of the transistor **1**A to the transistor **9**A and the transistor **1**B to the transistor **9**B. In FIG. **3**, the horizontal axis represents the effective channel length [nm] and the vertical axis represents DIBL [mV]. Note that DIBL of the transistors with a U-shaped structure (the transistor **1**A to the transistor **9**A) is plotted with black rhombuses. The DIBL of the transistors with a planar structure (the transistor **1**B to the transistor **9**B) is plotted with white triangles. On the basis of discussion similar to the Vsh results shown in FIG. **2**A, by employing the U-shaped structure, a transistor can be miniaturized and an increase in DIBL can be suppressed as compared with the planar structure.

(106) Next, in order to evaluate the controllability by the back gate of the transistors with a U-shaped structure and the transistors with a planar structure when they have the same effective channel length, the dependence of Id-Vg characteristics of the transistors on the back-gate voltage are calculated to obtain ∂Vsh/∂Vbg.

(107) Note that ∂Vsh/∂Vbg is the amount of change in Vsh when the back gate voltage Vbg changes by 1 V, and expressed in the unit of [V/V]. That is, the larger the absolute value of ∂Vsh/∂Vbg is, the larger the amount of change in Vsh with respect to the amount of change in potential applied to the back gate is. Thus, the larger the absolute value of ∂Vsh/∂Vbg is, the higher the controllability by the back gate is.

(108) The dependence of Id-Vg characteristics on the back gate voltage of the transistor **1**A to the transistor **9**A and the transistor **1**B to the transistor **9**B when drain voltage is 1.2 V and source voltage Vs=0.0 V were calculated to obtain ∂Vsh/∂Vbg. Note that the values of the parameters of the transistor **1**A to the transistor **9**A and the transistor **1**B to the transistor **9**B which were set in the calculation with the device simulator are shown in Table 1 to Table 3.

(109) FIG. **4** shows ∂Vsh/∂Vbg calculated from the transistor **1**A to the transistor **9**A and the transistor **1**B to the transistor **9**B. In FIG. **4**, the horizontal axis represents the effective channel length [nm] and the vertical axis represents ∂Vsh/∂Vbg [V/V] when Vbg=0 V. Note that the ∂Vsh/∂Vbg values of the transistors with a U-shaped structure (the transistor **1**A to the transistor **9**A) are plotted with black rhombuses. The ∂Vsh/∂Vbg values of the transistors with a planar structure (the transistor **1**B to the transistor **9**B) are plotted with white triangles.

(110) From FIG. **4**, it is found that the absolute value of ∂Vsh/∂Vbg of the transistor with a U-shaped structure is larger than that of the transistor with a planar structure when they have the same effective channel length. Accordingly, the controllability by the back gate of the transistor with a U-shaped structure is higher than that of the transistor with a planar structure when they have the same channel length.

(111) <Dependence of Electrical Characteristics of Transistor on Depth D**1** of Groove Portion>

(112) Next, the dependence of electrical characteristics of transistors on the depth D**1** of the groove

portion provided in the oxide **230***b* is evaluated by calculation with a device simulator. Specifically, the Vsh, the maximum values of gm, and the S values of the transistors are calculated. Note that structures of the transistors simulated in the device simulator calculation are the same as the structure of the transistor illustrated in FIG. **1**A. The values of parameters of the transistor **1**A to a transistor **10**A used in the calculation with a device simulator are shown in Table 1 and Table 3.
(113) Note that the depth D**1** of the transistor **10**A is 0 nm. That is, the transistor **10**A is a transistor with a planar structure in which a groove portion is not provided in the oxide **230***b*. The effective channel length at this time is equal to the length L**1**. When the thickness of the insulator **250** is 5 nm, the thickness of the conductor **260** is 15 nm.
(114) The Id-Vg characteristics of the transistor **1**A to the transistor **10**A when drain voltage Vd=0.1 V or 1.2 V, back gate voltage Vbg=0.0 V, and source voltage Vs=0.0 V are calculated to obtain the Vsh, the maximum value of gm, the S value, and DIBL.
(115) FIG. **5**A shows the calculated Vsh of each of the transistor **1**A to the transistor **10**A. In FIG. **5**A, the horizontal axis represents the depth D**1** [nm] and the vertical axis represents Vsh [mV]. Note that the values of Vsh at a drain voltage Vd of 0.1 V are plotted with black circles. The values of Vsh at a drain voltage Vd of 1.2 V are plotted with white squares. As can be seen from FIG. **5**A, as the depth D**1** is larger, Vsh is larger. Thus, when the transistor has a U-shaped structure, a decrease in Vsh can be suppressed.
(116) FIG. **5**B shows the calculated maximum value of gm of each of the transistor **1**A to the transistor **10**A. Note that the values of the maximum values of gm at a drain voltage Vd of 0.1 V are plotted with black circles. The maximum values of gm at a drain voltage Vd of 1.2 V are plotted with white squares. In FIG. **5**B, the horizontal axis represents the depth D**1** [nm] and the vertical axis represents the maximum value [S] of gm.
(117) FIG. **5**C shows the calculated S value of each of the transistor **1**A to the transistor **10**A. In FIG. **5**C, the horizontal axis represents the depth D**1** [nm] and the vertical axis represents the S value [mV/dec.]. Note that the S values at a drain voltage Vd of 0.1 V are plotted with black circles. The S values at a drain voltage Vd of 1.2 V are plotted with white squares. As can be seen from FIG. **5**C, as the depth D**1** is larger, the S value is smaller. Thus, when the transistor has a U-shaped structure, an increase in the S value can be suppressed.
(118) FIG. **6** shows the calculated DIBL of each of the transistor **1**A to the transistor **10**A. In FIG. **6**, the horizontal axis represents the depth D**1** [nm] and the vertical axis represents DIBL [mV]. As can be seen from FIG. **6**, as the depth D**1** is larger, DIBL is smaller. Thus, when the transistor has a U-shaped structure, an increase in DIBL can be suppressed.
(119) Accordingly, among transistors having the same area occupied by a transistor, the transistor with a U-shaped structure can ensure the effective channel length by increasing the depth L**1** of the groove portion provided in the oxide **230***b*; thus, the shift of Vsh in the negative direction, an increase in the S value, an increase in DIBL, and the like can be suppressed and the short-channel effect can be reduced.
(120) According to one embodiment of the present invention, a semiconductor device that can be miniaturized or highly integrated can be provided. A semiconductor device having favorable electrical characteristics can be provided. A semiconductor device with less variations in transistor characteristics can be provided. A highly reliable semiconductor device can be provided. A semiconductor device with a high on-state current can be provided. A semiconductor device with low power consumption can be provided.
(121) The structure, the method, and the like described above in this embodiment can be used in an appropriate combination with the structures, the methods, and the like described in the other embodiments and examples.
Embodiment 2
(122) In this embodiment, an example of a semiconductor device including a transistor **200** of one embodiment of the present invention and a manufacturing method thereof will be described. Note

that in the transistor **200** described in this embodiment, components having the same function as the components in the transistor described the above embodiment are denoted by the same reference numerals.

(123) <Structure Example 2 of Semiconductor Device>

(124) FIG. **7**A to FIG. **7**D are a top view and cross-sectional views of a semiconductor device including the transistor **200** of one embodiment of the present invention. FIG. **7**A is a top view of the semiconductor device. FIG. **7**B to FIG. **7**D are cross-sectional views of the semiconductor device. Here, FIG. **7**B is a cross-sectional view of a portion indicated by the dashed-dotted line A**1**-A**2** in FIG. **7**A, and is a cross-sectional view in the channel length direction of the transistor **200**. FIG. **7**C is a cross-sectional view of a portion indicated by the dashed-dotted line A**3**-A**4** in FIG. **7**A, and is a cross-sectional view in the channel width direction of the transistor **200**. FIG. **7**D is a cross-sectional view of a portion indicated by the dashed-dotted line A**5**-A**6** in FIG. **7**A. Note that for clarity of the drawing, some components are not illustrated in the top view of FIG. **7**A.

(125) The semiconductor device of one embodiment of the present invention includes an insulator **211** over a substrate (not illustrated), an insulator **212** over the insulator **211**, an insulator **214** over the insulator **212**, the transistor **200** over the insulator **214**, an insulator **280** over the transistor **200**, an insulator **282** over the insulator **280**, an insulator **283** over the insulator **282**, and an insulator **284** over the insulator **283**. The insulator **211**, the insulator **212**, the insulator **214**, the insulator **280**, the insulator **282**, the insulator **283**, and the insulator **284** function as interlayer films. A conductor **240** (a conductor **240***a* and a conductor **240***b*) that is electrically connected to the transistor **200** and functions as a plug is also included. Note that an insulator **241** (an insulator **241***a* and an insulator **241***b*) is provided in contact with side surfaces of the conductor **240** functioning as a plug. A conductor **246** (a conductor **246***a* and a conductor **246***b*) that is electrically connected to the conductor **240** and functions as a wiring is provided over the insulator **284** and the conductor **240**. An insulator **286** is provided over the conductor **246** and the insulator **284**.

(126) The insulator **241***a* is provided in contact with the inner wall of an opening in an insulator **272**, an insulator **273**, the insulator **280**, the insulator **282**, the insulator **283**, and the insulator **284**; a first conductor of the conductor **240***a* is provided in contact with a side surface of the insulator **241***a*; and a second conductor of the conductor **240***a* is provided on the inner side thereof. The insulator **241***b* is provided in contact with the inner wall of an opening in the insulator **272**, the insulator **273**, the insulator **280**, the insulator **282**, the insulator **283**, and the insulator **284**; a first conductor of the conductor **240***b* is provided in contact with a side surface of the insulator **241***b*; and a second conductor of the conductor **240***b* is provided on the inner side thereof. Here, the level of a top surface of the conductor **240** and the level of a top surface of the insulator **284** in a region overlapping with the conductor **246** can be substantially the same. Note that although the transistor **200** has a structure in which the first conductor of the conductor **240** and the second conductor of the conductor **240** are stacked, the present invention is not limited thereto. For example, the conductor **240** may be provided as a single layer or to have a stacked-layer structure of three or more layers. In the case where a structure body has a stacked-layer structure, layers may be distinguished by ordinal numbers corresponding to the formation order.

(127) [Transistor **200**]

(128) As illustrated in FIG. **7**A to FIG. **7**D, the transistor **200** includes an insulator **216** over the insulator **214**; a conductor **205** (a conductor **205***a* and a conductor **205***b*) disposed so as to be embedded in the insulator **214** or the insulator **216**; an insulator **222** over the insulator **216** and the conductor **205**; an insulator **224** over the insulator **222**; an oxide **230***a* over the insulator **224**; an oxide **230***b* over the oxide **230***a*; an oxide **243** (an oxide **243***a* and an oxide **243***b*) over the oxide **230***b*; an oxide **230***c*; a conductor **242***a* over the oxide **243***a*; a conductor **242***b* over the oxide **243***b*; an insulator **250** over the oxide **230***c*; a conductor **260** (a conductor **260***a* and a conductor **260***b*) that is positioned over the insulator **250** and overlaps with part of the oxide **230***c*; an insulator **272** in contact with part of a top surface of the insulator **224**, part of a side surface of the oxide **230***a*,

part of a side surface of the oxide **230***b*, part of a side surface of the oxide **243**, a side surface of the oxide **242***a*, a top surface of the conductor **242***a*, a side surface of the conductor **242***b*, and a top surface of the conductor **242***b*; and an insulator **273** over the insulator **272**. The oxide **230***c* is in contact with the side surface of the conductor **242***a* and the side surface of the conductor **242***b*. Here, as illustrated in FIG. 7B, a top surface of the conductor **260** is positioned to be substantially aligned with a top surface of the insulator **250** and a top surface of the oxide **230***c*. The insulator **282** is in contact with the top surfaces of the conductor **260**, the insulator **250**, the oxide **230***c*, and the insulator **280**.

(129) An opening reaching the oxide **230***b* is provided in the insulator **280**, the insulator **273**, and the insulator **272**. The oxide **230***c*, the insulator **250**, and the conductor **260** are provided in the opening. In addition, in the channel length direction of the transistor **200**, the conductor **260**, the insulator **250**, and the oxide **230***c* are provided between the conductor **242***a* and the conductor **242***b*. The insulator **250** includes a region overlapping with a side surface of the conductor **260** and a region overlapping with a bottom surface of the conductor **260**. The oxide **230***c* in a region overlapping with the oxide **230***b* includes a region in contact with the oxide **230***b*, a region overlapping with the side surface of the conductor **260** with the insulator **250** therebetween, and a region overlapping with the bottom surface of the conductor **260** with the insulator **250** therebetween.

(130) In a cross-sectional view of the transistor **200** in the channel length direction, a groove portion is provided in the oxide **230***b*, the oxide **230***c*, the insulator **250**, and the conductor **260** (the conductor **260***a* and the conductor **260***b*) are embedded in the groove portion. At this time, the oxide **230***c* is disposed to cover an inner wall (a sidewall and a bottom surface) of the groove portion, the insulator **250** is disposed to cover the inner wall of the groove portion with the oxide **230***c* positioned therebetween, and the conductor **260** is disposed to fill the groove portion with the oxide **230***c* and the insulator **250** positioned therebetween. In the cross-sectional view of the transistor **200** in the channel length direction, a sidewall of the groove portion is substantially aligned with a sidewall of the opening.

(131) Note that when the conductor **242***a* and the conductor **242***b* are processed, specifically etched, an upper portion of the oxide **230***b* might be slightly etched. In one embodiment of the present invention, however, in order to lengthen the effective channel length, the oxide **230***b* is processed using an insulator provided over the conductor **242***a* and the conductor **242***b* as a mask to form a groove portion. The depth D**1** of the groove portion is preferably larger than the thickness (film thickness) of the conductor **242***a* and the conductor **242***b* or the thickness (film thickness) of the insulator **250**, for example. Typically, the depth D**1** of the groove portion is greater than or equal to 5 nm and less than or equal to 50 nm, preferably greater than or equal to 10 nm and less than or equal to 30 nm. Note that the depth D**1** of the groove portion depends on the thickness of the conductor **242***a* and the conductor **242***b*, the thickness of the insulator **250**, the distance between the conductor **242***a* and the conductor **242***b*, and the like, and thus is not limited to the above values.

(132) The oxide **230***c* has a projecting shape downward. In particular, in the case where the end portion of the bottom surface of the groove portion has a curvature, the oxide **230***c* has a curved shape projecting downward.

(133) Note that as illustrated in FIG. **7**B, in the cross-sectional view of the transistor **200** in the channel length direction, the oxide **230***b* has a hollow curved shape. On the other hand, as illustrated in FIG. **7**C, the oxide **230***b* has a projecting curved shape in a cross-sectional view in the channel width direction of the transistor **200**. That is, the oxide **230***b* in a region in contact with the oxide **230***c* and a region in the vicinity thereof can be said to form a saddle-like shape. Note that the groove portion in the oxide **230***b* refers to such regions of the oxide **230***b* in some cases for simple description.

(134) As illustrated in FIG. **7**C, a curved surface is preferably provided between the side surface of

the oxide **230***b* and the top surface of the oxide **230***b* in a cross-sectional view of the transistor **200** in the channel width direction. That is, an end portion of the side surface and an end portion of the top surface are preferably curved (such a shape is hereinafter also referred to as a rounded shape).
(135) The radius of curvature of the curved surface is preferably greater than 0 nm and less than the thickness of the oxide **230***b* in a region overlapping with the conductor **242**, or less than the half of the length of a region that does not have the curved surface. Specifically, the radius of curvature of the curved surface is greater than 0 nm and less than or equal to 20 nm, preferably greater than or equal to 1 nm and less than or equal to 15 nm, and further preferably greater than or equal to 2 nm and less than or equal to 10 nm. With such a shape, an electric field is prevented from concentrating between the side surface and the top surface, so that variations in the transistor characteristics can be suppressed. Furthermore, a reduction in the length of the region that does not have the curved surface can be prevented, and decreases in the on-state current and mobility of the transistor **200** can be inhibited. Accordingly, a semiconductor device having favorable electrical characteristics can be provided.
(136) In the transistor **200**, a metal oxide functioning as a semiconductor (hereinafter, also referred to as an oxide semiconductor) is preferably used as the oxide **230** (the oxide **230***a*, the oxide **230***b*, and the oxide **230***c*) including a channel formation region.
(137) The metal oxide functioning as a semiconductor has a band gap of preferably 2 eV or higher, further preferably 2.5 eV or higher. With use of a metal oxide having such a wide bandgap, the off-state current of the transistor can be reduced.
(138) The transistor in which a metal oxide is used in its channel formation region has an extremely low leakage current in a non-conduction state; thus, a semiconductor device with low power consumption can be provided. The metal oxide can be deposited by a sputtering method or the like, and thus can be used for a transistor included in a highly integrated semiconductor device.
(139) For example, as the oxide **230**, a metal oxide such as an In-M-Zn oxide containing indium, an element M, and zinc (the element M is one or more kinds selected from aluminum, gallium, yttrium, tin, copper, vanadium, beryllium, boron, titanium, iron, nickel, germanium, zirconium, molybdenum, lanthanum, cerium, neodymium, hafnium, tantalum, tungsten, magnesium, and the like) is preferably used. An In—Ga oxide or an In—Zn oxide may be used for the oxide **230**.
(140) The oxide **230** preferably includes the oxide **230***a* positioned over the insulator **224**, the oxide **230***b* positioned over the oxide **230***a*, and the oxide **230***c* that is positioned over the oxide **230***b* and is at least partly in contact with the groove portion in the oxide **230***b*. Including the oxide **230***a* below the oxide **230***b* makes it possible to inhibit diffusion of impurities into the oxide **230***b* from the components formed below the oxide **230***a*. Moreover, including the oxide **230***c* over the oxide **230***b* makes it possible to inhibit diffusion of impurities into the oxide **230***b* from the components formed above the oxide **230***c*.
(141) Although a structure in which the oxide **230** has a three-layer stacked structure of the oxide **230***a*, the oxide **230***b*, and the oxide **230***c* in the transistor **200** is described, the present invention is not limited thereto. For example, the oxide **230** may be a single layer of the oxide **230***b* or have a two-layer structure of the oxide **230***a* and the oxide **230***b*, a two-layer structure of the oxide **230***b* and the oxide **230***c*, or a stacked-layer structure including four or more layers. Alternatively, each of the oxide **230***a*, the oxide **230***b*, and the oxide **230***c* may have a stacked-layer structure.
(142) Furthermore, the oxide **230***a* and the oxide **230***b* preferably contain the same element, other than oxygen, as its main component, and the oxide **230***b* and the oxide **230***c* preferably contain the same element, other than oxygen, as its main component. By this, the density of defect states at the interface between the oxide **230***a* and the oxide **230***b* and the interface between the oxide **230***b* and the oxide **230***c* can be made low. Thus, the influence of interface scattering on carrier conduction is small, and the transistor **200** can have high on-state current and excellent frequency characteristics.
(143) Note that the oxide **230** preferably has a stacked-layer structure using oxides with different chemical compositions. Specifically, the atomic ratio of the element M to metal elements of main

components in the metal oxide used for the oxide **230***a* is preferably greater than the atomic ratio of the element M to metal elements of main components in the metal oxide used for the oxide **230***b*. Moreover, the atomic ratio of the element M to In in the metal oxide used as the oxide **230***a* is preferably greater than the atomic ratio of the element M to In in the metal oxide used as the oxide **230***b*. Furthermore, the atomic ratio of In to the element M in the metal oxide used as the oxide **230***b* is preferably greater than the atomic ratio of In to the element Min the metal oxide used as the oxide **230***a*. A metal oxide that can be used as the oxide **230***a* or the oxide **230***b* can be used as the oxide **230***c*.
(144) Note that in order to increase the on-state current of the transistor **200**, an In—Zn oxide is preferably used as the oxide **230**. In the case where an In—Zn oxide is used as the oxide **230**, for example, a stacked-layer structure in which an In—Zn oxide is used as the oxide **230***a* and In-M-Zn oxides are used as the oxide **230***b* and the oxide **230***c*, or a stacked-layer structure in which an In-M-Zn oxide is used as the oxide **230***a* and an In—Zn oxide is used as one of the oxide **230***b* and the oxide **230***c* can be employed.
(145) The oxide **230***b* and the oxide **230***c* preferably have crystallinity. For example, a CAAC-OS (c-axis aligned crystalline oxide semiconductor) described later is preferably used. An oxide having crystallinity, such as a CAAC-OS, has a dense structure with small amounts of impurities and defects (e.g., oxygen vacancies) and high crystallinity. This can inhibit oxygen extraction from the oxide **230***b* by the source electrode or the drain electrode. This can reduce oxygen extraction from the oxide **230***b* even when heat treatment is performed; thus, the transistor **200** is stable with respect to high temperatures in a manufacturing process (what is called thermal budget).
(146) In addition, a CAAC-OS is preferably used for the oxide **230***c*; the c-axis of a crystal included in the oxide **230***c* is preferably aligned in a direction substantially perpendicular to the formation surface or the top surface of the oxide **230***c*. The CAAC-OS has a property of making oxygen move easily in the direction perpendicular to the c-axis. Thus, oxygen contained in the oxide **230***c* can be efficiently supplied to the oxide **230***b*.
(147) The conduction band minimum of each of the oxide **230***a* and the oxide **230***c* is preferably closer to the vacuum level than the conduction band minimum of the oxide **230***b*. In other words, the electron affinity of each of the oxide **230***a* and the oxide **230***c* is preferably smaller than the electron affinity of the oxide **230***b*. In that case, a metal oxide that can be used for the oxide **230***a* is preferably used for the oxide **230***c*. At this time, the oxide **230***b* serves as a main carrier path.
(148) The conduction band minimum gradually changes at a junction portion of the oxide **230***a*, the oxide **230***b*, and the oxide **230***c*. In other words, the conduction band minimum at a junction portion of the oxide **230***a*, the oxide **230***b*, and the oxide **230***c* continuously changes or is continuously connected. To obtain this, the density of defect states in a mixed layer formed at an interface between the oxide **230***a* and the oxide **230***b* and an interface between the oxide **230***b* and the oxide **230***c* is preferably made low.
(149) Specifically, when the oxide **230***a* and the oxide **230***b* or the oxide **230***b* and the oxide **230***c* contain the same element as a main component in addition to oxygen, a mixed layer with a low density of defect states can be formed. For example, an In—Ga—Zn oxide, a Ga—Zn oxide, gallium oxide, or the like may be used for the oxide **230***a* and the oxide **230***c* in the case where the oxide **230***b* is an In—Ga—Zn oxide.
(150) Specifically, as the oxide **230***a*, a metal oxide with In:Ga:Zn=1:3:4 [atomic ratio] or a composition in the vicinity thereof, or In:Ga:Zn=1:1:0.5 [atomic ratio] or a composition in the vicinity thereof may be used.
(151) For the oxide **230***b*, a metal oxide with a composition of In:Ga:Zn=1:1:1 [atomic ratio] or in the vicinity thereof or a composition of In:Ga:Zn=4:2:3 [atomic ratio] or in the vicinity thereof may be used. Alternatively, for the oxide **230***b*, a metal oxide with a composition of In:Ga:Zn=5:1:3 [atomic ratio] or in the vicinity thereof or a composition of In:Ga:Zn=10:1:3 [atomic ratio] or in the vicinity thereof may be used. Alternatively, for the oxide **230***b*, an In—Zn oxide (for example, with

a composition of In:Zn=2:1 [atomic ratio] or in the vicinity thereof, a composition of In:Zn=5:1 [atomic ratio] or in the vicinity thereof, or a composition of In:Zn=10:1 [atomic ratio] or in the vicinity thereof) may be used. Alternatively, for the oxide **230***b*, an indium oxide may be used.
(152) For the oxide **230***c*, a metal oxide with a composition of In:Ga:Zn=1:3:4 [atomic ratio] or in the vicinity thereof, a composition of Ga:Zn=2:1 [atomic ratio] or in the vicinity thereof, or a composition of Ga:Zn=2:5 [atomic ratio] or in the vicinity thereof may be used. The oxide **230***c* may be formed to have a single-layer structure or a stacked-layer structure with a material that can be used for the oxide **230***b*.
(153) The proportion of indium in the film for the oxide **230***b* and the oxide **230***c* is preferably increased, in which case the on-state current, the field-effect mobility, or the like of the transistor can be increased. Note that the vicinity of the composition includes ±30% of an intended atomic ratio.
(154) When the metal oxide is deposited by a sputtering method, the above atomic ratio is not limited to the atomic ratio of the deposited metal oxide and may be the atomic ratio of a sputtering target used for depositing the metal oxide.
(155) When the oxide **230***a* and the oxide **230***c* have the above structure, the density of defect states at the interface between the oxide **230***a* and the oxide **230***b* and the interface between the oxide **230***b* and the oxide **230***c* can be made low. Thus, the influence of interface scattering on carrier conduction is small, and the transistor **200** can have high on-state current and excellent frequency characteristics.
(156) The insulator **211**, the insulator **212**, the insulator **214**, the insulator **272**, the insulator **273**, the insulator **282**, the insulator **283**, the insulator **284**, and the insulator **286** preferably function as barrier insulating films that inhibit impurities such as water and hydrogen from diffusing into the transistor **200** from the substrate side or from above the transistor **200**. Thus, for each of the insulator **211**, the insulator **212**, the insulator **214**, the insulator **272**, the insulator **273**, the insulator **282**, the insulator **283**, the insulator **284**, and the insulator **286**, an insulating material having a function of inhibiting diffusion of impurities such as hydrogen atoms, hydrogen molecules, water molecules, nitrogen atoms, nitrogen molecules, nitrogen oxide molecules (e.g., N.sub.2O, NO, or NO.sub.2), or copper atoms (through which the impurities are less likely to pass) is preferably used. Alternatively, it is preferable to use an insulating material having a function of inhibiting diffusion of oxygen (e.g., at least one of an oxygen atom, an oxygen molecule, and the like) (through which the above oxygen is less likely to pass).
(157) For example, silicon nitride or the like is preferably used for the insulator **211**, the insulator **212**, the insulator **283**, and the insulator **284**, and aluminum oxide or the like is preferably used for the insulator **214**, the insulator **272**, the insulator **273**, and the insulator **282**. Accordingly, impurities such as water and hydrogen can be inhibited from being diffused to the transistor **200** side from the substrate side through the insulator **211**, the insulator **212** and the insulator **214**. Alternatively, oxygen contained in the insulator **224** or the like can be inhibited from being diffused to the substrate side through the insulator **211**, the insulator **212**, and the insulator **214**. Furthermore, impurities such as water and hydrogen can be inhibited from being diffused into the transistor **200** side from the insulator **280** and the conductor **246** and the like, which are placed above the insulator **273**, through the insulator **272** and the insulator **273**. In this manner, the transistor **200** is preferably surrounded by the insulator **211**, the insulator **212**, the insulator **214**, the insulator **272**, the insulator **273**, the insulator **282**, the insulator **283**, and the insulator **284** having a function of inhibiting diffusion of oxygen and impurities such as water and hydrogen.
(158) The resistivities of the insulator **211** the insulator **284**, and the insulator **286** are preferably low in some cases. For example, by setting the resistivities of the insulator **211**, the insulator **284**, and the insulator **286** to approximately 1×10.sup.13 Ωcm, the insulator **211**, the insulator **284**, and the insulator **286** can sometimes reduce charge up of the conductor **205**, the conductor **242**, the conductor **260**, or the conductor **246** in treatment using plasma or the like in the manufacturing

process of a semiconductor device. The resistivities of the insulator **211**, the insulator **284**, and the insulator **286** are preferably higher than or equal to 1×10.sup.10 Ωcm and lower than or equal to 1×10.sup.15 Ωcm.
(159) Note that in the case where the insulator **212** and the insulator **284** are deposited by a chemical vapor deposition (CVD) method using a compound gas which does not contain a hydrogen atom or whose hydrogen atom content is small, the insulator **211** and the insulator **284** are not necessarily provided.
(160) The insulator **216** and the insulator **280** preferably have a lower dielectric constant than the insulator **214**. When a material with a low dielectric constant is used for the interlayer film, the parasitic capacitance generated between wirings can be reduced. For example, silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, silicon oxide to which fluorine is added, silicon oxide to which carbon is added, silicon oxide to which carbon and nitrogen are added, or porous silicon oxide is used as appropriate for the insulator **216** and the insulator **280**.
(161) The conductor **205** is provided to overlap with the oxide **230** and the conductor **260**. Furthermore, the conductor **205** is preferably provided to be embedded in the insulator **214** or the insulator **216**.
(162) As illustrated in FIG. **7**A, the conductor **205** is preferably provided to be larger than a region of the oxide **230** that does not overlap with the conductor **242***a* or the conductor **242***b*. As illustrated in FIG. **7**C, it is particularly preferable that the conductor **205** extend to a region outside an end portion of the oxide **230** that intersects with the channel width direction. That is, the conductor **205** and the conductor **260** preferably overlap with each other with the insulators therebetween on an outer side of the side surface of the oxide **230** in the channel width direction. Since the above-described structure is included, the channel formation region of the oxide **230** can be electrically surrounded by the electric field of the conductor **260** functioning as the first gate electrode and the electric field of the conductor **205** functioning as the second gate electrode. In this specification, a transistor structure in which a channel formation region is electrically surrounded by electric fields of the first gate and the second gate is referred to as a surrounded channel (S-channel) structure.
(163) In this specification and the like, the S-channel structure refers to a transistor structure in which a channel formation region is electrically surrounded by the electric fields of a pair of gate electrodes. Furthermore, in this specification and the like, the S-channel structure has a feature in that the side surface and the vicinity of the oxide **230** in contact with the conductor **242***a* and the conductor **242***b* functioning as a source electrode and a drain electrode are of I-type like the channel formation region. The side surface and the vicinity of the oxide **230** in contact with the conductor **242***a* and the conductor **242***b* are in contact with the insulator **280** and thus can be of I-type like the channel formation region. Note that in this specification and the like, “I-type” can be equated with “highly purified intrinsic” to be described later. The S-channel structure disclosed in this specification and the like is different from a Fin-type structure and a planar structure. With the S-channel structure, resistance to a short-channel effect can be enhanced, that is, a transistor in which a short-channel effect is less likely to occur can be provided.
(164) Furthermore, as shown in FIG. **7**C, the conductor **205** extends to function as a wiring as well. However, without limitation to this structure, a structure where a conductor functioning as a wiring is provided below the conductor **205** may be employed. In addition, the conductor **205** does not necessarily have to be provided in each transistor. For example, the conductor **205** may be shared by a plurality of transistors.
(165) Although the transistor **200** having a structure in which the conductor **205** has a stacked structure of the conductor **205***a* and the conductor **205***b* is illustrated, the present invention is not limited thereto. For example, the conductor **205** may have a single-layer structure or a stacked-layer structure of three or more layers. In the case where a structure body has a stacked-layer structure, layers may be distinguished by ordinal numbers corresponding to the formation order.

(166) Here, for the conductor **205***a*, a conductive material having a function of inhibiting diffusion of impurities such as a hydrogen atom, a hydrogen molecule, a water molecule, a nitrogen atom, a nitrogen molecule, a nitrogen oxide molecule (N_2O, NO, NO_2, or the like), and a copper atom is preferably used. Alternatively, it is preferable to use a conductive material having a function of inhibiting diffusion of oxygen (e.g., at least one of oxygen atoms, oxygen molecules, and the like).
(167) When a conductive material having a function of inhibiting oxygen diffusion is used for the conductor **205***a*, the conductivity of the conductor **205***b* can be inhibited from being lowered because of oxidation. As a conductive material having a function of inhibiting diffusion of oxygen, for example, tantalum, tantalum nitride, ruthenium, or ruthenium oxide is preferably used. Thus, the conductor **205***a* is a single layer or a stacked layer of the above conductive materials. For example, the conductor **205***a* may be a stack of tantalum, tantalum nitride, ruthenium, or ruthenium oxide and titanium or titanium nitride.
(168) Moreover, the conductor **205***b* is preferably formed using a conductive material containing tungsten, copper, or aluminum as its main component. Note that the conductor **205***b* is illustrated as a single layer but may have a stacked-layer structure, for example, a stack of any of the above conductive materials and titanium or titanium nitride.
(169) The insulator **222** and the insulator **224** function as a gate insulator.
(170) It is preferable that the insulator **222** have a function of inhibiting diffusion of hydrogen (e.g., at least one of a hydrogen atom, a hydrogen molecule, and the like). In addition, it is preferable that the insulator **222** have a function of inhibiting diffusion of oxygen (e.g., at least one of an oxygen atom, an oxygen molecule, and the like). For example, the insulator **222** preferably has a function of further inhibiting diffusion of one or both of hydrogen and oxygen as compared to the insulator **224**.
(171) For the insulator **222**, an insulator containing an oxide of one or both of aluminum and hafnium, which is an insulating material, is preferably used. In particular, it is preferable that aluminum oxide, hafnium oxide, an oxide containing aluminum and hafnium (hafnium aluminate), or the like be used as the insulator. In the case where the insulator **222** is formed using such a material, the insulator **222** functions as a layer that inhibits release of oxygen from the oxide **230** to the substrate side and diffusion of impurities such as hydrogen from the periphery of the transistor **200** into the oxide **230**. Thus, providing the insulator **222** can inhibit diffusion of impurities such as hydrogen inside the transistor **200** and inhibit generation of oxygen vacancies in the oxide **230**. Moreover, the conductor **205** can be inhibited from reacting with oxygen contained in the insulator **224** and the oxide **230**.
(172) Alternatively, aluminum oxide, bismuth oxide, germanium oxide, niobium oxide, silicon oxide, titanium oxide, tungsten oxide, yttrium oxide, or zirconium oxide may be added to the above insulator, for example. Alternatively, these insulators may be subjected to nitriding treatment. A stack of silicon oxide, silicon oxynitride, or silicon nitride over these insulators may be used as the insulator **222**.
(173) A single layer or stacked layers of an insulator containing what is called a high-k material such as aluminum oxide, hafnium oxide, tantalum oxide, zirconium oxide, lead zirconate titanate (PZT), strontium titanate ($SrTiO_3$), or $(Ba,Sr)TiO_3$ (BST) may be used as the insulator **222**. In the case where the insulator **222** has a stacked-layer structure, a three-layer structure with zirconium oxide, aluminum oxide, and zirconium oxide in this order, or a four-layer structure with zirconium oxide, aluminum oxide, zirconium oxide, and aluminum oxide in this order can be employed, for example. For the insulator **222**, a compound containing hafnium and zirconium may be employed. When the semiconductor is miniaturized and highly integrated, a dielectric used for a gate insulator and a capacitor becomes thin, which might cause a problem of leak current of a transistor and a capacitor. When a high-k material is used as an insulator functioning as the dielectric used for the gate insulator and the capacitor, a gate potential during operation of the

transistor can be lowered and the capacitance of the capacitor can be ensured while the physical thickness is kept.

(174) It is preferable that oxygen be released from the insulator **224** in contact with the oxide **230** by heating. Silicon oxide, silicon oxynitride, or the like is used as appropriate for the insulator **224**, for example. When an insulator containing oxygen is provided in contact with the oxide **230**, oxygen vacancies in the oxide **230** can be reduced and the reliability of the transistor **200** can be improved.

(175) As the insulator **224**, specifically, an oxide material from which part of oxygen is released by heating, in other words, an insulating material including an excess-oxygen region is preferably used. An oxide film that releases oxygen by heating is an oxide film in which the amount of released oxygen converted into oxygen molecules is greater than or equal to 1.0×10.sup.18 molecules/cm.sup.3, preferably greater than or equal to 1.0×10.sup.19 molecules/cm.sup.3, further preferably greater than or equal to 2.0×10.sup.19 molecules/cm.sup.3 or greater than or equal to 3.0×10.sup.20 molecules/cm.sup.3 in TDS (Thermal Desorption Spectroscopy) analysis. Note that the temperature of the film surface in the TDS analysis is preferably within the range of from 100° C. to 700° C., or from 100° C. to 400° C.

(176) One or more of heat treatment, microwave treatment, and RF treatment may be performed in a state in which the insulator including an excess-oxygen region and the oxide **230** are in contact with each other. By the treatment, water or hydrogen in the oxide **230** can be removed. For example, in the oxide **230**, dehydrogenation can be performed when a reaction in which a bond of a defect where hydrogen enters an oxygen vacancy (V.sub.OH) is cut occurs, i.e., a reaction of "V.sub.OH.fwdarw.V.sub.O+H" occurs. Some hydrogen generated at this time is bonded to oxygen to be H.sub.2O, and removed from the oxide **230** or an insulator near the oxide **230** in some cases. Part of hydrogen is diffused into or gettered by the conductor **242** in some cases.

(177) For the microwave treatment, for example, an apparatus including a power supply that generates high-density plasma or an apparatus including a power supply that applies RF to the substrate side is suitably used. For example, the use of a gas containing oxygen and high-density plasma enables high-density oxygen radicals to be produced, and RF application to the substrate side allows the oxygen radicals generated by the high-density plasma to be efficiently introduced into the oxide **230** or an insulator near the oxide **230**. The pressure in the microwave treatment is higher than or equal to 133 Pa, preferably higher than or equal to 200 Pa, further preferably higher than or equal to 400 Pa. As a gas introduced into an apparatus for performing the microwave treatment, for example, oxygen and argon are used and the oxygen flow rate (O.sub.2/(O.sub.2+Ar)) is lower than or equal to 50%, preferably higher than or equal to 10% and lower than or equal to 30%.

(178) In a manufacturing process of the transistor **200**, the heat treatment is preferably performed with the surface of the oxide **230** exposed. The heat treatment is performed at higher than or equal to 100° C. and lower than or equal to 450° C., preferably higher than or equal to 350° C. and lower than or equal to 400° C., for example. Note that the heat treatment is performed in a nitrogen gas or inert gas atmosphere, or an atmosphere containing an oxidizing gas at 10 ppm or more, 1% or more, or 10% or more. For example, the heat treatment is preferably performed in an oxygen atmosphere. This provides oxygen to the oxide **230**, and reduces oxygen vacancies. The heat treatment may be performed under reduced pressure. Alternatively, the heat treatment may be performed in such a manner that heat treatment is performed in a nitrogen gas or inert gas atmosphere, and then another heat treatment is performed in an atmosphere containing an oxidizing gas at 10 ppm or more, 1% or more, or 10% or more in order to compensate for released oxygen. Alternatively, the heat treatment may be performed in such a manner that heat treatment is performed in an atmosphere containing an oxidizing gas at 10 ppm or more, 1% or more, or 10% or more, and then another heat treatment is performed in a nitrogen gas or inert gas atmosphere.

(179) Note that the oxygen adding treatment performed on the oxide **230** can promote a reaction in

which oxygen vacancies in the oxide **230** are filled with supplied oxygen, i.e., a reaction of "V.sub.O+O.fwdarw.null". Furthermore, hydrogen remaining in the oxide **230** reacts with supplied oxygen, so that the hydrogen can be removed as H.sub.2O (dehydration). This can inhibit recombination of hydrogen remaining in the oxide **230** with oxygen vacancies and formation of V.sub.OH.

(180) Note that the insulator **222** and the insulator **224** may each have a stacked-layer structure of two or more layers. In such cases, without limitation to a stacked-layer structure formed of the same material, a stacked-layer structure formed of different materials may be employed.

(181) The conductor **242** (the conductor **242***a* and the conductor **242***b*) is provided over the oxide **230***b*. The conductor **242***a* and the conductor **242***b* function as a source electrode and a drain electrode of the transistor **200**.

(182) For the conductor **242** (the conductor **242***a* and the conductor **242***b*), for example, a nitride containing tantalum, a nitride containing titanium, a nitride containing molybdenum, a nitride containing tungsten, a nitride containing tantalum and aluminum, a nitride containing titanium and aluminum, or the like is preferably used. In one embodiment of the present invention, a nitride containing tantalum is particularly preferable. As another example, ruthenium oxide, ruthenium nitride, an oxide containing strontium and ruthenium, or an oxide containing lanthanum and nickel may be used. These materials are preferable because they are conductive materials that are not easily oxidized or materials that maintain the conductivity even when absorbing oxygen.

(183) There is a curved surface between the side surface of the conductor **242** and the top surface of the conductor **242** in some cases. That is, the end portion of the side surface and the end portion of the top surface are curved in some cases. The curvature radius of the curved surface at an end portion of the conductor **242** is greater than or equal to 3 nm and less than or equal to 10 nm, preferably greater than or equal to 5 nm and less than or equal to 6 nm, for example. When the end portions are not angular, the coverage with films in later deposition steps is improved.

(184) The oxide **243** (the oxide **243***a* and the oxide **243***b*) preferably has a function of inhibiting penetration of oxygen. It is preferable to provide the oxide **243** having a function of inhibiting transmission of oxygen between the conductor **242** and the oxide **230***b*, which functions as the source electrode and the drain electrode, in which case the electrical resistance between the conductor **242** and the oxide **230***b* is reduced. Such a structure improves the electrical characteristics of the transistor **200** and the reliability of the transistor **200**.

(185) A metal oxide containing the element M may be used as the oxide **243**. In particular, aluminum, gallium, yttrium, or tin is preferably used as the element M. The concentration of the element M in the oxide **243** is preferably higher than that in the oxide **230***b*. Alternatively, gallium oxide may be used as the oxide **243**. Alternatively, a metal oxide such as an In-M-Zn oxide may be used as the oxide **243**. Specifically, the atomic ratio of the element M to In in the metal oxide used as the oxide **243** is preferably greater than the atomic ratio of the element M to In in the metal oxide used as the oxide **230***b*. The thickness of the oxide **243** is preferably greater than or equal to 0.5 nm and less than or equal to 5 nm, further preferably greater than or equal to 1 nm and less than or equal to 3 nm, still further preferably greater than or equal to 1 nm and less than or equal to 2 nm. The oxide **243** preferably has crystallinity. In the case where the oxide **243** has crystallinity, release of oxygen from the oxide **230** can be favorably inhibited. When the oxide **243** has a hexagonal crystal structure, for example, release of oxygen from the oxide **230** can sometimes be inhibited.

(186) The insulator **272** is provided in contact with a top surface of the conductor **242** and preferably functions as a barrier layer. With this structure, absorption of excess oxygen contained in the insulator **280** by the conductor **242** can be inhibited. Furthermore, by inhibiting oxidation of the conductor **242**, an increase in the contact resistance between the transistor **200** and a wiring can be inhibited. Consequently, the transistor **200** can have favorable electrical characteristics and reliability.

(187) Thus, the insulator **272** preferably has a function of inhibiting diffusion of oxygen. For example, the insulator **272** preferably has a function of further inhibiting diffusion of oxygen as compared to the insulator **280**. An insulator containing an oxide of one or both of aluminum and hafnium is preferably deposited as the insulator **272**, for example. An insulator containing aluminum nitride may be used as the insulator **272**, for example.
(188) In addition, oxygen can be supplied to the insulator **224** at the time of forming the insulator **272** in some cases. The insulator **224** is sealed with the insulator **272** and the insulator **273**; thus, oxygen supplied to the insulator **224** is inhibited from diffusing to the outside and can be efficiently supplied to the oxide **230**. Moreover, hydrogen in the insulator **224** may be absorbed by the insulator **273**, which is preferable.
(189) Note that an insulator functioning as a barrier layer may be provided between the top surface of the conductor **242** and the insulator **280** without providing the insulator **272** and the insulator **273**. With this structure, absorption of excess oxygen contained in the insulator **280** by the conductor **242** can be inhibited. Furthermore, by inhibiting oxidation of the conductor **242**, an increase in the contact resistance between the transistor **200** and a wiring can be inhibited. Consequently, the transistor **200** can have favorable electrical characteristics and reliability.
(190) Thus, the insulator preferably has a function of inhibiting diffusion of oxygen. For example, the insulator preferably has a function of further inhibiting diffusion of oxygen as compared to the insulator **280**.
(191) An insulator containing an oxide of one or both of aluminum and hafnium is preferably deposited as the insulator, for example. In particular, aluminum oxide is preferably deposited by an atomic layer deposition (ALD) method. By an ALD method, a dense film with a smaller number of defects such as cracks and pinholes or with a uniform thickness can be formed. An insulator containing aluminum nitride may be used as the insulator, for example.
(192) The insulator **250** functions as a gate insulator. The insulator **250** is preferably positioned in contact with at least part of the oxide **230***c*. For the insulator **250**, silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, silicon oxide to which fluorine is added, silicon oxide to which carbon is added, silicon oxide to which carbon and nitrogen are added, porous silicon oxide, or the like can be used. In particular, silicon oxide and silicon oxynitride, which have thermal stability, are preferable.
(193) Like the insulator **224**, the insulator **250** is preferably formed using an insulator that releases oxygen by heating. When an insulator from which oxygen is released by heating is provided as the insulator **250** in contact with at least part of the oxide **230***c*, oxygen can be efficiently supplied to the channel formation region of the oxide **230***b* and oxygen defects in the channel formation region of the oxide **230***b* can be reduced. Thus, a transistor that has stable electrical characteristics with less variations in electrical characteristics and improved reliability can be provided. Furthermore, as in the insulator **224**, the concentration of impurities such as water and hydrogen in the insulator **250** is preferably reduced. The thickness of the insulator **250** is preferably greater than or equal to 1 nm and less than or equal to 20 nm.
(194) Although the insulator **250** is a single layer in FIG. **7**B, a stacked-layer structure of two or more layers may be employed. In the case where the insulator **250** has a stacked-layer structure including two layers, it is preferable that a lower layer of the insulator **250** be formed using an insulator from which oxygen is released by heating and an upper layer of the insulator **250** be formed using an insulator having a function of inhibiting diffusion of oxygen. With such a structure, oxygen contained in the lower layer of the insulator **250** can be inhibited from diffusing into the conductor **260**. That is, a reduction in the amount of oxygen supplied to the oxide **230** can be inhibited. In addition, oxidation of the conductor **260** due to oxygen from the lower layer of the insulator **250** can be inhibited. For example, the lower layer of the insulator **250** can be formed using the above-described material that can be used for the insulator **250**, and the upper layer of the insulator **250** can be formed using a material similar to that for the insulator **222**.

(195) In the case where silicon oxide, silicon oxynitride, or the like is used for the lower layer of the insulator **250**, the upper layer of the insulator **250** may be formed using an insulating material that is a high-k material having a high relative dielectric constant. The gate insulator having a stacked-layer structure of the lower layer of the insulator **250** and the upper layer of the insulator **250** can be thermally stable and can have a high dielectric constant. Thus, a gate potential that is applied during operation of the transistor can be reduced while the physical thickness of the gate insulator is maintained. Furthermore, the equivalent oxide thickness (EOT) of the insulator functioning as the gate insulator can be reduced.
(196) Specifically, a metal oxide containing one kind or two or more kinds selected from hafnium, aluminum, gallium, yttrium, zirconium, tungsten, titanium, tantalum, nickel, germanium, magnesium, and the like can be used as the upper layer of the insulator **250**. Alternatively, the metal oxide that can be used for the oxide **230** can be used. In particular, an insulator containing an oxide of one or both of aluminum and hafnium is preferably used.
(197) Furthermore, a metal oxide may be provided between the insulator **250** and the conductor **260**. The metal oxide preferably inhibits diffusion of oxygen from the insulator **250** into the conductor **260**. Providing the metal oxide that inhibits diffusion of oxygen inhibits diffusion of oxygen from the insulator **250** into the conductor **260**. That is, a reduction in the amount of oxygen supplied to the oxide **230** can be inhibited. In addition, oxidation of the conductor **260** due to oxygen from the insulator **250** can be inhibited.
(198) The metal oxide preferably has a function of part of the first gate electrode. For example, a metal oxide that can be used for the oxide **230** can be used as the metal oxide. In that case, when the conductor **260***a* is deposited by a sputtering method, the metal oxide can have a reduced electric resistance value to be a conductor. This can be referred to as an OC (Oxide Conductor) electrode.
(199) With the above metal oxide, the on-state current of the transistor **200** can be increased without a reduction in the influence of the electric field from the conductor **260**. Since the distance between the conductor **260** and the oxide **230** is kept by the physical thicknesses of the insulator **250** and the metal oxide, leakage current between the conductor **260** and the oxide **230** can be reduced. Moreover, when the stacked-layer structure of the insulator **250** and the metal oxide is provided, the physical distance between the conductor **260** and the oxide **230** and the intensity of electric field applied to the oxide **230** from the conductor **260** can be easily adjusted as appropriate.
(200) The conductor **260** functions as a first gate electrode of the transistor **200**. The conductor **260** preferably includes the conductor **260***a* and the conductor **260***b* positioned over the conductor **260***a*. For example, the conductor **260***a* is preferably positioned to cover a bottom surface and a side surface of the conductor **260***b*. Moreover, as shown in FIG. **7**B, a top surface of the conductor **260** is substantially aligned with the top surface of the insulator **250** and the top surface of the oxide **230***c*. Note that although the conductor **260** has a two-layer structure of the conductor **260***a* and the conductor **260***b* in FIG. **7**B, the conductor **260** can have a single-layer structure or a stacked-layer structure of three or more layers.
(201) For the conductor **260***a*, a conductive material having a function of inhibiting diffusion of impurities such as a hydrogen atom, a hydrogen molecule, a water molecule, a nitrogen atom, a nitrogen molecule, a nitrogen oxide molecule, and a copper atom is preferably used. Alternatively, it is preferable to use a conductive material having a function of inhibiting diffusion of oxygen (e.g., at least one of oxygen atoms, oxygen molecules, and the like).
(202) In addition, when the conductor **260***a* has a function of inhibiting diffusion of oxygen, the conductivity of the conductor **260***b* can be inhibited from being lowered because of oxidation due to oxygen contained in the insulator **250**. As a conductive material having a function of inhibiting diffusion of oxygen, for example, tantalum, tantalum nitride, ruthenium, or ruthenium oxide is preferably used.
(203) The conductor **260** also functions as a wiring and thus is preferably formed using a conductor having high conductivity. For example, a conductive material containing tungsten, copper, or

aluminum as its main component can be used for the conductor **260***b*. The conductor **260***b* may have a stacked-layer structure, for example, a stacked-layer structure of any of the above conductive materials and titanium or titanium nitride.

(204) In the transistor **200**, the conductor **260** is formed in a self-aligned manner to fill an opening formed in the insulator **280** and the like. The formation of the conductor **260** in this manner allows the conductor **260** to be positioned certainly in a region between the conductor **242***a* and the conductor **242***b* without alignment.

(205) As illustrated in FIG. **7**C in the channel width direction of the transistor **200**, the level of the bottom surface of the conductor **260** in a region where the conductor **260** and the oxide **230***b* do not overlap with each other is preferably lower than the level of the bottom surface of the oxide **230***b*. When the conductor **260** functioning as the gate electrode covers the side and top surfaces of the channel formation region of the oxide **230***b* with the insulator **250** and the like therebetween, the electric field of the conductor **260** is likely to affect the entire channel formation region of the oxide **230***b*. Thus, the on-state current of the transistor **200** can be increased and the frequency characteristics of the transistor **200** can be improved. When the bottom surface of the insulator **222** is a benchmark, the difference between the level of the bottom surface of the conductor **260** in a region where the conductor **260**, the oxide **230***a*, and the oxide **230***b* do not overlap with each other and the level of the bottom surface of the oxide **230***b* is greater than or equal to 0 nm and less than or equal to 100 nm, preferably greater than or equal to 3 nm and less than or equal to 50 nm, further preferably greater than or equal to 5 nm and less than or equal to 20 nm.

(206) The insulator **280** is provided over the insulator **273**. In addition, a top surface of the insulator **280** may be planarized.

(207) The insulator **280** functioning as an interlayer film preferably has a low dielectric constant. When a material with a low dielectric constant is used for the interlayer film, the parasitic capacitance generated between wirings can be reduced. The insulator **280** is preferably formed using a material similar to that used for the insulator **216**, for example. In particular, silicon oxide and silicon oxynitride, which have thermal stability, are preferable. Materials such as silicon oxide, silicon oxynitride, and porous silicon oxide, in each of which a region containing oxygen released by heating can be easily formed, are particularly preferable.

(208) The concentration of impurities such as water and hydrogen in the insulator **280** is preferably reduced. Moreover, the insulator **280** preferably has a low hydrogen concentration and includes an excess-oxygen region or excess oxygen, and may be formed using a material similar to that for the insulator **216**, for example. The insulator **280** may have a stacked-layer structure of the above materials; silicon oxide formed by a sputtering method and silicon oxynitride formed by a CVD method stacked thereover. Furthermore, silicon nitride may be stacked thereover.

(209) The insulator **282** or the insulator **283** preferably functions as barrier insulating films that inhibit impurities such as water and hydrogen from diffusing into the insulator **280** from above. The insulator **282** or the insulator **283** preferably functions as barrier insulating films for inhibiting passage of oxygen. As the insulator **282** and the insulator **283**, for example, an insulator such as aluminum oxide, silicon nitride, or silicon nitride oxide may be used. The insulator **282** may be formed using aluminum oxide that has high blocking property against oxygen and the insulator **283** may be formed using silicon nitride that has high blocking property against hydrogen, for example.

(210) For the conductor **240***a* and the conductor **240***b*, a conductive material containing tungsten, copper, or aluminum as its main component is preferably used. The conductor **240***a* and the conductor **240***b* may each have a stacked-layer structure.

(211) In the case where the conductor **240** has a stacked-layer structure, a conductive material having a function of inhibiting transmission of impurities such as water and hydrogen is preferably used for a conductor in contact with the insulator **284**, the insulator **283**, the insulator **282**, the insulator **280**, the insulator **273**, and the insulator **272**. For example, tantalum, tantalum nitride, titanium, titanium nitride, ruthenium, ruthenium oxide, or the like is preferably used. The

conductive material having a function of inhibiting passage of impurities such as water and hydrogen may be used as a single layer or stacked layers. The use of the conductive material can prevent oxygen added to the insulator **280** from being absorbed by the conductor **240***a* and the conductor **240***b*. Moreover, impurities such as water and hydrogen contained in a layer above the insulator **284** can be inhibited from entering the oxide **230** through the conductor **240***a* and the conductor **240***b*.

(212) For the insulator **241***a* and the insulator **241***b*, for example, an insulator such as silicon nitride, aluminum oxide, or silicon nitride oxide may be used. Since the insulator **241***a* and the insulator **241***b* are provided in contact with the insulator **273** and the insulator **272**, impurities such as water and hydrogen contained in the insulator **280** or the like can be inhibited from entering the oxide **230** through the conductor **240***a* and the conductor **240***b*. In particular, silicon nitride is suitable because of having a high blocking property against hydrogen. In addition, oxygen contained in the insulator **280** can be prevented from being absorbed by the conductor **240***a* and the conductor **240***b*.

(213) The conductor **246** (the conductor **246***a* and the conductor **246***b*) functioning as a wiring may be provided in contact with a top surface of the conductor **240***a* and a top surface of the conductor **240***b*. The conductor **246** is preferably formed using a conductive material containing tungsten, copper, or aluminum as its main component. Furthermore, the conductor may have a stacked-layer structure and may be a stack of titanium or titanium nitride and any of the above conductive materials, for example. Note that the conductor may be formed to be embedded in an opening provided in an insulator.

(214) The insulator **286** is provided over the conductor **246** and the insulator **284**. Accordingly, the top surface of the conductor **246** and the side surface of the conductor **246** are in contact with the insulator **286**, and the bottom surface of the conductor **246** is in contact with the insulator **284**. In other words, the conductor **246** can be surrounded by the insulator **284** and the insulator **286**. The structure can inhibit transmission of oxygen from the outside and oxidation of the conductor **246**. Furthermore, the structure is preferable because impurities such as water and hydrogen can be prevented from diffusing from the conductor **246** to the outside.

(215) <Material for Semiconductor Device>

(216) Materials that can be used for the semiconductor device are described below.

(217) «Substrate»

(218) As a substrate where the transistor **200** is formed, an insulator substrate, a semiconductor substrate, or a conductor substrate is used, for example. Examples of the insulator substrate include a glass substrate, a quartz substrate, a sapphire substrate, a stabilized zirconia substrate (an yttria-stabilized zirconia substrate or the like), and a resin substrate. Examples of the semiconductor substrate include a semiconductor substrate including silicon or germanium, and a compound semiconductor substrate including silicon carbide, silicon germanium, gallium arsenide, indium phosphide, zinc oxide, or gallium oxide. Another example is a semiconductor substrate in which an insulator region is included in the semiconductor substrate, e.g., an SOI (Silicon On Insulator) substrate. Examples of the conductor substrate include a graphite substrate, a metal substrate, an alloy substrate, and a conductive resin substrate. Other examples include a substrate including a metal nitride and a substrate including a metal oxide. Other examples include an insulator substrate provided with a conductor or a semiconductor, a semiconductor substrate provided with a conductor or an insulator, and a conductor substrate provided with a semiconductor or an insulator. Alternatively, these substrates provided with elements may be used. Examples of the element provided for the substrate include a capacitor, a resistor, a switching element, a light-emitting element, and a memory element.

(219) «Insulator»

(220) Examples of an insulator include an oxide, a nitride, an oxynitride, a nitride oxide, a metal oxide, a metal oxynitride, and a metal nitride oxide, each of which has an insulating property.

(221) As miniaturization and high integration of transistors progress, for example, a problem such as leakage current may arise because of a thinner gate insulator. When a high-k material is used as the insulator functioning as a gate insulator, the voltage during operation of the transistor can be lowered while the physical thickness of the gate insulator is maintained. In contrast, when a material with a low relative dielectric constant is used as the insulator functioning as an interlayer film, parasitic capacitance generated between wirings can be reduced. Accordingly, a material is preferably selected depending on the function of an insulator.
(222) Examples of the insulator with a high relative dielectric constant include gallium oxide, hafnium oxide, zirconium oxide, an oxide containing aluminum and hafnium, an oxynitride containing aluminum and hafnium, an oxide containing silicon and hafnium, an oxynitride containing silicon and hafnium, and a nitride containing silicon and hafnium.
(223) Examples of the insulator with a low dielectric constant include silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, silicon oxide to which fluorine is added, silicon oxide to which carbon is added, silicon oxide to which carbon and nitrogen are added, porous silicon oxide, and a resin.
(224) When a transistor using a metal oxide is surrounded by insulators having a function of inhibiting passage of oxygen and impurities such as hydrogen, the electrical characteristics of the transistor can be stable. As the insulator having a function of inhibiting passage of oxygen and impurities such as hydrogen, a single layer or stacked layers of an insulator containing, for example, boron, carbon, nitrogen, oxygen, fluorine, magnesium, aluminum, silicon, phosphorus, chlorine, argon, gallium, germanium, yttrium, zirconium, lanthanum, neodymium, hafnium, or tantalum is used. Specifically, as the insulator having a function of inhibiting passage of oxygen and impurities such as hydrogen, a metal oxide such as aluminum oxide, magnesium oxide, gallium oxide, germanium oxide, yttrium oxide, zirconium oxide, lanthanum oxide, neodymium oxide, hafnium oxide, or tantalum oxide; or a metal nitride such as aluminum nitride, silicon nitride oxide, or silicon nitride can be used.
(225) The insulator functioning as the gate insulator is preferably an insulator including a region containing oxygen released by heating. For example, when a structure is employed in which silicon oxide or silicon oxynitride including a region containing oxygen released by heating is in contact with the oxide **230**, oxygen vacancies included in the oxide **230** can be filled.
(226) «Conductor»
(227) For the conductor, it is preferable to use a metal element selected from aluminum, chromium, copper, silver, gold, platinum, tantalum, nickel, titanium, molybdenum, tungsten, hafnium, vanadium, niobium, manganese, magnesium, zirconium, beryllium, indium, ruthenium, iridium, strontium, and lanthanum; an alloy containing any of the above metal elements; an alloy containing a combination of the above metal elements; or the like. For example, it is preferable to use tantalum nitride, titanium nitride, tungsten, a nitride containing titanium and aluminum, a nitride containing tantalum and aluminum, ruthenium oxide, ruthenium nitride, an oxide containing strontium and ruthenium, an oxide containing lanthanum and nickel, or the like. Tantalum nitride, titanium nitride, a nitride containing titanium and aluminum, a nitride containing tantalum and aluminum, ruthenium oxide, ruthenium nitride, an oxide containing strontium and ruthenium, and an oxide containing lanthanum and nickel are preferable because they are oxidation-resistant conductive materials or materials that retain their conductivity even after absorbing oxygen. Furthermore, a semiconductor having high electrical conductivity, typified by polycrystalline silicon containing an impurity element such as phosphorus, or silicide such as nickel silicide may be used.
(228) A stack including a plurality of conductive layers formed of the above materials may be used. For example, a stacked-layer structure combining a material containing the above metal element and a conductive material containing oxygen may be employed. A stacked-layer structure combining a material containing the above metal element and a conductive material containing nitrogen may be employed. A stacked-layer structure combining a material containing the above

metal element, a conductive material containing oxygen, and a conductive material containing nitrogen may be employed.
(229) Note that when an oxide is used for the channel formation region of the transistor, a stacked-layer structure combining a material containing the above metal element and a conductive material containing oxygen is preferably used for the conductor functioning as the gate electrode. In that case, the conductive material containing oxygen is preferably provided on the channel formation region side. When the conductive material containing oxygen is provided on the channel formation region side, oxygen released from the conductive material is easily supplied to the channel formation region.
(230) It is particularly preferable to use, for the conductor functioning as the gate electrode, a conductive material containing oxygen and a metal element contained in a metal oxide where the channel is formed. Alternatively, a conductive material containing the above metal element and nitrogen may be used. For example, a conductive material containing nitrogen, such as titanium nitride or tantalum nitride, may be used. Alternatively, indium tin oxide, indium oxide containing tungsten oxide, indium zinc oxide containing tungsten oxide, indium oxide containing titanium oxide, indium tin oxide containing titanium oxide, indium zinc oxide, or indium tin oxide to which silicon is added may be used. Furthermore, indium gallium zinc oxide containing nitrogen may be used. With use of such a material, hydrogen contained in the metal oxide where the channel is formed can be trapped in some cases. Alternatively, hydrogen entering from an external insulator or the like can be trapped in some cases.
(231) «Metal Oxide»
(232) The oxide **230** is preferably formed using a metal oxide functioning as a semiconductor (an oxide semiconductor). A metal oxide that can be used for the oxide **230** of the present invention is described below.
(233) The metal oxide preferably contains at least indium or zinc. In particular, indium and zinc are preferably contained. Furthermore, aluminum, gallium, yttrium, tin, or the like is preferably contained in addition to them. Furthermore, one or more kinds selected from boron, titanium, iron, nickel, germanium, zirconium, molybdenum, lanthanum, cerium, neodymium, hafnium, tantalum, tungsten, magnesium, and the like may be contained.
(234) Here, the case where the metal oxide is an In-M-Zn oxide containing indium, the element M, and zinc is considered. Note that the element M is aluminum, gallium, yttrium, or tin. Examples of other elements that can be used as the element M include boron, titanium, iron, nickel, germanium, zirconium, molybdenum, lanthanum, cerium, neodymium, hafnium, tantalum, tungsten, and magnesium. Note that it is sometimes acceptable to use a plurality of the above-described elements in combination as the element M.
(235) Note that in this specification and the like, a metal oxide containing nitrogen is also referred to as a metal oxide in some cases. A metal oxide containing nitrogen may be referred to as a metal oxynitride.
(236) [Structure of Metal Oxide]
(237) Oxide semiconductors (metal oxides) can be classified into a single crystal oxide semiconductor and a non-single-crystal oxide semiconductor. Examples of a non-single-crystal oxide semiconductor include a CAAC-OS, a polycrystalline oxide semiconductor, an nc-OS (nanocrystalline oxide semiconductor), an amorphous-like oxide semiconductor (a-like OS), and an amorphous oxide semiconductor.
(238) The CAAC-OS has c-axis alignment, a plurality of nanocrystals are connected in the a-b plane direction, and its crystal structure has distortion. Note that the distortion refers to a portion where the direction of a lattice arrangement changes between a region with a regular lattice arrangement and another region with a regular lattice arrangement in a region where the plurality of nanocrystals are connected.
(239) The nanocrystal is basically a hexagon but is not always a regular hexagon and is a non-

regular hexagon in some cases. Furthermore, a pentagonal or heptagonal lattice arrangement, for example, is included in the distortion in some cases. Note that it is difficult to observe a clear grain boundary even in the vicinity of distortion in the CAAC-OS. That is, formation of a crystal grain boundary is found to be inhibited by the distortion of a lattice arrangement. This is because the CAAC-OS can tolerate distortion owing to a low density of arrangement of oxygen atoms in the a-b plane direction, an interatomic bond length changed by substitution of a metal element, and the like. Note that a crystal structure in which a clear grain boundary is observed is what is called polycrystal. It is highly probable that the grain boundary becomes a recombination center and traps carriers and thus decreases the on-state current and field-effect mobility of a transistor. Thus, the CAAC-OS in which no clear grain boundary is observed is one of crystalline oxides having a crystal structure suitable for a semiconductor layer of a transistor. Note that Zn is preferably contained to form the CAAC-OS. For example, an In—Zn oxide and an In—Ga—Zn oxide are suitable because they can inhibit generation of a grain boundary as compared with an In oxide.
(240) The CAAC-OS tends to have a layered crystal structure (also referred to as a layered structure) in which a layer containing indium and oxygen (hereinafter, an In layer) and a layer containing the element M, zinc, and oxygen (hereinafter, an (M, Zn) layer) are stacked. Note that indium and the element M can be replaced with each other, and when the element Min the (M, Zn) layer is replaced with indium, the layer can also be referred to as an (In, M, Zn) layer. Furthermore, when indium in the In layer is replaced with the element M, the layer can be referred to as an (In, M) layer.
(241) The CAAC-OS is a metal oxide with high crystallinity. On the other hand, a clear crystal grain boundary cannot be observed in the CAAC-OS; thus, it can be said that a reduction in electron mobility due to the crystal grain boundary is less likely to occur. Entry of impurities, formation of defects, or the like might decrease the crystallinity of a metal oxide, which means that the CAAC-OS is a metal oxide having small amounts of impurities and defects (e.g., oxygen vacancies). Thus, a metal oxide including a CAAC-OS is physically stable. Therefore, the metal oxide including a CAAC-OS is resistant to heat and has high reliability.
(242) In the nc-OS, a microscopic region (e.g., a region with a size greater than or equal to 1 nm and less than or equal to 10 nm, in particular, a region with a size greater than or equal to 1 nm and less than or equal to 3 nm) has a periodic atomic arrangement. Furthermore, there is no regularity of crystal orientation between different nanocrystals in the nc-OS. Thus, the orientation in the whole film is not observed. Accordingly, the nc-OS cannot be distinguished from an a-like OS or an amorphous oxide semiconductor by some analysis methods.
(243) Note that an In—Ga—Zn oxide (hereinafter, IGZO) that is a kind of metal oxide containing indium, gallium, and zinc has a stable structure in some cases by being formed of the above-described nanocrystals. In particular, crystals of IGZO tend not to grow in the air and thus, a stable structure is obtained when IGZO is formed of smaller crystals (e.g., the above-described nanocrystals) rather than larger crystals (here, crystals with a size of several millimeters or several centimeters).
(244) An a-like OS is a metal oxide having a structure between those of the nc-OS and an amorphous oxide semiconductor. The a-like OS includes a void or a low-density region. That is, the a-like OS has low crystallinity compared with the nc-OS and the CAAC-OS.
(245) An oxide semiconductor (metal oxide) can have various structures which show different properties. Two or more of the amorphous oxide semiconductor, the polycrystalline oxide semiconductor, the a-like OS, the nc-OS, and the CAAC-OS may be included in an oxide semiconductor of one embodiment of the present invention.
(246) As an oxide semiconductor other than the above, a CAC (Cloud-Aligned Composite)-OS may be used.
(247) A CAC-OS has a conducting function in part of the material and has an insulating function in another part of the material; as a whole, the CAC-OS has a function of a semiconductor. Note that

in the case where the CAC-OS is used in an active layer of a transistor, the conducting function is a function that allows electrons (or holes) serving as carriers to flow, and the insulating function is a function that does not allow electrons serving as carriers to flow. By the complementary action of the conducting function and the insulating function, a switching function (On/Off function) can be given to the CAC-OS. In the CAC-OS, separation of the functions can maximize both of the functions.
(248) In addition, the CAC-OS includes conductive regions and insulating regions. The conductive regions have the above-described conducting function, and the insulating regions have the above-described insulating function. In some cases, the conductive regions and the insulating regions in the material are separated at the nanoparticle level. In some cases, the conductive regions and the insulating regions are unevenly distributed in the material. Furthermore, in some cases, the conductive regions are observed to be coupled in a cloud-like manner with their boundaries blurred.
(249) In the CAC-OS, the conductive regions and the insulating regions each have a size greater than or equal to 0.5 nm and less than or equal to 10 nm, preferably greater than or equal to 0.5 nm and less than or equal to 3 nm, and are dispersed in the material, in some cases.
(250) The CAC-OS is composed of components having different band gaps. For example, the CAC-OS is composed of a component having a wide gap due to the insulating region and a component having a narrow gap due to the conductive region. In the case of the structure, when carriers flow, carriers mainly flow in the component having a narrow gap. Furthermore, the component having a narrow gap complements the component having a wide gap, and carriers also flow in the component having a wide gap in conjunction with the component having a narrow gap. Therefore, in the case where the above-described CAC-OS is used in a channel formation region of a transistor, the transistor in the on state can achieve high current driving capability, that is, high on-state current and high field-effect mobility.
(251) In other words, the CAC-OS can also be referred to as a matrix composite or a metal matrix composite.
(252) Oxide semiconductors might be classified in a manner different from the above-described one when classified in terms of the crystal structure. Here, the classification of the crystal structures of an oxide semiconductor is explained with FIG. **8**A. FIG. **8**A is a diagram showing the classification of crystal structures of an oxide semiconductor, typically IGZO (a metal oxide containing In, Ga, and Zn).
(253) As shown in FIG. **8**A, IGZO is roughly classified into Amorphous, Crystalline, and Crystal. Amorphous includes completely amorphous. Crystalline includes CAAC, nc, and CAC. Crystal includes single crystal and poly crystal.
(254) Note that the structure shown in the thick frame in FIG. **8**A is a structure that belongs to a new crystalline phase. This structure is positioned in a boundary region between Amorphous and Crystal. In other words, Amorphous, which is energetically unstable, and Crystalline are completely different structures.
(255) A crystal structure of a film or a substrate can be analyzed with an X-ray diffraction (XRD) spectrum. Here, XRD spectra of quartz glass and IGZO, which has a crystal structure classified into Crystalline (also referred to as crystalline IGZO), are shown in FIG. **8**B and FIG. **8**C. In FIG. **8**B and FIG. **8**C, the horizontal axis represents 2θ [deg.], and the vertical axis represents intensity [a.u.]. FIG. **8**B shows an XRD spectrum of quartz glass and FIG. **8**C shows an XRD spectrum of crystalline IGZO. Note that the crystalline IGZO shown in FIG. **8**C has a composition of In:Ga:Zn=4:2:3 [atomic ratio]. Furthermore, the crystalline IGZO shown in FIG. **8**C has a thickness of 500 nm.
(256) As indicated by arrows in FIG. **8**B, the XRD spectrum of the quartz glass shows a substantially bilaterally symmetrical peak. In contrast, as indicated by arrows in FIG. **8**C, the XRD spectrum of the crystalline IGZO shows a bilaterally asymmetrical peak. The bilaterally

asymmetrical peak of the XRD spectrum clearly shows the existence of crystal. In other words, the structure cannot be regarded as Amorphous unless it has a bilaterally symmetrical peak in the XRD spectrum.
(257) [Impurity]
(258) Here, the influence of each impurity in the metal oxide will be described.
(259) Entry of the impurities into the oxide semiconductor causes formation of defect states or oxygen vacancies in some cases. Thus, when impurities enter a channel formation region of the oxide semiconductor, the electrical characteristics of a transistor using the oxide semiconductor are likely to vary and its reliability is degraded in some cases. Moreover, when the channel formation region includes oxygen vacancies, the transistor tends to have normally-on characteristics (characteristics in that a channel exists without voltage application to a gate electrode and current flows in a transistor).
(260) In contrast, a transistor using a metal oxide is likely to have normally-on characteristics owing to an impurity and an oxygen vacancy in the metal oxide that affect the electrical characteristics. In the case where the transistor is driven in the state where excess oxygen exceeding the proper amount is included in the metal oxide, the valence of the excess oxygen atoms is changed and the electrical characteristics of the transistor are changed, so that reliability is decreased in some cases.
(261) Thus, a metal oxide having a low carrier concentration is preferably used for a channel formation region of a transistor of one embodiment of the present invention. In order to reduce the carrier concentration of the metal oxide, the concentration of impurities in the metal oxide is reduced so that the density of defect states can be reduced. In this specification and the like, a state with a low impurity concentration and a low density of defect states is referred to as a highly purified intrinsic or substantially highly purified intrinsic state. Note that in this specification and the like, the case where the carrier concentration of the metal oxide in the channel formation region is lower than or equal to 1×10.sup.16 cm.sup.−3 is defined as a substantially highly purified intrinsic state.
(262) The carrier concentration of the metal oxide in the channel formation region is preferably lower than or equal to 1×10.sup.18 cm.sup.−3, further preferably lower than or equal to 1×10.sup.17 cm.sup.−3, still further preferably lower than or equal to 1×10.sup.16 cm.sup.−3, yet further preferably lower than 1×10.sup.13 cm.sup.−3, and yet still further preferably lower than 1×10.sup.12 cm.sup.−3. Note that the lower limit of the carrier concentration of the metal oxide in the channel formation region is not particularly limited and can be, for example, 1×10.sup.−9 cm.sup.3.
(263) Examples of impurities in a metal oxide include hydrogen, nitrogen, alkali metal, alkaline earth metal, iron, nickel, and silicon. In particular, hydrogen contained in a metal oxide reacts with oxygen bonded to a metal atom to be water, and thus forms oxygen vacancies in the metal oxide in some cases. If the channel formation region in the metal oxide includes oxygen vacancies, the transistor sometimes has normally-on characteristics. Moreover, in the case where hydrogen enters an oxygen vacancy in the metal oxide, the oxygen vacancy and the hydrogen are bonded to each other to form V.sub.OH in some cases. In some cases, a defect in which hydrogen has entered an oxygen vacancy (V.sub.OH) functions as a donor and generates an electron serving as a carrier. In other cases, bonding of part of hydrogen to oxygen bonded to a metal atom generates electrons serving as carriers. Thus, a transistor using a metal oxide containing a large amount of hydrogen is likely to have normally-on characteristics. Moreover, hydrogen in a metal oxide easily moves by stress such as heat and an electric field; thus, the reliability of a transistor may be low when the metal oxide contains a plenty of hydrogen.
(264) In one embodiment of the present invention, V.sub.OH in the oxide **230** is preferably reduced as much as possible so that the oxide **230** becomes a highly purified intrinsic or substantially highly purified intrinsic oxide. It is important to remove impurities such as moisture and hydrogen in a

metal oxide (sometimes described as dehydration or dehydrogenation treatment) and to compensate for oxygen vacancies by supplying oxygen to the metal oxide (sometimes described as oxygen supplying treatment) to obtain a metal oxide whose V.sub.OH is reduced enough. When a metal oxide in which impurities such as V.sub.OH are sufficiently reduced is used for a channel formation region of a transistor, stable electrical characteristics can be given.

(265) A defect in which hydrogen has entered an oxygen vacancy (V.sub.OH) can function as a donor in the metal oxide. However, it is difficult to evaluate the defects quantitatively. Thus, the metal oxide is sometimes evaluated by not its donor concentration but its carrier concentration. Therefore, in this specification and the like, the carrier concentration in a state where an electric field is assumed to be not applied is sometimes used, instead of the donor concentration, as the parameter of the metal oxide. That is, "carrier concentration" in this specification and the like can be replaced with "donor concentration" in some cases. In addition, "carrier concentration" in this specification and the like can be replaced with "carrier density".

(266) Therefore, hydrogen in the metal oxide is preferably reduced as much as possible. Specifically, the hydrogen concentration of the metal oxide obtained by secondary ion mass spectrometry (SIMS) is set lower than 1×10.sup.20 atoms/cm.sup.3, preferably lower than 1×10.sup.19 atoms/cm.sup.3, further preferably lower than 5×10.sup.18 atoms/cm.sup.3, still further preferably lower than 1×10.sup.18 atoms/cm.sup.3. When a metal oxide with a sufficiently low concentration of impurities such as hydrogen is used for a channel formation region of a transistor, the transistor can have stable electrical characteristics.

(267) The above-described defect states may include a trap state. Charges trapped by the trap states in the metal oxide take a long time to be released and may behave like fixed charges. Thus, a transistor whose channel formation region includes a metal oxide having a high density of trap states has unstable electrical characteristics in some cases.

(268) If the impurities exist in the channel formation region of the oxide semiconductor, the crystallinity of the channel formation region may decrease, and the crystallinity of an oxide provided in contact with the channel formation region may decrease. Low crystallinity of the channel formation region tends to result in deterioration in stability or reliability of the transistor. Moreover, if the crystallinity of the oxide provided in contact with the channel formation region is low, an interface state may be formed and the stability or reliability of the transistor may deteriorate.

(269) Therefore, the reduction in concentration of impurities in and around the channel formation region of the oxide semiconductor is effective in improving the stability or reliability of the transistor. Examples of impurities include hydrogen, nitrogen, an alkali metal, an alkaline earth metal, iron, nickel, and silicon.

(270) Specifically, the concentration of the above impurities obtained by SIMS is lower than or equal to 1×10.sup.18 atoms/cm.sup.3, preferably lower than or equal to 2×10.sup.16 atoms/cm.sup.3 in and around the channel formation region of the oxide semiconductor. Alternatively, the concentration of the above impurities obtained by element analysis using energy dispersive X-ray spectroscopy (EDX) is lower than or equal to 1.0 atomic % in and around the channel formation region of the oxide semiconductor. When an oxide containing the element M is used as the oxide semiconductor, the concentration ratio of the impurities to the element M is lower than 0.10, preferably lower than 0.05 in and around the channel formation region of the oxide semiconductor. Here, the concentration of the element M used in the calculation of the concentration ratio may be a concentration in a region whose concertation of the impurities is calculated or may be a concentration in the oxide semiconductor.

(271) A metal oxide with a low impurity concentration has a low density of defect states and thus has a low density of trap states in some cases.

(272) In a transistor using an oxide semiconductor, when impurities and oxygen vacancies exist in a channel formation region of the oxide semiconductor, the resistance of the oxide semiconductor is

reduced in some cases. In addition, the electrical characteristics are easily changed, which might decrease the reliability.
(273) For example, silicon has a larger bonding energy with oxygen than indium and zinc have. For example, in the case where an In-M-Zn oxide is used as an oxide semiconductor and silicon enters the oxide semiconductor, oxygen contained in the oxide semiconductor is trapped by silicon, so that oxygen vacancies might be formed in the vicinity of indium or zinc.
(274) In the transistor using an oxide semiconductor in a channel formation region, when a low-resistance region is formed in the channel formation region, leakage current (a parasitic channel) between a source electrode and a drain electrode of the transistor is likely to be generated in the low-resistance region. Due to the parasitic channel, the transistor is likely to have poor characteristics, such as normally-on of the transistor, an increase in leakage current, or fluctuations (shift) in threshold voltage caused by stress application. When the processing accuracy of transistors is low, the parasitic channels vary between transistors, so that variations in transistor characteristics occur.
(275) Therefore, the impurities and oxygen vacancies are preferably reduced as much as possible in and around the channel formation region of the oxide semiconductor.
(276) «Other Semiconductor Materials»
(277) A semiconductor material that can be used for the oxide **230** is not limited to the above metal oxides. A semiconductor material which has a band gap (a semiconductor material that is not a zero-gap semiconductor) can be used for the oxide **230**. For example, a single element semiconductor such as silicon, a compound semiconductor such as gallium arsenide, or a layered material functioning as a semiconductor (also referred to as an atomic layered material or a two-dimensional material) is preferably used as a semiconductor material. In particular, a layered material functioning as a semiconductor is preferably used as a semiconductor material.
(278) Here, in this specification and the like, the layered material generally refers to a group of materials having a layered crystal structure. In the layered crystal structure, layers formed by covalent bonding or ionic bonding are stacked with bonding such as the Van der Waals force, which is weaker than covalent bonding or ionic bonding. The layered material has high electrical conductivity in a monolayer, that is, high two-dimensional electrical conductivity. When a material that functions as a semiconductor and has high two-dimensional electrical conductivity is used for a channel formation region, the transistor can have a high on-state current.
(279) Examples of the layered material include graphene, silicene, and chalcogenide. Chalcogenide is a compound containing chalcogen. Chalcogen is a general term of elements belonging to Group 16, which includes oxygen, sulfur, selenium, tellurium, polonium, and livermorium. Examples of chalcogenide include transition metal chalcogenide and chalcogenide of Group 13 elements.
(280) As the oxide **230**, a transition metal chalcogenide functioning as a semiconductor is preferably used, for example. Specific examples of the transition metal chalcogenide which can be used for the oxide **230** include molybdenum sulfide (typically MoS_2), molybdenum selenide (typically $MoSe_2$), molybdenum telluride (typically $MoTe_2$), tungsten sulfide (typically WS_2), tungsten selenide (typically WSe_2), tungsten telluride (typically WTe_2), hafnium sulfide (typically HfS_2), hafnium selenide (typically $HfSe_2$), zirconium sulfide (typically ZrS_2), zirconium selenide (typically $ZrSe_2$).
(281) <Manufacturing Method of Semiconductor Device>
(282) Next, a method for manufacturing a semiconductor device that is one embodiment of the present invention, which is illustrated in FIG. **7**A to FIG. **7**D, is described with reference to FIG. **9**A to FIG. **16**D.
(283) FIG. **9**A, FIG. **10**A, FIG. **11**A, FIG. **12**A, FIG. **13**A, FIG. **14**A, FIG. **15**A, and FIG. **16**A are top views. FIG. **9**B, FIG. **10**B, FIG. **11**B, FIG. **12**B, FIG. **13**B, FIG. **14**B, FIG. **15**B, and FIG. **16**B are cross-sectional views corresponding to portions indicated by the dashed-dotted lines A**1**-A**2** in FIG. **9**A, FIG. **10**A, FIG. **11**A, FIG. **12**A, FIG. **13**A, FIG. **14**A, FIG. **15**A, and FIG. **16**A,

respectively, and are also cross-sectional views of the transistor **200** in the channel length direction. Furthermore, FIG. **9**C, FIG. **10**C, FIG. **11**C, FIG. **12**C, FIG. **13**C, FIG. **14**C, FIG. **15**C, and FIG. **16**C are cross-sectional views corresponding to portions indicated by the dashed-dotted lines A**3**-A**4** in FIG. **9**A, FIG. **10**A, FIG. **11**A, FIG. **12**A, FIG. **13**A, FIG. **14**A, FIG. **15**A, and FIG. **16**A, respectively, and are also cross-sectional views of the transistor **200** in the channel width direction. Moreover, FIG. **9**D, FIG. **10**D, FIG. **11**D, FIG. **12**D, FIG. **13**D, FIG. **14**D, FIG. **15**D, and FIG. **16**D are cross-sectional views corresponding to portions indicated by the dashed-dotted lines A**5**-A**6** in FIG. **9**A, FIG. **10**A, FIG. **11**A, FIG. **12**A, FIG. **13**A, FIG. **14**A, FIG. **15**A, and FIG. **16**A, respectively. Note that for simplification of the drawing, some components are not illustrated in the top views of FIG. **9**A, FIG. **10**A, FIG. **11**A, FIG. **12**A, FIG. **13**A, FIG. **14**A, FIG. **15**A, and FIG. **16**A.
(284) First, a substrate (not illustrated) is prepared, and the insulator **211** is deposited over the substrate. The insulator **211** can be deposited by a sputtering method, a CVD method, a molecular beam epitaxy (MBE) method, a pulsed laser deposition (PLD) method, an ALD method, or the like.
(285) Note that the CVD method can be classified into a plasma enhanced CVD (PECVD) method using plasma, a thermal CVD (TCVD) method using heat, a photo CVD method using light, and the like. Moreover, the CVD method can be classified into a metal CVD (MCVD) method and a metal organic CVD (MOCVD) method depending on a source gas to be used.
(286) By a plasma CVD method, a high-quality film can be obtained at a relatively low temperature. Furthermore, a thermal CVD method is a deposition method that does not use plasma and thus enables less plasma damage to an object to be processed. For example, a wiring, an electrode, an element (a transistor, a capacitor, or the like), or the like included in a semiconductor device might be charged up by receiving electric charge from plasma. In that case, accumulated electric charge might break the wiring, the electrode, the element, or the like included in the semiconductor device. In contrast, such plasma damage does not occur in the case of a thermal CVD method, which does not use plasma, and thus the yield of the semiconductor device can be increased. In addition, a thermal CVD method does not cause plasma damage during deposition, so that a film with few defects can be obtained.
(287) As the ALD method, a thermal ALD method, in which a precursor and a reactant react with each other only by a thermal energy, a PEALD (Plasma Enhanced ALD) method, in which a reactant excited by plasma is used, and the like can be used.
(288) In an ALD method, one atomic layer can be deposited at a time using self-regulating characteristics of atoms. Thus, the ALD method has advantages such as deposition of an extremely thin film, deposition on a component with a high aspect ratio, deposition of a film with a small number of defects such as pinholes, deposition with good coverage, and low-temperature deposition. The PEALD (Plasma Enhanced ALD) method is sometimes preferable because deposition at lower temperature is possible by using plasma. Note that a precursor used in the ALD method sometimes contains impurities such as carbon. Thus, in some cases, a film provided by the ALD method contains impurities such as carbon in a larger amount than a film provided by another deposition method. Note that impurities can be quantified by X-ray photoelectron spectroscopy (XPS).
(289) Unlike a film formation method in which particles ejected from a target or the like are deposited, a CVD method and an ALD method are film deposition methods in which a film is deposited by reaction at a surface of an object. Thus, a CVD method and an ALD method are film deposition methods that enable favorable step coverage almost regardless of the shape of an object. In particular, an ALD method has excellent step coverage and excellent thickness uniformity and thus is suitable for covering a surface of an opening with a high aspect ratio, for example. On the other hand, an ALD method has a relatively low deposition rate, and thus is preferably used in combination with another film deposition method with a high deposition rate, such as a CVD method, in some cases.

(290) Each of a CVD method and an ALD method enables the composition of a film that is to be deposited to be controlled with a flow rate ratio of source gases. For example, by each of a CVD method and an ALD method, a film with a certain composition can be deposited depending on the flow rate ratio of the source gases. Moreover, with each of a CVD method and an ALD method, by changing the flow rate ratio of the source gases while depositing the film, a film whose composition is continuously changed can be formed. In the case where the film is deposited while changing the flow rate ratio of the source gases, as compared to the case where the film is deposited using a plurality of deposition chambers, the time taken for the deposition can be shortened because the time taken for transfer and pressure adjustment is omitted. Thus, the productivity of the semiconductor device can be increased in some cases.
(291) In this embodiment, for the insulator **211**, silicon nitride is deposited by a CVD method.
(292) Next, the insulator **212** is deposited over the insulator **211**. The insulator **212** can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. In this embodiment, for the insulator **212**, silicon nitride is deposited by a sputtering method.
(293) When an insulator through which copper is less likely to pass, such as silicon nitride, is used for the insulator **211** and the insulator **212** in such a manner, even in the case where a metal that is likely to diffuse, such as copper, is used for a conductor in a layer (not illustrated) below the insulator **211**, diffusion of the metal into an upper portion through the insulator **211** and the insulator **212** can be inhibited. The use of an insulator through which impurities such as water and hydrogen are less likely to pass, such as silicon nitride, can inhibit diffusion of impurities such as water and hydrogen contained in a layer under the insulator **211**.
(294) Next, the insulator **214** is deposited over the insulator **212**. The insulator **214** can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. In this embodiment, aluminum oxide is used for the insulator **214**.
(295) It is preferable that the hydrogen concentration of the insulator **212** be lower than the hydrogen concentration of the insulator **211**, and the hydrogen concentration of the insulator **214** be lower than the hydrogen concentration of the insulator **212**. The insulator **212** formed using silicon nitride by a sputtering method can have lower hydrogen concentration than the insulator **211** formed using silicon nitride by a CVD method. The insulator **214** formed using aluminum oxide can have lower hydrogen concentration than the insulator **212**.
(296) The transistor **200** is formed over the insulator **214** in a later step. It is preferable that a film near the transistor **200** have a relatively low hydrogen concentration and a film with a relatively high hydrogen concentration be away from the transistor **200**.
(297) Next, the insulator **216** is deposited over the insulator **214**. The insulator **216** can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. In this embodiment, silicon oxide or silicon oxynitride is used for the insulator **216**. The insulator **216** is preferably deposited by the deposition method using the gas in which hydrogen atoms are reduced or removed. Thus, the hydrogen concentration of the insulator **216** can be reduced.
(298) Then, an opening reaching the insulator **214** is formed in the insulator **216**. A groove and a slit, for example, are included in the category of the opening. A region where an opening is formed may be referred to as an opening portion. Wet etching can be used for the formation of the opening; however, dry etching is preferably used for microfabrication. As the insulator **214**, it is preferable to select an insulator that functions as an etching stopper film used in forming the groove by etching the insulator **216**. For example, in the case where silicon oxide or silicon oxynitride is used as the insulator **216** in which the groove is to be formed, silicon nitride, aluminum oxide, or hafnium oxide is preferably used as the insulator **214**.
(299) As a dry etching apparatus, a capacitively coupled plasma (CCP) etching apparatus including parallel plate electrodes can be used. The capacitively coupled plasma etching apparatus including

the parallel plate electrodes may have a structure in which a high-frequency voltage is applied to one of the parallel plate electrodes. Alternatively, a structure may be employed in which different high-frequency voltages are applied to one of the parallel plate electrodes. Alternatively, a structure may be employed in which high-frequency voltages with the same frequency are applied to the parallel plate electrodes. Alternatively, a structure may be employed in which high-frequency voltages with different frequencies are applied to the parallel plate electrodes. Alternatively, a dry etching apparatus including a high-density plasma source can be used. As the dry etching apparatus including a high-density plasma source, an inductively coupled plasma (ICP) etching apparatus or the like can be used, for example.

(300) After the formation of the opening, a conductive film to be the conductor **205***a* is deposited. The conductive film preferably includes a conductor that has a function of inhibiting passage of oxygen. For example, tantalum nitride, tungsten nitride, or titanium nitride can be used. Alternatively, a stacked-layer film of the conductor having a function of inhibiting passage of oxygen and tantalum, tungsten, titanium, molybdenum, aluminum, copper, or a molybdenum-tungsten alloy can be used. The conductive film can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like.

(301) In this embodiment, the conductive film to be the conductor **205***a* has a multilayer structure. First, tantalum nitride is deposited by a sputtering method, and titanium nitride is stacked over the tantalum nitride. When such metal nitrides are used for a lower layer of the conductor **205***b*, even in the case where a metal that is likely to diffuse, such as copper, is used for a conductive film to be a conductor **205***b* described below, outward diffusion of the metal from the conductor **205***a* can be inhibited.

(302) Next, a conductive film to be the conductor **205***b* is deposited. The conductive film can be deposited by a plating method, a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. In this embodiment, for the conductive film, a low-resistance conductive material such as copper is deposited.

(303) Next, CMP treatment is performed, thereby removing part of the conductive film to be the conductor **205***a* and part of the conductive film to be the conductor **205***b* to expose the insulator **216**. As a result, the conductor **205***a* and the conductor **205***b* remain only in the opening portion. Thus, the conductor **205** whose top surface is flat can be formed (see FIG. **9**A to FIG. **9**C). Note that the insulator **216** is partly removed by the CMP treatment in some cases.

(304) Although the conductor **205** is embedded in the opening in the insulator **216** in the above description, this embodiment is not limited to this structure. For example, the surface of the conductor **205** may be exposed in the following manner: the conductor **205** is formed over the insulator **214**, the insulator **216** is formed over the conductor **205**, and the insulator **216** is subjected to the CMP treatment so that the insulator **216** is partly removed.

(305) Next, the insulator **222** is deposited over the insulator **216** and the conductor **205**. An insulator containing an oxide of one or both of aluminum and hafnium is preferably deposited as the insulator **222**. The insulator containing an oxide of one or both of aluminum and hafnium has a barrier property against oxygen, hydrogen, and water. When the insulator **222** has a barrier property against hydrogen and water, hydrogen and water contained in components provided around the transistor **200** are inhibited from diffusing into the transistor **200** through the insulator **222**, and generation of oxygen vacancies in the oxide **230** can be inhibited.

(306) The insulator **222** can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like.

(307) Sequentially, heat treatment is preferably performed. The heat treatment is performed at a temperature higher than or equal to 250° C. and lower than or equal to 650° C., preferably higher than or equal to 300° C. and lower than or equal to 500° C., further preferably higher than or equal to 320° C. and lower than or equal to 450° C. Note that the heat treatment is performed in a nitrogen gas or inert gas atmosphere, or an atmosphere containing an oxidizing gas at 10 ppm or

more, 1% or more, or 10% or more. The heat treatment may be performed under reduced pressure. Alternatively, the heat treatment may be performed in such a manner that heat treatment is performed in a nitrogen gas or inert gas atmosphere, and then another heat treatment is performed in an atmosphere containing an oxidizing gas at 10 ppm or more, 1% or more, or 10% or more in order to compensate for released oxygen.

(308) In this embodiment, the heat treatment is performed in such a manner that treatment is performed at 400° C. in a nitrogen atmosphere for one hour after the deposition of the insulator **222**, and then another treatment is successively performed at 400° C. in an oxygen atmosphere for one hour. By the heat treatment, impurities such as water and hydrogen contained in the insulator **222** can be removed, for example. The heat treatment can also be performed after the deposition of the insulator **224**, for example.

(309) Next, the insulator **224** is deposited over the insulator **222**. The insulator **224** can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. In this embodiment, for the insulator **224**, silicon oxide or a silicon oxynitride film is deposited by a CVD method. The insulator **224** is preferably deposited by the deposition method using a gas in which hydrogen atoms are reduced or removed. Thus, the concentration of hydrogen in the insulator **224** can be reduced. As described above, the hydrogen concentration of the insulator **224** is preferably reduced because the insulator **224** is in contact with the oxide **230***a* in a later step.

(310) Here, plasma treatment containing oxygen may be performed under reduced pressure so that an excess-oxygen region can be formed in the insulator **224**. For the plasma treatment with oxygen, an apparatus including a power source for generating high-density plasma using a microwave is preferably used, for example. Alternatively, a power source for applying an RF (Radio Frequency) to a substrate side may be included. The use of high-density plasma enables high-density oxygen radicals to be produced, and RF application to the substrate side allows the oxygen radicals generated by the high-density plasma to be efficiently introduced into the insulator **224**. Alternatively, after plasma treatment with an inert gas is performed using this apparatus, plasma treatment with oxygen may be performed to compensate for released oxygen. Note that impurities such as water and hydrogen contained in the insulator **224** can be removed by selecting the conditions for the plasma treatment appropriately. In that case, the heat treatment does not need to be performed.

(311) Here, after aluminum oxide is deposited over the insulator **224** by a sputtering method, for example, the aluminum oxide may be subjected to CMP treatment until the insulator **224** is reached. The CMP treatment can planarize and smooth the surface of the insulator **224**. When the CMP treatment is performed on the aluminum oxide placed over the insulator **224**, it is easy to detect the endpoint of the CMP treatment. Although part of the insulator **224** is polished by the CMP treatment and the thickness of the insulator **224** is reduced in some cases, the thickness can be adjusted when the insulator **224** is deposited. Planarizing and smoothing the surface of the insulator **224** can prevent deterioration of the coverage with an oxide deposited later and a decrease in the yield of the semiconductor device in some cases. The deposition of aluminum oxide over the insulator **224** by a sputtering method is preferred because oxygen can be added to the insulator **224**.

(312) Next, an oxide film **230**A and an oxide film **230**B are deposited in this order over the insulator **224** (see FIG. **9**A to FIG. **9**D). Note that it is preferable to deposit the oxide film **230**A and the oxide film **230**B successively without exposure to the air. By the deposition without exposure to the air, impurities or moisture from the atmospheric environment can be prevented from being attached onto the oxide film **230**A and the oxide film **230**B, so that the vicinity of the interface between the oxide film **230**A and the oxide film **230**B can be kept clean.

(313) The oxide film **230**A and the oxide film **230**B can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like.

(314) For example, in the case where the oxide film **230**A and the oxide film **230**B are deposited by

a sputtering method, oxygen or a mixed gas of oxygen and a rare gas is used as a sputtering gas. Increasing the proportion of oxygen contained in the sputtering gas can increase the amount of excess oxygen in the deposited oxide films. In the case where the oxide films are deposited by a sputtering method, the above In-M-Zn oxide target or the like can be used.
(315) In particular, when the oxide film **230**A is deposited, part of oxygen contained in the sputtering gas is supplied to the insulator **224** in some cases. Thus, the proportion of oxygen contained in the sputtering gas is higher than or equal to 70%, preferably higher than or equal to 80%, further preferably 100%.
(316) In the case where the oxide film **230**B is formed by a sputtering method and the proportion of oxygen contained in the sputtering gas for deposition is higher than 30% and lower than or equal to 100%, preferably higher than or equal to 70% and lower than or equal to 100%, an oxygen-excess oxide semiconductor is formed. In a transistor using an oxygen-excess oxide semiconductor for its channel formation region, relatively high reliability can be obtained. Note that one embodiment of the present invention is not limited thereto. In the case where the oxide film **230**B is formed by a sputtering method and the proportion of oxygen contained in the sputtering gas for deposition is higher than or equal to 1% and lower than or equal to 30%, preferably higher than or equal to 5% and lower than or equal to 20%, an oxygen-deficient oxide semiconductor is formed. A transistor in which an oxygen-deficient oxide semiconductor is used for its channel formation region can have relatively high field-effect mobility. Furthermore, when the deposition is performed while the substrate is heated, the crystallinity of the oxide film can be improved.
(317) In this embodiment, the oxide film **230**A is formed by a sputtering method using an oxide target with In:Ga:Zn=1:3:4 [atomic ratio]. In addition, the oxide film **230**B is formed by a sputtering method using an oxide target with In:Ga:Zn=4:2:4.1 [atomic ratio], In:Ga:Zn=5:1:3 [atomic ratio], In:Ga:Zn=10:1:3 [atomic ratio], In:Zn=2:1 [atomic ratio], In:Zn=5:1 [atomic ratio], In:Zn=10:1 [atomic ratio], or indium. Note that each of the oxide films is formed to have characteristics required for the oxide **230***a* and the oxide **230***b* by selecting the deposition condition and the atomic ratio as appropriate.
(318) Next, an oxide film **243**A is deposited over the oxide film **230**B (see FIG. **9**A to FIG. **9**D). The oxide film **243**A can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. The atomic ratio of Ga to In in the oxide film **243**A is preferably greater than the atomic ratio of Ga to In in the oxide film **230**B. In this embodiment, the oxide film **243**A is formed by a sputtering method using an oxide target with In:Ga:Zn=1:3:4 [atomic ratio].
(319) Note that the insulator **222**, the insulator **224**, the oxide film **230**A, the oxide film **230**B, and the oxide film **243**A are preferably deposited without exposure to the air. For example, a multi-chamber deposition apparatus is used.
(320) Next, heat treatment may be performed. For the heat treatment, the above-described heat treatment conditions can be used. Through the heat treatment, impurities such as water and hydrogen in the oxide film **230**A, the oxide film **230**B, and the oxide film **243**A can be removed, for example. In this embodiment, treatment is performed at 400° C. in a nitrogen atmosphere for one hour, and treatment is successively performed at 400° C. in an oxygen atmosphere for one hour.
(321) Next, a conductive film **242**A is deposited over the oxide film **243**A (see FIG. **9**A to FIG. **9**D). The conductive film **242**A can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. Note that heat treatment may be performed before the formation of the conductive film **242**A. This heat treatment may be performed under reduced pressure, and the conductive film **242**A may be successively formed without exposure to the air. The treatment enables removal of moisture and hydrogen adsorbed onto the surface of the oxide film **243**A and the like, and further enables reductions in the moisture concentration and the hydrogen concentration of the oxide film **230**A, the oxide film **230**B, and the oxide film **243**A. The

heat treatment is preferably performed at a temperature higher than or equal to 100° C. and lower than or equal to 400° C. In this embodiment, the heat treatment is performed at 200° C.

(322) Next, the oxide film **230**A, the oxide film **230**B, the oxide film **243**A, and the conductive film **242**A are processed into island shapes by a lithography method to form the oxide **230***a*, the oxide **230***b*, an oxide layer **243**B, and a conductive layer **242**B (see FIG. **10**A to FIG. **10**D). A dry etching method or a wet etching method can be used for the processing. Processing by a dry etching method is suitable for microfabrication. The oxide film **230**A, the oxide film **230**B, the oxide film **243**A, and the conductive film **242**A may be processed under different conditions. Note that in this step, the thickness of a region of the insulator **224** which does not overlap with the oxide **230***a* becomes small in some cases.

(323) Note that in the lithography method, first, a resist is exposed to light through a mask. Next, a region exposed to light is removed or left using a developer, so that a resist mask is formed. Then, etching treatment through the resist mask is conducted, whereby a conductor, a semiconductor, an insulator, or the like can be processed into a desired shape. The resist mask is formed by, for example, exposure of the resist to KrF excimer laser light, ArF excimer laser light, EUV (Extreme Ultraviolet) light, or the like. Alternatively, a liquid immersion technique may be employed in which a gap between a substrate and a projection lens is filled with liquid (e.g., water) in light exposure. Alternatively, an electron beam or an ion beam may be used instead of the light. Note that a mask is unnecessary in the case of using an electron beam or an ion beam. Note that the resist mask can be removed by dry etching treatment such as ashing, wet etching treatment, wet etching treatment after dry etching treatment, or dry etching treatment after wet etching treatment.

(324) In addition, a hard mask formed of an insulator or a conductor may be used instead of the resist mask. In the case where a hard mask is used, a hard mask with a desired shape can be formed by forming an insulating film or a conductive film to be the hard mask material over the conductive film **242**A, forming a resist mask thereover, and then etching the hard mask material. The etching of the conductive film **242**A and the like may be performed after removing the resist mask or with the resist mask remaining. In the latter case, the resist mask sometimes disappears during the etching. The hard mask may be removed by etching after the etching of the conductive film **242**A and the like. Meanwhile, the hard mask is not necessarily removed when the hard mask material does not affect subsequent steps or can be utilized in the subsequent steps.

(325) Here, the oxide **230***a*, the oxide **230***b*, the oxide layer **243**B and the conductive layer **242**B are formed so as to at least partly overlap with the conductor **205**. It is preferable that the side surfaces of the oxide **230***a*, the oxide **230***b*, the oxide layer **243**B, and the conductive layer **242**B be substantially perpendicular to a top surface of the insulator **222**. When the side surfaces of the oxide **230***a*, the oxide **230***b*, the oxide layer **243**B, and the conductive layer **242**B are substantially perpendicular to the top surface of the insulator **222**, a plurality of transistors **200** can be provided in a smaller area and at a higher density. Alternatively, a structure may be employed in which the angle formed by the side surfaces of the oxide **230***a*, the oxide **230***b*, the oxide layer **243**B, and the conductive layer **242**B and the top surface of the insulator **222** is a small angle. In that case, the angle formed by the side surfaces of the oxide **230***a*, the oxide **230***b*, the oxide layer **243**B, and the conductive layer **242**B and the top surface of the insulator **222** is preferably greater than or equal to 60° and less than 70°. With such a shape, coverage with the insulator **272** and the like can be improved in a later step, so that defects such as voids can be reduced.

(326) There is a curved surface between the side surface of the conductive layer **242**B and a top surface of the conductive layer **242**B. That is, an end portion of the side surface and an end portion of the top surface are preferably curved. The curvature radius of the curved surface at the end portion of the conductive layer **242**B is greater than or equal to 3 nm and less than or equal to 10 nm, preferably greater than or equal to 5 nm and less than or equal to 6 nm, for example. When the end portions are not angular, the coverage with films in later deposition steps is improved.

(327) Next, the insulator **272** is deposited over the insulator **224**, the oxide **230***a*, the oxide **230***b*,

the oxide layer **243**B, and the conductive layer **242**B (see FIG. **11**B to FIG. **11**D). The insulator **272** can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. In this embodiment, for the insulator **272**, aluminum oxide is deposited by a sputtering method. When an aluminum oxide film is deposited by a sputtering method, oxygen can be injected into the insulator **224**.

(328) Next, the insulator **273** is deposited over the insulator **272**. The insulator **273** can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. In this embodiment, for the insulator **273**, silicon nitride is deposited by a sputtering method (see FIG. **11**B to FIG. **11**D).

(329) Next, an insulating film to be the insulator **280** is deposited over the insulator **273**. The insulating film can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. For example, as the insulator, a silicon oxide film may be deposited by a sputtering method, and a silicon oxide film may be deposited thereover by a PEALD method or a thermal ALD method. The insulating film is preferably deposited by the deposition method using the gas in which hydrogen atoms are reduced or removed. Thus, the hydrogen concentration of the insulator **280** can be reduced. Note that the heat treatment may be performed before the insulating film is deposited. The heat treatment may be performed under reduced pressure, and the insulating films may be successively formed without exposure to the air. The treatment can remove moisture and hydrogen adsorbed onto the surface of the insulator **273** and the like, and further can reduce the moisture concentration and the hydrogen concentration of the oxide **230***a*, the oxide **230***b*, the oxide layer **243**B, and the insulator **224**. The conditions for the above-described heat treatment can be employed.

(330) Next, the insulating film is subjected to CMP treatment, so that the insulator **280** having a flat top surface is formed (see FIG. **11**B to FIG. **11**D). Note that in a manner similar to that of the insulator **224**, aluminum oxide may be deposited over the insulator **280** by a sputtering method, for example, and the aluminum oxide may be subjected to CMP treatment until the insulator **280** is exposed.

(331) Then, part of the insulator **280**, part of the insulator **273**, part of the insulator **272**, part of the conductive layer **242**B, and part of the oxide layer **243**B are processed to form an opening reaching the oxide **230***b*. The opening is preferably formed to overlap with the conductor **205**. The conductor **242***a*, the conductor **242***b*, the oxide **243***a*, and the oxide **243***b* are formed by the formation of the opening (see FIG. **12**A to FIG. **12**D).

(332) When the opening is formed, an upper portion of the oxide **230***b* is slightly removed in some cases. In one embodiment of the present invention, however, in order to lengthen the effective channel length, part of the oxide **230***b* is processed using an insulator provided over the conductor **242***a* and the conductor **242***b* as a mask to form a groove portion in the oxide **230***b*. Note that the groove portion may be formed in the same step as the formation of the opening or in a step different from the formation of the opening, in accordance with the depth of the groove portion.

(333) The part of the insulator **280**, the part of the insulator **273**, the part of the insulator **272**, the part of the conductive layer **242**B, the part of the oxide layer **243**B, and the part of the oxide **230***b* can be processed by a dry etching method or a wet etching method. Processing by a dry etching method is suitable for microfabrication. The processing may be performed under different conditions. For example, part of the insulator **280** may be processed by a dry etching method, part of the insulator **273** may be processed by a wet etching method, and part of the insulator **272** may be processed by a dry etching method, part of the oxide layer **243**B, part of the conductive layer **242**B, and part of the oxide **230***b* may be processed by a dry etching method. Processing of part of the oxide layer **243**B and part of the conductive layer **242**B and processing of part of the oxide **230***b* may be performed under different conditions.

(334) In some cases, the treatment such as the dry etching performed thus far causes impurities due to an etching gas or the like to be attached to the surfaces or diffused to the inside of the oxide

230*a*, the oxide **230***b*, and the like. Examples of the impurities include fluorine and chlorine.
(335) In order to remove the above impurities and the like, cleaning treatment is performed. Examples of the cleaning method include wet cleaning using a cleaning solution, plasma treatment using plasma, and cleaning by heat treatment, and any of these cleanings may be performed in appropriate combination.
(336) As the wet cleaning, cleaning treatment may be performed using an aqueous solution in which ammonia water, oxalic acid, phosphoric acid, hydrofluoric acid, or the like is diluted with carbonated water or pure water; pure water; carbonated water; or the like. Alternatively, ultrasonic cleaning using such an aqueous solution, pure water, or carbonated water may be performed. Alternatively, such cleaning methods may be performed in combination as appropriate.
(337) In that case, the thickness of the insulator **224** in a region which overlaps with the opening and does not overlap with the oxide **230***b* is reduced in some cases.
(338) By the processing such as dry etching or the cleaning treatment, the thickness of the insulator **224** in a region that overlaps with the opening and does not overlap with the oxide **230***b* might become smaller than the thickness of the insulator **224** in a region that overlaps with the oxide **230***b*.
(339) Next, an oxide film **230**C is deposited (see FIG. **13**A to FIG. **13**D). Heat treatment may be performed before deposition of the oxide film **230**C, and it is preferable that the heat treatment be performed under reduced pressure and that the oxide film **230**C be successively deposited without exposure to the air. Preferably, the heat treatment is performed in an atmosphere containing oxygen. The treatment can remove moisture and hydrogen adsorbed onto the surface of the oxide **230***b* and the like, and further can reduce the moisture concentration and the hydrogen concentration of the oxide **230***a* and the oxide **230***b*. The heat treatment is preferably performed at a temperature higher than or equal to 100° C. and lower than or equal to 400° C. In this embodiment, the heat treatment is performed at 200° C.
(340) Here, it is preferable that the oxide film **230**C be provided in contact with at least the inner wall of the groove portion formed in the oxide **230***b*, part of the side surface of the oxide **243**, part of the side surface of the oxide **242**, part of the side surface of the insulator **272**, part of the side surface of the insulator **273**, and part of the side surface of the insulator **280**. When the conductor **242** is surrounded by the oxide **243**, the insulator **272**, the insulator **273**, and the oxide film **230**C, a decrease in the conductivity of the conductor **242** due to oxidation can be inhibited in subsequent steps.
(341) The oxide film **230**C can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. The oxide film **230**C is deposited by a deposition method similar to that for the oxide film **230**A or the oxide film **230**B depending on characteristics required for the oxide film **230**C. In this embodiment, the oxide film **230**C is deposited by a sputtering method using an oxide target with In:Ga:Zn=1:3:4 [atomic ratio], In:Ga:Zn=4:2:4.1 [atomic ratio], In:Ga:Zn=5:1:3 [atomic ratio], In:Ga:Zn=10:1:3 [atomic ratio], In:Zn=2:1 [atomic ratio], In:Zn=5:1 [atomic ratio], or In:Zn=10:1 [atomic ratio] or indium.
(342) Note that the oxide film **230**C may have a stacked-layer structure. For example, deposition may be performed by a sputtering method using an oxide target with In:Ga:Zn=4:2:4.1 [atomic ratio], In:Ga:Zn=5:1:3 [atomic ratio], In:Ga:Zn=10:1:3 [atomic ratio], In:Zn=2:1 [atomic ratio], In:Zn=5:1 [atomic ratio], or In:Zn=10:1 [atomic ratio], or indium, and deposition may be successively performed using an oxide target with In:Ga:Zn=1:3:4 [atomic ratio], Ga:Zn=2:1 [atomic ratio], or Ga:Zn=2:5 [atomic ratio].
(343) Part of oxygen contained in the sputtering gas is sometimes supplied to the oxide **230***a* and the oxide **230***b* during the deposition of the oxide film **230**C. Alternatively, in the deposition of the oxide film **230**C, part of oxygen contained in the sputtering gas is supplied to the insulator **280** in some cases. Therefore, the proportion of oxygen in the sputtering gas for formation of the oxide film **230**C is preferably 70% or higher, further preferably 80% or higher, still further preferably

100%.

(344) Next, an insulating film **250**A is deposited (see FIG. **13**A to FIG. **13**D). The heat treatment may be performed before the insulating film **250**A is deposited. The heat treatment may be performed under a reduced pressure and the insulating film **250**A be successively deposited without exposure to the air. The heat treatment is preferably performed in an oxygen-containing atmosphere. Such treatment can remove moisture and hydrogen adsorbed on the surface of the oxide film **230**C or the like and can reduce the moisture concentration and the hydrogen concentration of the oxide **230***a*, the oxide **230***b*, and the oxide film **230**C. The temperature of the heat treatment is preferably higher than or equal to 100° C. and lower than or equal to 400° C.

(345) The insulating film **250**A can be formed by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. The insulating film **250**A is preferably deposited by a deposition method using the gas in which the number of hydrogen atoms is reduced or hydrogen atoms are removed. Thus, the hydrogen concentration of the insulating film **250**A can be reduced. The hydrogen concentration is preferably reduced because the insulating film **250**A becomes the insulator **250** that is in contact with the oxide **230***c* in a later step.

(346) Note that in the case where the insulator **250** has a two-layer stacked structure, an insulating film below the insulator **250** and an insulating film over the insulator **250** are preferably formed successively without being exposed to an atmospheric environment. The formation of films without being exposed to the atmospheric environment can prevent impurity or moisture from the atmospheric environment from being attached to the insulating film below the insulator **250** and the insulating film over the insulator **250**; thus, the vicinity of an interface between the insulating film below the insulator **250** and the insulating film over the insulator **250** can be kept clean.

(347) Here, after the insulating film **250**A is deposited, the microwave treatment may be performed in an atmosphere containing oxygen under reduced pressure. By performing the microwave treatment, an electric field by a microwave is applied to the insulating film **250**A, the oxide film **230**C, the oxide **230***b*, the oxide **230***a*, and the like, so that V.sub.OH in the oxide film **230**C, the oxide **230***b*, and the oxide **230***a* can be divided into V.sub.O and hydrogen. Some hydrogen divided at this time is bonded to oxygen and is removed as H.sub.2O from the insulating film **250**A, the oxide film **230**C, the oxide **230***b*, and the oxide **230***a* in some cases. Some hydrogen may be gettered by the conductor **242** (the conductor **242***a* and the conductor **242***b*). Performing the microwave treatment in such a manner can reduce the hydrogen concentration in the insulating film **250**A, the oxide film **230**C, the oxide **230***b*, and the oxide **230***a*. Furthermore, oxygen is supplied to V.sub.O that can exist after V.sub.OH in the oxide **230***a*, the oxide **230***b*, and the oxide film **230**C is divided into V.sub.O and hydrogen, so that V.sub.O can be repaired or filled.

(348) After the microwave treatment, heat treatment may be performed with the reduced pressure being maintained. Such treatment enables hydrogen in the insulating film **250**A, the oxide film **230**C, the oxide **230***b*, and the oxide **230***a* to be removed efficiently. Some hydrogen may be gettered by the conductor **242** (the conductor **242***a* and the conductor **242***b*). Alternatively, it is possible to repeat the step of performing microwave treatment and the step of performing heat treatment with the reduced pressure being maintained after the microwave treatment. The repetition of the heat treatment enables hydrogen in the insulating film **250**A, the oxide film **230**C, the oxide **230***b*, and the oxide **230***a* to be removed more efficiently. Note that the temperature of the heat treatment is preferably higher than or equal to 300° C. and lower than or equal to 500° C.

(349) Furthermore, microwave plasma treatment improves the film quality of the insulating film **250**A, whereby diffusion of hydrogen, water, an impurity, or the like can be inhibited. Accordingly, hydrogen, water, an impurity, or the like can be inhibited from being diffused into the oxide **230***a* and the oxide **230***b* through the insulator **250** in the following step such as deposition of a conductive film to be the conductor **260** or the following treatment such as heat treatment.

(350) Next, a conductive film **260**A and a conductive film **260**B are deposited in this order (see FIG. **14**A to FIG. **14**D). The conductive film **260**A and the conductive film **260**B can be deposited

by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. In this embodiment, the conductive film **260**A is deposited by an ALD method, and the conductive film **260**B is deposited by a CVD method.

(351) Subsequently, the oxide film **230**C, the insulating film **250**A, the conductive film **260**A, and the conductive film **260**B are polished by CMP treatment until the insulator **280** is exposed, whereby the oxide **230***c*, the insulator **250**, and the conductor **260** (the conductor **260***a* and the conductor **260***b*) are formed (see FIG. **15**A to FIG. **15**D). Accordingly, the oxide **230***c* is positioned to cover the inner wall (the sidewall and bottom surface) of the opening reaching the oxide **230***b* and the groove portion in the oxide **230***b*. The insulator **250** is positioned to cover the inner wall of the opening and the groove portion with the oxide **230***c* therebetween. The conductor **260** is positioned to fill the opening and the groove portion in the oxide **230***b* with the oxide **230***c* and the insulator **250** therebetween.

(352) Next, heat treatment may be performed. In this embodiment, treatment is performed at 400° C. in a nitrogen atmosphere for one hour. The heat treatment enables reductions in the moisture concentration and the hydrogen concentration of the insulator **250** and the insulator **280**. After the heat treatment, the insulator **282** may be successively deposited without exposure to the air.

(353) Next, the insulator **282** is formed over the oxide **230***c*, the insulator **250**, the conductor **260**, and the insulator **280** (see FIG. **16**B to FIG. **16**D). The insulator **282** can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. Aluminum oxide is preferably deposited as the insulator **282** by a sputtering method, for example. The insulator **282** is deposited by a sputtering method in an oxygen-containing atmosphere, whereby oxygen can be added to the insulator **280** during the deposition. At this time, the insulator **282** is preferably deposited while the substrate is being heated. It is preferable to form the insulator **282** in contact with the top surface of the conductor **260** because oxygen contained in the insulator **280** can be inhibited from being absorbed into the conductor **260** in a later heat treatment.

(354) Next, the insulator **283** is formed over the insulator **282** (see FIG. **16**B to FIG. **16**D). The insulator **283** can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. As the insulator **283**, silicon nitride or silicon nitride oxide is preferably deposited. In addition, the insulator **283** may have a multilayer structure. For example, silicon nitride may be deposited by a sputtering method and silicon nitride may be deposited by a CVD method over the silicon nitride.

(355) Next, heat treatment may be performed. In this embodiment, treatment is performed at 400° C. in a nitrogen atmosphere for one hour. By the heat treatment, oxygen added by the deposition of the insulator **282** is diffused to the insulator **280** and can be supplied to the oxide **230***a* and the oxide **230***b* through the oxide **230***c*. Note that the heat treatment is not necessarily performed after the deposition of the insulator **283** and may be performed after the deposition of the insulator **282**.

(356) Next, the insulator **284** may be deposited over the insulator **283**. The insulator **284** can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. Silicon nitride is preferably deposited as the insulator **284** by a sputtering method, for example.

(357) Next, openings reaching the conductor **242***a* and the conductor **242***b* are formed in the insulator **272**, the insulator **273**, the insulator **280**, the insulator **282**, the insulator **283**, and the insulator **284**. The openings are formed by a lithography method.

(358) Subsequently, an insulating film to be the insulator **241** (the insulator **241***a* and the insulator **241***b*) is deposited and subjected to anisotropic etching, so that the insulator **241** is formed. The insulating film can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. As the insulating film, an insulating film having a function of inhibiting passage of oxygen is preferably used. For example, silicon nitride is preferably deposited by a PEALD method. Silicon nitride is preferable because it has high blocking property against hydrogen.

(359) As an anisotropic etching for the insulating film to be the insulator **241**, a dry etching method may be performed, for example. When the insulator **241** is provided on the sidewall portions of the openings, passage of oxygen from the outside can be inhibited and oxidation of the conductor **240***a* and the conductor **240***b* to be formed next can be prevented. Furthermore, impurities such as water and hydrogen can be prevented from diffusing from the conductor **240***a* and the conductor **240***b* to the outside.
(360) Next, a conductive film to be the conductor **240***a* and the conductor **240***b* is formed. The conductive film desirably has a stacked-layer structure that includes a conductor having a function of inhibiting passage of impurities such as water and hydrogen. For example, a stacked layer of tantalum nitride, titanium nitride, or the like and tungsten, molybdenum, copper, or the like can be employed. The conductive film can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like.
(361) Next, CMP treatment is performed, thereby removing part of the conductive film to be the conductor **240***a* and the conductor **240***b* to expose the insulator **284**. As a result, the conductive film remains only in the openings, so that the conductor **240***a* and the conductor **240***b* having flat top surfaces can be formed (see FIG. **7**A to FIG. **7**D). Note that the insulator **284** is partly removed by the CMP treatment in some cases.
(362) Next, a conductive film to be the conductor **246** is formed. The conductive film can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like.
(363) Next, the conductive film to be the conductor **246** is processed by a lithography method, thereby forming the conductor **246***a* in contact with the top surface of the conductor **240***a* and the conductor **246***b* in contact with the top surface of the conductor **240***b* (see FIG. **7**A to FIG. **7**D). At this time, the insulator **284** in a region not overlapping with the conductor **246***a* and the conductor **246***b* is sometimes removed.
(364) Next, the insulator **286** is deposited over the conductor **246** and the insulator **284** (see FIG. **7**A to FIG. **7**D). The insulator **286** can be deposited by a sputtering method, a CVD method, an MBE method, a PLD method, an ALD method, or the like. The insulator **286** may have a multilayer structure. For example, silicon nitride may be deposited by a sputtering method and silicon nitride may be deposited by a CVD method over the silicon nitride.
(365) Through the above process, the semiconductor device including the transistor **200** illustrated in FIG. **7**A to FIG. **7**D can be manufactured. As shown in FIG. **9**A to FIG. **16**D, the transistor **200** can be fabricated with use of the method for manufacturing the semiconductor device described in this embodiment.
(366) <Modification Example of Semiconductor Device>
(367) An example of a semiconductor device of one embodiment of the present invention will be described below with reference to FIG. **17**A to FIG. **17**D.
(368) FIG. **17**A is a top view of the semiconductor device. FIG. **17**B is a cross-sectional view corresponding to a portion indicated by the dashed-dotted line A**1**-A**2** in FIG. **17**A. FIG. **17**C is a cross-sectional view corresponding to a portion indicated by the dashed-dotted line A**3**-A**4** in FIG. **17**A. FIG. **17**D is a cross-sectional view corresponding to a portion indicated by the dashed-dotted line A**5**-A**6** in FIG. **17**A. Note that for clarity of the drawing, some components are not illustrated in the top view of FIG. **17**A.
(369) Note that in the semiconductor devices illustrated in FIG. **17**A to FIG. **17**D, components having the same functions as the components included in the semiconductor device described in <Structure example 2 of semiconductor device> are denoted by the same reference numerals. Note that the materials described in detail in <Structure example 2 of semiconductor device> can also be used as materials for the semiconductor devices in this section.
(370) A semiconductor device illustrated in FIG. **17**A to FIG. **17**D is a modification example of the semiconductor device illustrated in FIG. **7**A to FIG. **7**D. The semiconductor device in FIG. **17**A to

FIG. **17**D is different from the semiconductor device in FIG. **7**A to FIG. **7**D in the shapes of the insulator **283** and the insulator **284**. An insulator **274** and an insulator **287** are included, which is also a difference. In addition, a structure in which the oxide **230***c* has a two-layer stacked structure is shown.
(371) In the semiconductor device illustrated in FIG. **17**A to FIG. **17**D, the insulator **212**, the insulator **214**, the insulator **216**, the insulator **222**, the insulator **224**, the insulator **272**, the insulator **273**, the insulator **280**, and the insulator **282** are patterned, and the insulator **287** is provided in contact with side surfaces of the insulator **212**, the insulator **214**, the insulator **216**, the insulator **222**, the insulator **224**, the insulator **272**, the insulator **273**, the insulator **280**, and the insulator **282**. The insulator **283** and the insulator **284** have a structure covering the insulator **212**, the insulator **214**, the insulator **216**, the insulator **222**, the insulator **224**, the insulator **272**, the insulator **273**, the insulator **280**, the insulator **282**, and the insulator **287**. That is, the insulator **283** is in contact with a top surface of the insulator **282**, a top surface and a side surface of the insulator **287**, and a top surface of the insulator **211**, and the insulator **284** is in contact with a top surface and a side surface of the insulator **283**. Accordingly, the insulator **212**, the insulator **214**, the insulator **216**, the insulator **222**, the insulator **224**, the insulator **272**, the insulator **273**, the insulator **280**, the insulator **282**, and the insulator **287** in addition to the oxide **230** and the like are isolated from the outside by the insulator **283**, the insulator **284**, and the insulator **211**. In other words, the transistor **200** is placed in a region sealed by the insulator **283**, the insulator **284**, and the insulator **211**.
(372) For example, the insulator **212**, the insulator **214**, the insulator **287**, and the insulator **282** are preferably formed with a material having functions of capturing hydrogen and fixing hydrogen, and the insulator **211**, the insulator **283**, and the insulator **284** are preferably formed with a material having a function of suppressing diffusion of hydrogen and oxygen. Typically, aluminum oxide can be used for the insulator **212**, the insulator **214**, the insulator **287**, and the insulator **282**. Typically, silicon nitride can be used for the insulator **211**, the insulator **283**, and the insulator **284**.
(373) With the above structure, entry of hydrogen contained in the region outside the sealed region into the sealed region can be inhibited.
(374) The transistor **200** illustrated in FIG. **17**A to FIG. **17**D shows a structure where the insulator **211**, the insulator **283**, and the insulator **284** each have a single layer; however, the present invention is not limited thereto. For example, a structure in which the insulator **211**, the insulator **283**, and the insulator **284** each have a stacked structure including two or more layers may be employed.
(375) The insulator **274** functions as an interlayer film. The dielectric constant of the insulator **274** is preferably lower than that of the insulator **214**. When a material with a low dielectric constant is used for the interlayer film, the parasitic capacitance generated between wirings can be reduced. The insulator **274** can be provided using a material similar to that for the insulator **280**, for example.
(376) The transistor **200** illustrated in FIG. **17**A to FIG. **17**D shows a structure in which the oxide **230***c* has a stacked structure of an oxide **230***c***1** and an oxide **230***c***2**.
(377) The oxide **230***c***2** preferably contains at least one of the metal elements contained in the metal oxide used as the oxide **230***c***1**, and further preferably contains all of these metal elements. For example, it is preferable that an In—Ga—Zn oxide or an IN—Zn oxide be used as the oxide **230***c***1**, and an In—Ga—Zn oxide, a Ga—Zn oxide, or gallium oxide be used as the oxide **230***c***2**. Accordingly, the density of defect states at the interface between the oxide **230***c***1** and the oxide **230***c***2** can be decreased.
(378) The conduction band minimum of each of the oxide **230***a* and the oxide **230***c***2** is preferably closer to the vacuum level than the conduction band minimum of each of the oxide **230***b* and the oxide **230***c***1**. In other words, the electron affinity of each of the oxide **230***a* and the oxide **230***c***2** is preferably smaller than the electron affinity of each of the oxide **230***b* and the oxide **230***c***1**. In that case, it is preferable that a metal oxide that can be used as the oxide **230***a* be used as the oxide

230*c***2**, and a metal oxide that can be used as the oxide **230***b* be used as the oxide **230***c***1**. At this time, not only the oxide **230***b* but also the oxide **230***c***1** serves as a main carrier path in some cases. The metal oxide that can be used as the oxide **230***b* is used for the oxide **230***c***1**, whereby an increase in the effective channel length on the top surface of the channel formation region can be inhibited and a decrease in the on-state current of the transistor **200** can be inhibited.

(379) Specifically, an In—Zn oxide or a metal oxide with a composition of In:Ga:Zn=4:2:3 [atomic ratio] or in the vicinity thereof, a composition of In:Ga:Zn=5:1:6 [atomic ratio] or in the vicinity thereof, or a composition of In:Ga:Zn=5:1:3 [atomic ratio] or in the vicinity thereof, a composition of In:Ga:Zn=10:1:3 [atomic ratio] or in the vicinity thereof may be used for the oxide **230***c***1**; a gallium oxide or a metal oxide with a composition of In:Ga:Zn=1:3:4 [atomic ratio] or in the vicinity thereof, a composition of Ga:Zn=2:1 [atomic ratio] or in the vicinity thereof, a composition of Ga:Zn=2:5 [atomic ratio] or in the vicinity thereof may be used for the oxide **230***c***2**.

(380) The oxide **230***c***2** is preferably a metal oxide that inhibits diffusion or passage of oxygen, compared to the oxide **230***c***1**. Providing the oxide **230***c***2** between the insulator **250** and the oxide **230***c***1** can inhibit diffusion of oxygen contained in the insulator **280** into the insulator **250**. Accordingly, the oxygen can be efficiently supplied to the oxide **230***b* through the oxide **230***c***1**.

(381) When the atomic ratio of In to the metal element of the main component in the metal oxide used as the oxide **230***c***2** is lower than the atomic ratio of In to the metal element of the main component in the metal oxide used as the oxide **230***c***1**, the diffusion of In into the insulator **250** side can be inhibited. Since the insulator **250** functions as a gate insulator, the transistor exhibits poor characteristics when In enters the insulator **250** and the like. Thus, the oxide **230***c***2** provided between the oxide **230***c***1** and the insulator **250** allows the semiconductor device to have high reliability.

(382) Note that the oxide **230***c***1** may be provided for each of the transistors **200**. Accordingly, the oxide **230***c***1** of the transistor **200** is not necessarily in contact with the oxide **230***c***1** of another transistor **200** adjacent to the transistor **200**. Furthermore, the oxide **230***c***1** of the transistor **200** may be apart from the oxide **230***c***1** of another transistor **200** adjacent to the transistor **200**. In other words, a structure in which the oxide **230***c***1** is not located between the transistor **200** and another transistor **200** adjacent to the transistor **200** may be employed.

(383) When the above structure is employed for the semiconductor device where a plurality of transistors **200** are located in the channel width direction, the oxide **230***c* can be independently provided for each transistor **200**. Accordingly, generation of a parasitic transistor between the transistor **200** and another transistor **200** adjacent to the transistor **200** can be prevented, and generation of the leakage path can be prevented. Thus, a semiconductor device that has favorable electrical characteristics and can be miniaturized or highly integrated can be provided.

(384) For example, when a side end portion of the oxide **230***c***1** of the transistor **200** faces a side end portion of the oxide **230***c***1** of another transistor **200** adjacent to the transistor **200** and a distance between the side end portions in the channel width direction of the transistor **200** is denoted by W.sub.1, W.sub.1 is made greater than 0 nm. In the channel width direction of the transistor **200**, when a side end portion of the oxide **230***a* of the transistor **200** faces a side end portion of the oxide **230***a* of another transistor **200** adjacent to the transistor **200** and the distance between the side end portions is denoted by W.sub.2, a value of a ratio of W.sub.1 to W.sub.2 (W.sub.1/W.sub.2) is preferably greater than 0 and less than 1, further preferably greater than or equal to 0.1 and less than or equal to 0.9, still further preferably greater than or equal to 0.2 and less than or equal to 0.8. Note that W.sub.2 may be a distance between a side end portion of the oxide **230***b* of the transistor **200** and a side end portion of the oxide **230***b* of another transistor **200** adjacent to the transistor **200** when the end portions face each other.

(385) By a reduction in the ratio of W.sub.1 to W.sub.2 (W.sub.1/W.sub.2), even when misalignment of a region where the oxide **230***c***1** is not located between the transistor **200** and another transistor **200** adjacent to the transistor **200** occurs, the oxide **230***c***1** of the transistor **200**

can be apart from the oxide **230***c***1** of another transistor **200** adjacent to the transistor **200**.
(386) By an increase in the ratio of W.sub.1 to W.sub.2 (W.sub.1/W.sub.2), even when the interval between the transistor **200** and another transistor **200** adjacent to the transistor **200** is decreased, the width of the minimum feature size can be secured, and further miniaturization and higher integration of the semiconductor device can be achieved.
(387) Note that each of the conductor **260**, the insulator **250**, and the oxide **230***c***2** may be shared by adjacent transistors **200**. In other words, the conductor **260** of the transistor **200** includes a region continuous with the conductor **260** of another transistor **200** adjacent to the transistor **200**. In addition, the insulator **250** of the transistor **200** includes a region continuous with the insulator **250** of another transistor **200** adjacent to the transistor **200**. In addition, the oxide **230***c***2** of the transistor **200** includes a region continuous with the oxide **230***c***2** of another transistor **200** adjacent to the transistor **200**.
(388) In the above structure, the oxide **230***c***2** includes a region in contact with the insulator **224** between the transistor **200** and another transistor **200** adjacent to the transistor **200**.
(389) Note that like the oxide **230***c***1**, the oxide **230***c***2** of the transistor **200** may be apart from the oxide **230***c***2** of another transistor **200** adjacent to the transistor **200**. In that case, the insulator **250** includes a region in contact with the insulator **224** between the transistor **200** and another transistor **200** adjacent to the transistor **200**.
(390) <Application Example Semiconductor Device>
(391) An example of a semiconductor device including the transistor **200** of one embodiment of the present invention which is different from the semiconductor device described in the above <Structure example 2 of semiconductor device> and <Modification example of semiconductor device> will be described below with reference to FIG. **18**A and FIG. **18**B. Note that in the semiconductor device illustrated in FIG. **18**A and FIG. **18**B, components having the same functions as the components in the semiconductor device described in <Modification example of semiconductor device> (see FIG. **17**A to FIG. **17**D) are denoted by the same reference numerals. Note that also in this section, the materials described in detail in <Structure example 2 of semiconductor device> and <Modification example of semiconductor device> can be used as the materials for the transistor **200**.
(392) FIG. **18**A and FIG. **18**B each illustrate a structure in which a plurality of transistors (a transistor **200_1** to a transistor **200_***n*) are sealed with the insulator **283**, the insulator **284**, and the insulator **211**. Note that although the plurality of transistors appear to be arranged in the channel length direction in FIG. **18**A and FIG. **18**B, the present invention is not limited thereto. The plurality of transistors may be arranged in the channel width direction or in a matrix. Depending on the design, the transistors may be arranged without regularity.
(393) As illustrated in FIG. **18**A, a portion where the insulator **283** is in contact with the insulator **211** (hereinafter, sometimes referred to as a sealing portion **265**) is formed outside the plurality of transistors (the transistor **200_1** to the transistor **200_***n*). The sealing portion **265** is formed to surround the plurality of transistors (also referred to as a transistor group). With such a structure, the plurality of transistors can be surrounded by the insulator **283** and the insulator **211**. As described above, a plurality of transistor groups surrounded by the sealing portion **265** are provided over a substrate.
(394) A dicing line (sometimes referred to as a scribe line, a dividing line, or a cutting line) may be provided to overlap with the sealing portion **265**. The above substrate is divided at the dicing line, so that the transistor group surrounded by the sealing portion **265** is taken out as one chip.
(395) Although FIG. **18**A shows an example in which the plurality of transistors (the transistor **200_1** to the transistor **200_***n*) are surrounded by one sealing portion **265**, the present invention is not limited thereto. As illustrated in FIG. **18**B, the plurality of transistors may be surrounded by a plurality of sealing portions. In FIG. **18**B, the plurality of transistors are surrounded by a sealing portion **265***a* and are further surrounded by an outer sealing portion **265***b*.

(396) When the plurality of transistors the transistor (**200_1** to the transistor **200_***n*) are surrounded by the plurality of sealing portions in this manner, a portion where the insulator **283** is in contact with the insulator **211** increases, which further can improve adhesion between the insulator **283** and the insulator **211**. Accordingly, the plurality of transistors can be sealed more surely.
(397) In that case, a dicing line may be provided to overlap with the sealing portion **265***a* or the sealing portion **265***b*, or may be provided between the sealing portion **265***a* and the sealing portion **265***b*.
(398) According to one embodiment of the present invention, a semiconductor device having a high on-state current can be provided. According to one embodiment of the present invention, a semiconductor device with less variations in transistor characteristics can be provided. Furthermore, according to one embodiment of the present invention, a highly reliable semiconductor device can be provided. Furthermore, according to one embodiment of the present invention, a semiconductor device having favorable electrical characteristics can be provided. Furthermore, according to one embodiment of the present invention, a semiconductor device that can be miniaturized or highly integrated can be provided. Furthermore, according to one embodiment of the present invention, a semiconductor device with low power consumption can be provided.
(399) The structure, the method, and the like described above in this embodiment can be used in an appropriate combination with the structures, the methods, and the like described in the other embodiments and examples.
Embodiment 3
(400) In this embodiment, one embodiment of a semiconductor device will be described with reference to FIG. **19** and FIG. **25**.
(401) [Storage Device **1**]
(402) FIG. **19** illustrates an example of a semiconductor device (storage device) of one embodiment of the present invention. In the semiconductor device of one embodiment of the present invention, the transistor **200** is provided above a transistor **300**, and a capacitor **100** is provided above the transistor **300** and the transistor **200**. The transistor **200** described in the above embodiment can be used as the transistor **200** described in the above embodiment.
(403) The transistor **200** is a transistor whose channel is formed in a semiconductor layer containing an oxide semiconductor. Since the transistor **200** has a low off-state current, a storage device including the transistor **200** can retain stored data for a long time. In other words, such a storage device does not require refresh operation or has an extremely low frequency of the refresh operation, which leads to a sufficient reduction in power consumption of the storage device.
(404) In the semiconductor device illustrated in FIG. **19**, a wiring **1001** is electrically connected to a source of the transistor **300**, and a wiring **1002** is electrically connected to a drain of the transistor **300**. A wiring **1003** is electrically connected to one of the source and the drain of the transistor **200**, a wiring **1004** is electrically connected to the first gate of the transistor **200**, and a wiring **1006** is electrically connected to the second gate of the transistor **200**. A gate of the transistor **300** and the other of the source and the drain of the transistor **200** are electrically connected to one electrode of the capacitor **100**. A wiring **1005** is electrically connected to the other electrode of the capacitor **100**.
(405) Furthermore, by arranging the storage devices illustrated in FIG. **19** in a matrix, a memory cell array can be formed.
(406) <Transistor **300**>
(407) The transistor **300** is provided over a substrate **311** and includes a conductor **316** functioning as a gate, an insulator **315** functioning as a gate insulator, a semiconductor region **313** formed of part of the substrate **311**, and a low-resistance region **314***a* and a low-resistance region **314***b* functioning as the source region and the drain region. The transistor **300** may be a p-channel transistor or an n-channel transistor.

(408) Here, in the transistor **300** illustrated in FIG. **19**, the semiconductor region **313** (part of the substrate **311**) in which a channel is formed has a convex shape. Furthermore, the conductor **316** is provided so as to cover a side surface and the top surface of the semiconductor region **313** with the insulator **315** positioned therebetween. Note that a material adjusting the work function may be used for the conductor **316**. Such a transistor **300** is also referred to as a FIN-type transistor because it utilizes a convex portion of the semiconductor substrate. Note that an insulator functioning as a mask for forming the convex portion may be placed in contact with an upper portion of the convex portion. Furthermore, although the case where the convex portion is formed by processing part of the semiconductor substrate is described here, a semiconductor film having a convex shape may be formed by processing an SOI substrate.
(409) Note that the transistor **300** illustrated in FIG. **19** is an example and the structure is not limited thereto; an appropriate transistor may be used in accordance with a circuit configuration or a driving method.
(410) <Capacitor **100**>
(411) The capacitor **100** is provided above the transistor **200**. The capacitor **100** includes a conductor **110** functioning as a first electrode, a conductor **120** functioning as a second electrode, and an insulator **130** functioning as a dielectric. Here, as the insulator **130**, an insulator that can be used as the insulator **286** described in the above embodiment is preferably used.
(412) For example, a conductor **112** and the conductor **110** over the conductor **246** can be formed at the same time. Note that the conductor **112** functions as a plug or a wiring that is electrically connected to the capacitor **100**, the transistor **200**, or the transistor **300**.
(413) The conductor **112** and the conductor **110** illustrated in FIG. **19** each have a single-layer structure; however, the structure is not limited thereto, and a stacked-layer structure of two or more layers may be employed. For example, between a conductor having a barrier property and a conductor having high conductivity, a conductor that is highly adhesive to the conductor having a barrier property and the conductor having high conductivity may be formed.
(414) For the insulator **130**, for example, silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, aluminum oxide, aluminum oxynitride, aluminum nitride oxide, aluminum nitride, hafnium oxide, hafnium oxynitride, hafnium nitride oxide, hafnium nitride, or the like is used, and a stacked layer or a single layer can be provided.
(415) For example, for the insulator **130**, a stacked-layer structure using a material with high dielectric strength such as silicon oxynitride and a high dielectric constant (high-k) material is preferably used. In the capacitor **100** having such a structure, a sufficient capacitance can be ensured owing to the high dielectric constant (high-k) insulator, and the dielectric strength can be increased owing to the insulator with high dielectric strength, so that the electrostatic breakdown of the capacitor **100** can be inhibited.
(416) As an insulator of a high dielectric constant (high-k) material (a material having a high relative dielectric constant), gallium oxide, hafnium oxide, zirconium oxide, an oxide containing aluminum and hafnium, an oxynitride containing aluminum and hafnium, an oxide containing silicon and hafnium, an oxynitride containing silicon and hafnium, a nitride containing silicon and hafnium, and the like can be given.
(417) Examples of a material with high dielectric strength (a material having a low relative dielectric constant) include silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, silicon oxide to which fluorine is added, silicon oxide to which carbon is added, silicon oxide to which carbon and nitrogen are added, porous silicon oxide, and a resin.
(418) <Wiring Layer>
(419) A wiring layer provided with an interlayer film, a wiring, a plug, and the like may be provided between the structure bodies. A plurality of wiring layers can be provided in accordance with the design. Here, a plurality of conductors functioning as plugs or wirings are collectively denoted by the same reference numeral in some cases. Furthermore, in this specification and the

like, a wiring and a plug electrically connected to the wiring may be a single component. That is, there are cases where part of a conductor functions as a wiring and another part of the conductor functions as a plug.
(420) For example, an insulator **320**, an insulator **322**, an insulator **324**, and the insulator **326** are stacked over the transistor **300** in this order as interlayer films. A conductor **328**, a conductor **330**, and the like that are electrically connected to the capacitor **100** or the transistor **200** are embedded in the insulator **320**, the insulator **322**, the insulator **324**, and the insulator **326**. Note that the conductor **328** and the conductor **330** function as plugs or wirings.
(421) The insulator functioning as an interlayer film may function as a planarization film that covers an uneven shape thereunder. For example, the top surface of the insulator **322** may be planarized by planarization treatment using a chemical mechanical polishing (CMP) method or the like to improve planarity.
(422) A wiring layer may be provided over the insulator **326** and the conductor **330**. For example, in FIG. **19**, an insulator **350**, an insulator **352**, and an insulator **354** are provided to be stacked in this order. Furthermore, a conductor **356** is formed in the insulator **350**, the insulator **352**, and the insulator **354**. The conductor **356** functions as a plug or a wiring.
(423) Similarly, a conductor **218**, a conductor (conductor **205**) included in the transistor **200**, and the like are embedded in an insulator **210**, the insulator **211**, the insulator **212**, the insulator **214**, and the insulator **216**. Note that the conductor **218** functions as a plug or a wiring that is electrically connected to the capacitor **100** or the transistor **300**. In addition, an insulator **150** is provided over the conductor **120** and the insulator **130**.
(424) Here, like the insulator **241** described in the above embodiment, an insulator **217** is provided in contact with the side surface of the conductor **218** functioning as a plug. The insulator **217** is provided in contact with the inner wall of the opening formed in the insulator **210**, the insulator **211**, the insulator **212**, the insulator **214**, and the insulator **216**. That is, the insulator **217** is provided between the conductor **218** and the insulator **210**, the insulator **211**, the insulator **212**, the insulator **214**, and the insulator **216**. Note that the conductor **205** and the conductor **218** can be formed in parallel; thus, the insulator **217** is sometimes formed in contact with the side surface of the conductor **205**.
(425) As the insulator **217**, an insulator such as silicon nitride, aluminum oxide, or silicon nitride oxide may be used. Since the insulator **217** is provided in contact with the insulator **210**, the insulator **211**, the insulator **212**, the insulator **214**, and the insulator **222**, the entry of impurities such as water and hydrogen into the oxide **230** through the conductor **218** from the insulator **210**, the insulator **216**, or the like can be inhibited. In particular, silicon nitride is suitable because of having a high blocking property against hydrogen. Moreover, oxygen contained in the insulator **210** or the insulator **216** can be prevented from being absorbed into the conductor **218**.
(426) The insulator **217** can be formed in a manner similar to that of the insulator **241**. For example, silicon nitride is deposited by a PEALD method and an opening reaching the conductor **356** is formed by anisotropic etching.
(427) As an insulator that can be used as an interlayer film, an insulating oxide, an insulating nitride, an insulating oxynitride, an insulating nitride oxide, an insulating metal oxide, an insulating metal oxynitride, an insulating metal nitride oxide, or the like is given.
(428) For example, when a material having a low relative dielectric constant is used for the insulator functioning as an interlayer film, the parasitic capacitance between wirings can be reduced. Accordingly, a material is preferably selected depending on the function of an insulator.
(429) For example, for the insulator **150**, the insulator **210**, the insulator **352**, the insulator **354**, or the like, an insulator having a low relative dielectric constant is preferably used. For example, the insulator preferably includes silicon nitride oxide, silicon nitride, silicon oxide to which fluorine is added, silicon oxide to which carbon is added, silicon oxide to which carbon and nitrogen are added, porous silicon oxide, a resin, or the like. Alternatively, the insulator preferably has a

stacked-layer structure of a resin and silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, silicon oxide to which fluorine is added, silicon oxide to which carbon is added, silicon oxide to which carbon and nitrogen are added, or porous silicon oxide. When silicon oxide or silicon oxynitride, which is thermally stable, is combined with a resin, the stacked-layer structure can have thermal stability and a low relative dielectric constant. Examples of the resin include polyester, polyolefin, polyamide (e.g., nylon and aramid), polyimide, polycarbonate, and acrylic.
(430) When a transistor using an oxide semiconductor is surrounded by insulators having a function of inhibiting passage of oxygen and impurities such as hydrogen, the electrical characteristics of the transistor can be stable. Thus, the insulator having a function of inhibiting passage of oxygen and impurities such as hydrogen can be used for the insulator **214**, the insulator **211**, the insulator **212**, the insulator **350**, and the like.
(431) As the insulator having a function of inhibiting passage of oxygen and impurities such as hydrogen, a single layer or stacked layers of an insulator containing, for example, boron, carbon, nitrogen, oxygen, fluorine, magnesium, aluminum, silicon, phosphorus, chlorine, argon, gallium, germanium, yttrium, zirconium, lanthanum, neodymium, hafnium, or tantalum is used. Specifically, as the insulator having a function of inhibiting passage of oxygen and impurities such as hydrogen, a metal oxide such as aluminum oxide, magnesium oxide, gallium oxide, germanium oxide, yttrium oxide, zirconium oxide, lanthanum oxide, neodymium oxide, hafnium oxide, or tantalum oxide; silicon nitride oxide; or silicon nitride can be used.
(432) As the conductors that can be used for a wiring or a plug, a material containing one or more kinds of metal elements selected from aluminum, chromium, copper, silver, gold, platinum, tantalum, nickel, titanium, molybdenum, tungsten, hafnium, vanadium, niobium, manganese, magnesium, zirconium, beryllium, indium, ruthenium, and the like can be used. Furthermore, a semiconductor having high electrical conductivity, typified by polycrystalline silicon containing an impurity element such as phosphorus, or silicide such as nickel silicide may be used.
(433) For example, for the conductor **328**, the conductor **330**, the conductor **356**, the conductor **218**, the conductor **112**, and the like, a single-layer structure or a stacked-layer structure using a conductive material such as a metal material, an alloy material, a metal nitride material, or a metal oxide material that is formed using the above materials can be used. It is preferable to use a high-melting-point material that has both heat resistance and conductivity, such as tungsten or molybdenum, and it is preferable to use tungsten. Alternatively, it is preferable to use a low-resistance conductive material such as aluminum or copper. The use of a low-resistance conductive material can reduce wiring resistance.
(434) <Wiring or Plug in Layer Provided with Oxide Semiconductor>
(435) In the case where an oxide semiconductor is used in the transistor **200**, an insulator including an excess oxygen region is provided in the vicinity of the oxide semiconductor in some cases. In that case, an insulator having a barrier property is preferably provided between the insulator including the excess-oxygen region and a conductor provided in the insulator including the excess-oxygen region.
(436) For example, the insulator **241** is preferably provided between the conductor **240** and the insulator **224** and the insulator **280** that include excess oxygen in FIG. **19**. Since the insulator **241** is provided in contact with the insulator **222**, the insulator **272**, the insulator **273**, the insulator **282**, the insulator **283**, and the insulator **284**, the insulator **224** and the transistor **200** can be sealed by the insulators having a barrier property.
(437) That is, the insulator **241** can inhibit excess oxygen contained in the insulator **224** and the insulator **280** from being absorbed by the conductor **240**. In addition, diffusion of hydrogen, which is an impurity, into the transistor **200** through the conductor **240** can be inhibited when the insulator **241** is provided.
(438) Note that an insulating material having a function of inhibiting diffusion of oxygen and impurities such as water and hydrogen is preferably used for the insulator **241**. For example, silicon

nitride, silicon nitride oxide, aluminum oxide, hafnium oxide, or the like is preferably used. In particular, silicon nitride is preferably used because silicon nitride has a high blocking property against hydrogen. Other than that, a metal oxide such as magnesium oxide, gallium oxide, germanium oxide, yttrium oxide, zirconium oxide, lanthanum oxide, neodymium oxide, or tantalum oxide can be used, for example.
(439) As in the above embodiment, the transistor **200** is preferably sealed with the insulator **211**, the insulator **212**, the insulator **214**, the insulator **287**, the insulator **282**, the insulator **283**, and the insulator **284**. Such a structure can inhibit entry of hydrogen contained in the insulator **274**, the insulator **150**, or the like into the insulator **280** or the like.
(440) Note that the conductor **240** penetrates the insulator **284**, the insulator **283**, and the insulator **282**, and the conductor **218** penetrates the insulator **214**, the insulator **212**, and the insulator **211**; however, as described above, the insulator **241** is provided in contact with the conductor **240**, and the insulator **217** is provided in contact with the conductor **218**. This can reduce the amount of hydrogen entering the inside of the insulator **211**, the insulator **212**, the insulator **214**, the insulator **287**, the insulator **282**, the insulator **283**, and the insulator **284** through the conductor **240** and the conductor **218**. In this manner, the transistor **200** is sealed more surely with the insulator **211**, the insulator **212**, the insulator **214**, the insulator **287**, the insulator **282**, the insulator **283**, the insulator **284**, the insulator **241**, and the insulator **217**, so that impurities such as hydrogen contained in the insulator **274** or the like can be inhibited from entering from the outside.
(441) As described in the above embodiment, the insulator **216**, the insulator **224**, the insulator **280**, the insulator **250**, and the insulator **274** are preferably formed by a deposition method using the gas in which the number of hydrogen atoms is reduced or hydrogen atoms are removed. This can reduce the hydrogen concentration of the insulator **216**, the insulator **224**, the insulator **280**, the insulator **250**, and the insulator **274**.
(442) In this manner, the hydrogen concentration of silicon-based insulating films near the transistor **200** can be reduced; thus, the hydrogen concentration of the oxide **230** can be reduced.
(443) <Dicing Line>
(444) Here, a dicing line (also referred to as a scribe line, a dividing line, or a cutting line in some cases) that is provided when a large-sized substrate is divided into semiconductor elements so that a plurality of semiconductor devices are each formed in a chip form is described below. Examples of a dividing method include the case where a groove (a dicing line) for dividing the semiconductor elements is formed on the substrate, and then the substrate is cut along the dicing line to divide (split) it into a plurality of semiconductor devices.
(445) Here, for example, as illustrated in FIG. **19**, a region where the insulator **283** is in contact with the insulator **211** preferably overlaps with the dicing line. That is, an opening is formed in the insulator **282**, the insulator **280**, the insulator **273**, the insulator **272**, the insulator **224**, the insulator **222**, the insulator **216**, the insulator **214**, and the insulator **212** in the vicinity of a region to be the dicing line that is provided on the outer edge of a memory cell including the plurality of transistors **200**.
(446) That is, in the opening provided in the insulator **282**, the insulator **280**, the insulator **273**, the insulator **272**, the insulator **224**, the insulator **222**, the insulator **216**, the insulator **214**, and the insulator **212**, the insulator **211** is in contact with the insulator **283**. Alternatively, an opening may be provided in the insulator **282**, the insulator **280**, the insulator **273**, the insulator **272**, the insulator **224**, the insulator **222**, the insulator **216**, and the insulator **214**, and the insulator **212** and the insulator **283** may be in contact with each other. For example, the insulator **212** and the insulator **283** may be formed using the same material and the same method. When the insulator **212** and the insulator **283** are formed using the same material and the same method, the adhesion therebetween can be increased. For example, silicon nitride is preferably used.
(447) With such a structure, the transistor **200** can be enclosed with the insulator **211**, the insulator **212**, the insulator **214**, the insulator **287**, the insulator **282**, the insulator **283**, and the insulator **284**.

At least one of the insulator **211**, the insulator **212**, the insulator **214**, the insulator **287**, the insulator **282**, the insulator **283**, and the insulator **284** has a function of inhibiting diffusion of oxygen, hydrogen, and water; thus, even when the substrate is divided into circuit regions each of which is provided with the semiconductor elements in this embodiment to form a plurality of chips, the entry and diffusion of impurities such as hydrogen or water from the direction of a side surface of the divided substrate to the transistor **200** can be inhibited.

(448) Furthermore, in the structure, excess oxygen in the insulator **280** and the insulator **224** can be inhibited from being diffused to the outside. Accordingly, excess oxygen in the insulator **280** and the insulator **224** is efficiently supplied to the oxide where the channel is formed in the transistor **200**. The oxygen can reduce oxygen vacancies in the oxide where the channel is formed in the transistor **200**. Thus, the oxide where the channel is formed in the transistor **200** can be an oxide semiconductor with a low density of defect states and stable characteristics. That is, variations in the electrical characteristics of the transistor **200** can be suppressed and the reliability thereof can be improved.

(449) Note that although the capacitor **100** of the storage device illustrated in FIG. **19** is a planar capacitor, the capacitor **100** of the storage device described in this embodiment is not limited thereto. For example, the capacitor **100** may be a cylindrical capacitor as illustrated in FIG. **20**. Note that the structure below and including the insulator **150** of a storage device illustrated in FIG. **20** is similar to that of the semiconductor device illustrated in FIG. **19**.

(450) The capacitor **100** illustrated in FIG. **20** includes the insulator **150** over the insulator **130**, an insulator **142** over the insulator **150**, a conductor **115** in an opening formed in the insulator **150** and the insulator **142**, an insulator **145** over the conductor **115** and the insulator **142**, a conductor **125** over the insulator **145**, and an insulator **152** over the conductor **125** and the insulator **145**. Here, at least part of the conductor **115**, the insulator **145**, and the conductor **125** is provided in the opening formed in the insulator **150** and the insulator **142**.

(451) The conductor **115** functions as a lower electrode of the capacitor **100**, the conductor **125** functions as an upper electrode of the capacitor **100**, and the insulator **145** functions as a dielectric of the capacitor **100**. The upper electrode and the lower electrode of the capacitor **100** face each other with the dielectric therebetween, along the side surface as well as the bottom surface of the opening in the insulator **150** and the insulator **142**; thus, the capacitance per unit area can be increased. Accordingly, the deeper the opening is, the larger the capacitance of the capacitor **100** can be. Increasing the capacitance per unit area of the capacitor **100** in this manner enhances miniaturization and integration of the semiconductor device.

(452) An insulator that can be used as the insulator **280** may be used as the insulator **152**. The insulator **142** preferably functions as an etching stopper at the time of forming the opening in the insulator **150** and is formed using an insulator that can be used for the insulator **214**.

(453) The shape of the opening formed in the insulator **150** and the insulator **142** when seen from above may be a quadrangular shape, a polygonal shape other than a quadrangular shape, a polygonal shape with rounded corners, or a circular shape such as an elliptical shape. Here, the area where the opening and the transistor **200** overlap with each other is preferably larger in the top view. Such a structure can reduce the area occupied by the semiconductor device including the capacitor **100** and the transistor **200**.

(454) The conductor **115** is provided in contact with the opening formed in the insulator **142** and the insulator **150**. It is preferable that a top surface of the conductor **115** be substantially aligned with a top surface of the insulator **142**. Furthermore, a bottom surface of the conductor **115** is in contact with the conductor **110** in an opening formed in the insulator **130**. The conductor **115** is preferably deposited by an ALD method, a CVD method, or the like and is deposited using a conductor that can be used for the conductor **205**, for example.

(455) The insulator **145** is positioned to cover the conductor **115** and the insulator **142**. The insulator **145** is preferably deposited by an ALD method or a CVD method, for example. The

insulator **145** can be formed to have a stacked-layer structure or a single-layer structure using, for example, silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, zirconium oxide, aluminum oxide, aluminum oxynitride, aluminum nitride oxide, aluminum nitride, hafnium oxide, hafnium oxynitride, hafnium nitride oxide, hafnium nitride, or the like. As the insulator **145**, an insulating film in which zirconium oxide, aluminum oxide, and zirconium oxide are stacked in this order can be used, for instance.
(456) The insulator **145** is preferably formed using a material with high dielectric strength, such as silicon oxynitride, or a high dielectric constant (high-k) material. The insulator **145** may have a stacked-layer structure using a material with high dielectric strength and a high dielectric (high-k) material.
(457) As the insulator using a high dielectric constant (high-k) material (a material having a high dielectric constant), gallium oxide, hafnium oxide, zirconium oxide, an oxide containing aluminum and hafnium, an oxynitride containing aluminum and hafnium, an oxide containing silicon and hafnium, an oxynitride containing silicon and hafnium, a nitride containing silicon and hafnium, or the like can be given. The use of such a high-k material enables sufficient capacitance of the capacitor **100** to be ensured even if the insulator **145** has a large thickness. The insulator **145** having a large thickness can inhibit leakage current generated between the conductor **115** and the conductor **125**.
(458) Examples of a material with high dielectric strength include silicon oxide, silicon oxynitride, silicon nitride oxide, silicon nitride, silicon oxide to which fluorine is added, silicon oxide to which carbon is added, silicon oxide to which carbon and nitrogen are added, porous silicon oxide, and a resin. For example, it is possible to use an insulating film in which silicon nitride deposited by an ALD method, silicon oxide deposited by a PEALD method, and silicon nitride deposited by an ALD method are stacked in this order. The use of such an insulator having high dielectric strength can increase the dielectric strength and inhibit electrostatic breakdown of the capacitor **100**.
(459) The conductor **125** is provided to fill the opening formed in the insulator **142** and the insulator **150**. The conductor **125** is electrically connected to the wiring **1005** through a conductor **140** and a conductor **153**. The conductor **125** is preferably formed by an ALD method, a CVD method, or the like and is formed using a conductor that can be used as the conductor **205**, for example.
(460) The conductor **153** is provided over an insulator **154** and is covered with an insulator **156**. The conductor **153** is formed using a conductor that can be used for the conductor **112**, and the insulator **156** is formed using an insulator that can be used for the insulator **152**. Here, the conductor **153** is in contact with a top surface of the conductor **140** and functions as a terminal of the capacitor **100**, the transistor **200**, or the transistor **300**.
(461) [Storage Device **2**]
(462) FIG. **21** illustrates an example of a storage device using the semiconductor device of one embodiment of the present invention. The storage device illustrated in FIG. **21** includes a transistor **400** in addition to the semiconductor device that includes the transistor **200**, the transistor **300**, and the capacitor **100** in FIG. **19**.
(463) The transistor **400** can control a second gate voltage of the transistor **200**. For example, a first gate and a second gate of the transistor **400** are diode-connected to a source of the transistor **400**, and the source of the transistor **400** is connected to the second gate of the transistor **200**. When a negative potential of the second gate of the transistor **200** is held in this structure, a first gate-source voltage and a second gate-source voltage of the transistor **400** are 0 V. In the transistor **400**, a drain current when the second gate voltage and the first gate voltage are 0 V is extremely low; thus, the negative potential of the second gate of the transistor **200** can be held for a long time even without power supply to the transistor **200** and the transistor **400**. Accordingly, the storage device including the transistor **200** and the transistor **400** can retain stored data for a long time.
(464) Thus, in FIG. **21**, the wiring **1001** is electrically connected to the source of the transistor **300**.

The wiring **1002** is electrically connected to the drain of the transistor **300**. The wiring **1003** is electrically connected to one of the source and the drain of the transistor **200**. The wiring **1004** is electrically connected to the first gate of the transistor **200**. The wiring **1006** is electrically connected to the second gate of the transistor **200**. The gate of the transistor **300** and the other of the source and the drain of the transistor **200** are electrically connected to one electrode of the capacitor **100**. The wiring **1005** is electrically connected to the other electrode of the capacitor **100**. A wiring **1007** is electrically connected to the source of the transistor **400**. A wiring **1008** is electrically connected to a first gate of the transistor **400**. A wiring **1009** is electrically connected to the second gate of the transistor **400**. A wiring **1010** is electrically connected to the drain of the transistor **400**. Here, the wiring **1006**, the wiring **1007**, the wiring **1008**, and the wiring **1009** are electrically connected to each other.
(465) When the storage devices in FIG. **21** are arranged in a matrix like the storage devices illustrated in FIG. **19**, a memory cell array can be formed. Note that one transistor **400** can control second gate voltages of the plurality of transistors **200**. For this reason, the number of the transistors **400** is preferably smaller than that of the transistors **200**. As in the storage device illustrated in FIG. **19**, the transistor **200** and the transistor **400** in the storage device illustrated in FIG. **21** can be sealed with the insulator **211**, the insulator **212**, the insulator **214**, the insulator **287**, the insulator **282**, the insulator **283**, and the insulator **284**.
(466) <Transistor **400**>
(467) The transistor **400** and the transistors **200** are formed in the same layer and thus can be fabricated in parallel. The transistor **400** includes a conductor **460** (a conductor **460***a* and a conductor **460***b*) functioning as a first gate, a conductor **405** functioning as a second gate, the insulator **222**, the insulator **224**, and an insulator **450** functioning as gate insulating layers, an oxide **430***c* including a channel formation region, a conductor **442***a*, an oxide **443***a*, an oxide **431***a*, and an oxide **431***b* functioning as a source, and a conductor **442***b*, an oxide **443***b*, an oxide **432***a*, and an oxide **432***b* functioning as a drain. As in the transistor **200**, conductors serving as plugs are provided in contact with the conductor **442***a* and the conductor **442***b*.
(468) The conductor **405** is formed in the same layer as the conductor **205**. The oxide **431***a* and the oxide **432***a* are formed in the same layer as the oxide **230***a*, and the oxide **431***b* and the oxide **432***b* are formed in the same layer as the oxide **230***b*. The conductor **442***a* and the conductor **442***b* are formed in the same layer as the conductor **242**. The oxide **443***a* and the oxide **443***b* are formed in the same layer as the oxide **243**. The oxide **430***c* is formed in the same layer as the oxide **230***c*. The insulator **450** is formed in the same layer as the insulator **250**. The conductor **460** is formed in the same layer as the conductor **260**.
(469) Note that the components formed in the same layer can be formed at the same time. For example, the oxide **430***c* can be formed by processing an oxide film to be the oxide **230***c*.
(470) In the oxide **430***c* functioning as an active layer of the transistor **400**, oxygen vacancies and impurities such as hydrogen and water are reduced, as in the oxide **230** or the like. Accordingly, the threshold voltage of the transistor **400** can be further increased, the off-state current can be reduced, and the drain current at the time when the second gate voltage and the first gate voltage are 0 V can be extremely low.
(471) [Storage Device **3**]
(472) FIG. **22** illustrates an example of a semiconductor device (a storage device) of one embodiment of the present invention.
(473) <Structure Example of Memory Device>
(474) FIG. **22** is a cross-sectional view of a semiconductor device including a memory device **290**. The memory device **290** illustrated in FIG. **22** includes a capacitor device **292** in addition to the transistor **200** illustrated in FIG. **7**A to FIG. **7**D. FIG. **22** corresponds to a cross-sectional view of the transistor **200** in the channel length direction.
(475) The capacitor device **292** includes the conductor **242***b*, the insulator **272** and the insulator **273**

provided over the conductor **242***b*, and a conductor **294** provided over the insulator **273**. In other words, the capacitor device **292** forms an MIM (Metal-Insulator-Metal) capacitor. Note that one of a pair of electrodes of the capacitor device **292**, i.e., the conductor **242***b*, can double as the source electrode or the drain electrode of the transistor. The dielectric layer of the capacitor device **292** can double as a protective layer provided in the transistor, i.e., the insulator **272** and the insulator **273**. Thus, the manufacturing process of the capacitor device **292** and that of the transistor **200** can share some of the steps, improving the productivity of the semiconductor device. One of the pair of electrodes of the capacitor device **292**, i.e., the conductor **242***b* serves as the source electrode or the drain electrode of the transistor **200**, in which case the area where the transistor **200** and the capacitor device **292** are disposed can be reduced.
(476) Note that for the conductor **294**, for example, a material that can be used for the conductor **242** may be used.
(477) <Modification Examples of Memory Device>
(478) Examples of a semiconductor device of one embodiment of the present invention including the transistor **200** and the capacitor device **292**, which are different from the one described above in <Structure example of memory device>, will be described below with reference to FIG. **23**A, FIG. **23**B, FIG. **24**, and FIG. **25**. Note that in the semiconductor device illustrated in FIG. **23**A, FIG. **23**B, FIG. **24**, and FIG. **25**, components having the same functions as the components in the semiconductor devices described in the above embodiments and <Structure example of memory device> are denoted by the same reference numerals. Note that in this section, for the materials of the transistor **200** and the capacitor device **292**, the materials described in the above embodiments and <Structure example of memory device> can be used.
(479) «Modification Example 1 of Memory Device»
(480) An example of a semiconductor device **600** of one embodiment of the present invention including a transistor **200***a*, a transistor **200***b*, a capacitor device **292***a*, and a capacitor device **292***b* is described with reference to FIG. **23**A.
(481) FIG. **23**A is a cross-sectional view in the channel length direction of the semiconductor device **600** including the transistor **200***a*, the transistor **200***b*, the capacitor device **292***a*, and the capacitor device **292***b*. The semiconductor device **600** has a line-symmetric structure with respect to the dashed-dotted line A**3**-A**4** as illustrated in FIG. **23**A. A conductor **242***c* functions as one of a source electrode and a drain electrode of the transistor **200***a* and one of a source electrode and a drain electrode of the transistor **200***b*. The conductor **240** functioning as a plug connects the conductor **246** functioning as a wiring, the transistor **200***a*, and the transistor **200***b*. The two transistors, the two capacitors, the wiring, and the plug are connected in the above manner, whereby a semiconductor device that can be miniaturized or highly integrated can be provided.
(482) For the structures and the effects of the transistor **200***a*, the transistor **200***b*, the capacitor device **292***a*, and the capacitor device **292***b*, the structure examples of the semiconductor devices illustrated in FIG. **7**A to FIG. **7**D and FIG. **22** can be referred to.
(483) «Modification Example 2 of Memory Device»
(484) In the above description, the transistor **200***a*, the transistor **200***b*, the capacitor device **292***a*, and the capacitor device **292***b* are given as examples of components of the semiconductor device; however, the semiconductor device of this embodiment is not limited thereto. For example, as illustrated in FIG. **23**B, a structure in which the semiconductor device **600** and a semiconductor device having a structure similar to that of the semiconductor device **600** are connected through a capacitor portion may be employed. In this specification, the semiconductor device including the transistor **200***a*, the transistor **200***b*, the capacitor device **292***a*, and the capacitor device **292***b* is referred to as a cell. For the structures of the transistor **200***a*, the transistor **200***b*, the capacitor device **292***a*, and the capacitor device **292***b*, the above description of the transistor **200***a*, the transistor **200***b*, the capacitor device **292***a*, and the capacitor device **292***b* can be referred to.
(485) FIG. **23**B is a cross-sectional view in which the semiconductor device **600** including the

transistor **200***a*, the transistor **200***b*, the capacitor device **292***a*, and the capacitor device **292***b* and a cell having a structure similar to that of the semiconductor device **600** are connected through a capacitor portion.
(486) As illustrated in FIG. **23**B, a conductor **294***b* functioning as one electrode of the capacitor device **292***b* included in the semiconductor device **600** also serves as one electrode of a capacitor device included in a semiconductor device **601** that has a structure similar to that of the semiconductor device **600**. Although not illustrated, the conductor **294***a* that functions as one electrode of the capacitor device **292***a* included in the semiconductor device **600** also serves one electrode of a capacitor device included in a semiconductor device on the left side of the semiconductor device **600**, that is, a semiconductor device adjacent to the semiconductor device **600** in the A**1** direction in FIG. **23**B. The same applies to a cell on the right side of the semiconductor device **601**, that is, a cell in the A**2** direction in FIG. **23**B. That is, a cell array (also referred to as a memory device layer) can be formed. With this structure of the cell array, the space between the adjacent cells can be reduced; thus, the projected area of the cell array can be reduced and high integration can be achieved. The cell arrays illustrated in FIG. **23**B are arranged in a matrix, whereby cell arrays arranged in a matrix can be formed.
(487) When the transistor **200***a*, the transistor **200***b*, the capacitor device **292***a*, and the capacitor device **292***b* are formed with the structure described in this embodiment as described above, the area of the cell can be reduced and the semiconductor device including the cell array can be miniaturized or highly integrated.
(488) Furthermore, the cell array may have a stacked-layer structure instead of a single-layer cell array. FIG. **24** is a cross-sectional view of a stacked structure of n layers of cell arrays **610**. As illustrated in FIG. **24**, by stacking a plurality of cell arrays (a cell array **610_1** to a cell array **610_***n*), the cells can be integrated without an increase in the area occupied by the cell arrays. In other words, a 3D cell array can be formed.
(489) «Modification Example 3 of Memory Device»
(490) FIG. **25** illustrates an example in which a memory unit **470** includes a transistor layer **413** including a transistor **200**T and four memory device layers **415** (a memory device layer **415_1** to a memory device layer **415_4**).
(491) Each of the memory device layer **415_1** to the memory device layer **415_4** includes a plurality of memory devices **420**.
(492) The memory device **420** is electrically connected to the memory device **420** included in a different memory device layer **415** and the transistor **200**T included in the transistor layer **413** via a conductor **424** and the conductor **205**.
(493) The memory unit **470** is sealed by the insulator **211**, the insulator **212**, the insulator **214**, the insulator **287**, the insulator **282**, the insulator **283**, and the insulator **284** (such a structure is referred to as a sealing structure below for convenience). The insulator **274** is provided near the insulator **284**. A conductor **440** is provided in the insulator **274**, the insulator **284**, the insulator **283**, and the insulator **211**, and is electrically connected to an element layer **411**.
(494) The insulator **280** is provided in the sealing structure. The insulator **280** has a function of releasing oxygen by heating. The insulator **280** includes an excess oxygen region.
(495) The insulator **211**, the insulator **283**, and the insulator **284** are suitably formed using a material having a high blocking property against hydrogen. The insulator **214**, the insulator **282**, and the insulator **287** are suitably formed using a material having a function of capturing or fixing hydrogen.
(496) Examples of the material having a high blocking property against hydrogen include silicon nitride and silicon nitride oxide. Examples of the material having a function of capturing or fixing hydrogen include aluminum oxide, hafnium oxide, and an oxide containing aluminum and hafnium (hafnium aluminate).
(497) A barrier property in this specification means a function of inhibiting diffusion of a particular

substance (also referred to as a function of less easily transmitting the substance). Alternatively, a barrier property in this specification means a function of capturing or fixing (also referred to as gettering) a particular substance.
(498) Materials for the insulator **211**, the insulator **212**, the insulator **214**, the insulator **287**, the insulator **282**, the insulator **283**, and the insulator **284** may have an amorphous or crystal structure, although the crystallinity of the materials is not limited thereto. For example, an amorphous aluminum oxide film is suitably used for the material having a function of capturing or fixing hydrogen. Amorphous aluminum oxide may capture or fix hydrogen more than aluminum oxide with high crystallinity.
(499) The following model can be given for the reaction of excess oxygen in the insulator **280** with respect to diffusion of hydrogen in an oxide semiconductor in contact with the insulator **280**.
(500) Hydrogen in the oxide semiconductor is diffused to another structure body through the insulator **280** which is in contact with the oxide semiconductor. The hydrogen reacts with the excess oxygen in the insulator **280**, which yields the OH bonding and diffuses in the insulator **280** as OH. The hydrogen atom having the OH bonding reacts with the oxygen atom bonded to an atom (such as a metal atom) in the insulator **282** in reaching a material which has a function of capturing or fixing hydrogen (typically the insulator **282**), and is trapped or fixed in the insulator **282**. The excess oxygen having the OH bonding probably remains as an excess oxygen in the insulator **280**. That is, the excess oxygen in the insulator **280** highly probably acts as a bridge in the hydrogen diffusion.
(501) A manufacturing process of the semiconductor device is one of important factors for the model.
(502) For example, the insulator **280** containing excess oxygen is formed over the oxide semiconductor, and then the insulator **282** is formed. After that, heat treatment is preferably performed. The heat treatment is performed at 350° C. or higher, preferably 400° C. or higher under an atmosphere containing oxygen, an atmosphere containing nitrogen, or a mixed atmosphere of oxygen and nitrogen. The heat treatment time is performed one hour or more, preferably four hours or more, further preferably eight hours or more.
(503) The heat treatment enables hydrogen in the oxide semiconductor to diffuse to the outside through the insulator **280**, the insulator **282**, and the insulator **287**. That is, the absolute amount of hydrogen in the oxide semiconductor and near the oxide semiconductor can be reduced.
(504) The insulator **283** and the insulator **284** are formed after the heat treatment. The insulator **283** and the insulator **284** have a high blocking property against hydrogen. Thus, hydrogen which has been diffused to the outside or hydrogen existing the outside can be prevented from entering the inside, specifically, the oxide semiconductor or insulator **280** side.
(505) The heat treatment is performed after the insulator **282** is formed in the above example; however, one embodiment of the present invention is not limited thereto. For example, after the formation of the transistor layer **413** or after the formation of the memory device layer **415_1** to the memory device layer **415_3**, the heat treatment may be performed. When hydrogen is diffused outward by the heat treatment, hydrogen is diffused above the transistor layer **413** or in the lateral direction. Similarly, in the case where heat treatment is performed after the formation of the memory device layer **415_1** to the memory device layer **415_3**, hydrogen is diffused upward or in the lateral direction.
(506) Through the above manufacturing process, the insulator **211** and the insulator **283** are bonded, whereby the sealing structure is formed.
(507) The above-described structure and manufacturing process enable a semiconductor device using an oxide semiconductor with reduced hydrogen concentration. Thus, a highly reliable semiconductor device can be provided. According to one embodiment of the present invention, a semiconductor device with favorable electrical characteristics can be provided.
(508) The structure, the method, and the like described above in this embodiment can be used in an

appropriate combination with the structures, the configurations, the methods, and the like described in the other embodiments and the example.

Embodiment 4

(509) In this embodiment, a storage device of one embodiment of the present invention including a transistor in which an oxide is used for a semiconductor (hereinafter referred to as an OS transistor in some cases) and a capacitor (hereinafter referred to as an OS memory device in some cases), is described with reference to FIG. **26**A and FIG. **26**B and FIG. **27**A to FIG. **27**H. The OS memory device includes at least a capacitor and an OS transistor that controls the charging and discharging of the capacitor. Since the OS transistor has an extremely low off-state current, the OS memory device has excellent retention characteristics and thus can function as a nonvolatile memory.

(510) <Structure Example of Storage Device>

(511) FIG. **26**A illustrates an example of the structure of an OS memory device. A storage device **1400** includes a peripheral circuit **1411** and a memory cell array **1470**. The peripheral circuit **1411** includes a row circuit **1420**, a column circuit **1430**, an output circuit **1440**, and a control logic circuit **1460**.

(512) The column circuit **1430** includes, for example, a column decoder, a precharge circuit, a sense amplifier, a write circuit, and the like. The precharge circuit has a function of precharging wirings. The sense amplifier has a function of amplifying a data signal read from a memory cell. Note that the wirings are connected to the memory cell included in the memory cell array **1470**, and will be described later in detail. The amplified data signal is output as a data signal RDATA to the outside of the storage device **1400** through the output circuit **1440**. The row circuit **1420** includes, for example, a row decoder and a word line driver circuit, and can select a row to be accessed.

(513) As power supply voltages from the outside, a low power supply voltage (VSS), a high power supply voltage (VDD) for the peripheral circuit **1411**, and a high power supply voltage (VIL) for the memory cell array **1470** are supplied to the storage device **1400**. Control signals (CE, WE, and RE), an address signal ADDR, and a data signal WDATA are also input to the storage device **1400** from the outside. The address signal ADDR is input to the row decoder and the column decoder, and the data signal WDATA is input to the write circuit.

(514) The control logic circuit **1460** processes the control signals (CE, WE, and RE) input from the outside, and generates control signals for the row decoder and the column decoder. The control signal CE is a chip enable signal, the control signal WE is a write enable signal, and the control signal RE is a read enable signal. Signals processed by the control logic circuit **1460** are not limited thereto, and other control signals may be input as necessary.

(515) The memory cell array **1470** includes a plurality of memory cells MC arranged in a matrix and a plurality of wirings. Note that the number of the wirings that connect the memory cell array **1470** to the row circuit **1420** depends on the structure of the memory cell MC, the number of the memory cells MC in a column, and the like. The number of the wirings that connect the memory cell array **1470** to the column circuit **1430** depends on the structure of the memory cell MC, the number of the memory cells MC in a row, and the like.

(516) Note that FIG. **26**A shows an example in which the peripheral circuit **1411** and the memory cell array **1470** are formed on the same plane; however, this embodiment is not limited thereto. For example, as shown in FIG. **26**B, the memory cell array **1470** may be provided over the peripheral circuit **1411** to partly overlap with the peripheral circuit **1411**. For example, the sense amplifier may be provided below the memory cell array **1470** so that they overlap with each other.

(517) FIG. **27**A to FIG. **27**H show structure examples of a memory cell which can be used to the memory cell MC.

(518) [DOSRAM]

(519) FIG. **27**A to FIG. **27**C each illustrate a circuit structure example of a memory cell of a DRAM. In this specification and the like, a DRAM using a memory cell including one OS transistor and one capacitor is referred to as DOSRAM (Dynamic Oxide Semiconductor Random

Access Memory) in some cases. A memory cell **1471** shown in FIG. **27**A includes a transistor M**1** and a capacitor CA. Note that the transistor M**1** includes a gate (also referred to as a top gate in some cases) and a back gate.
(520) A first terminal of the transistor M**1** is connected to a first terminal of the capacitor CA. A second terminal of the transistor M**1** is connected to a wiring BIL. A gate of the transistor M**1** is connected to a wiring WOL. A back gate of the transistor M**1** is connected to a wiring BGL. A second terminal of the capacitor CA is connected to a wiring CAL.
(521) The wiring BIL functions as a bit line, and the wiring WOL functions as a word line. The wiring CAL functions as a wiring for applying a predetermined potential to the second terminal of the capacitor CA. In the time of data writing and data reading, a low-level potential is preferably applied to the wiring CAL. The wiring BGL functions as a wiring for applying a potential to the back gate of the transistor M**1**. Applying a given potential to the wiring BGL can increase or decrease the threshold voltage of the transistor M**1**.
(522) Here, the memory cell **1471** illustrated in FIG. **27**A corresponds to the storage device illustrated in FIG. **22**. That is, the transistor M**1** corresponds to the transistor **200** and the capacitor CA corresponds to the capacitor device **292**.
(523) The memory cell MC is not limited to the memory cell **1471**, and the circuit structure can be changed. For example, like a memory cell **1472** in FIG. **27**B, a structure may be used in which the back gate of the transistor M**1** is connected not to the wiring BGL but to the wiring WOL in the memory cell MC. Alternatively, for example, the memory cell MC may be a memory cell including a single-gate transistor, that is, the transistor M**1** not including a back gate, as in a memory cell **1473** shown in FIG. **27**C.
(524) In the case where the semiconductor device described in the above embodiments is used in the memory cell **1471** and the like, the transistor **200** can be used as the transistor M**1**, and the capacitor **100** can be used as the capacitor CA. When an OS transistor is used as the transistor M**1**, the leakage current of the transistor M**1** can be extremely low. That is, with use of the transistor M**1**, written data can be retained for a long time, and thus the frequency of the refresh operation for the memory cell can be decreased. Alternatively, the refresh operation of the memory cell can be omitted. In addition, since the transistor M**1** has an extremely low leakage current, multi-level data or analog data can be retained in the memory cell **1471**, the memory cell **1472**, and the memory cell **1473**.
(525) In the DOSRAM, when the sense amplifier is provided below the memory cell array **1470** so that they overlap with each other as described above, the bit line can be shortened. Thus, the bit line capacitance can be small, and the storage capacitance of the memory cell can be reduced.
(526) [NOSRAM]
(527) FIGS. **27**D to **27**G each show a circuit structure example of a gain-cell memory cell including two transistors and one capacitor. A memory cell **1474** shown in FIG. **27**D includes a transistor M**2**, a transistor M**3**, and a capacitor CB. Note that the transistor M**2** includes a top gate (simply referred to as a gate in some cases) and a back gate. In this specification and the like, a storage device including a gain-cell memory cell using an OS transistor as the transistor M**2** is referred to as NOSRAM (Nonvolatile Oxide Semiconductor RAM) in some cases.
(528) A first terminal of the transistor M**2** is connected to a first terminal of the capacitor CB. A second terminal of the transistor M**2** is connected to a wiring WBL. A gate of the transistor M**2** is connected to the wiring WOL. A back gate of the transistor M**2** is connected to the wiring BGL. A second terminal of the capacitor CB is connected to the wiring CAL. A first terminal of the transistor M**3** is connected to a wiring RBL. A second terminal of the transistor M**3** is connected to a wiring SL. A gate of the transistor M**3** is connected to the first terminal of the capacitor CB.
(529) The wiring WBL functions as a write bit line, the wiring RBL functions as a read bit line, and the wiring WOL functions as a word line. The wiring CAL functions as a wiring for applying a predetermined potential to the second terminal of the capacitor CB. In the time of data writing, data

retaining, and data reading, a low-level potential is preferably applied to the wiring CAL. The wiring BGL functions as a wiring for applying a potential to the back gate of the transistor M**2**. By application of a given potential to the wiring BGL, the threshold voltage of the transistor M**2** can be increased or decreased.
(530) Here, the memory cell **1474** shown in FIG. **27**D corresponds to the storage device shown in FIG. **19**. That is, the transistor M**2**, the capacitor CB, the transistor M**3**, the wiring WBL, the wiring WOL, the wiring BGL, the wiring CAL, the wiring RBL, and the wiring SL correspond to the transistor **200**, the capacitor **100**, the transistor **300**, the wiring **1003**, the wiring **1004**, the wiring **1006**, the wiring **1005**, the wiring **1002**, and the wiring **1001**, respectively.
(531) The memory cell MC is not limited to the memory cell **1474**, and the circuit structure can be changed as appropriate. For example, like a memory cell **1475** in FIG. **27**E, a structure may be used in which the back gate of the transistor M**2** is connected not to the wiring BGL but to the wiring WOL in the memory cell MC. Alternatively, for example, like a memory cell **1476** in FIG. **27**F, the memory cell MC may be a memory cell including a single-gate transistor, that is, the transistor M**2** that does not include a back gate. Alternatively, for example, like a memory cell **1477** shown in FIG. **27**G, the memory cell MC may have a structure where the wiring WBL and the wiring RBL are combined into one wiring BIL.
(532) In the case where the semiconductor device described in the above embodiments is used in the memory cell **1474** and the like, the transistor **200** can be used as the transistor M**2**, the transistor **300** can be used as the transistor M**3**, and the capacitor **100** can be used as the capacitor CB. When an OS transistor is used as the transistor M**2**, the leakage current of the transistor M**2** can be extremely low. That is, with use of the transistor M**2**, written data can be retained for a long time, and thus the frequency of the refresh operation for the memory cell can be decreased. Alternatively, the refresh operation of the memory cell can be omitted. In addition, since the transistor M**2** has an extremely low leakage current, multi-level data or analog data can be retained in the memory cell **1474**. The same applies to the memory cell **1475** to the memory cell **1477**.
(533) Note that the transistor M**3** may be a transistor containing silicon in a channel formation region (hereinafter also referred to as a Si transistor in some cases). The conductivity type of the Si transistor may be of either an n-channel type or a p-channel type. The Si transistor has higher field-effect mobility than the OS transistor in some cases. Therefore, a Si transistor may be used as the transistor M**3** functioning as a reading transistor. Furthermore, the transistor M**2** can be provided to be stacked over the transistor M**3** when a Si transistor is used as the transistor M**3**; therefore, the area occupied by the memory cell can be reduced, leading to high integration of the storage device.
(534) The transistor M**3** may be an OS transistor. When an OS transistor is used as each of the transistor M**2** and the transistor M**3**, the circuit of the memory cell array **1470** can be formed using only n-channel transistors.
(535) In addition, FIG. **27**H shows an example of a gain-cell memory cell including three transistors and one capacitor. A memory cell **1478** shown in FIG. **27**H includes a transistor M**4** to a transistor M**6** and a capacitor CC. The capacitor CC is provided as appropriate. The memory cell **1478** is electrically connected to the wiring BIL, a wiring RWL, a wiring WWL, the wiring BGL, and a wiring GNDL. The wiring GNDL is a wiring for supplying a low-level potential. Note that the memory cell **1478** may be electrically connected to the wiring RBL and the wiring WBL instead of the wiring BIL.
(536) The transistor M**4** is an OS transistor including a back gate that is electrically connected to the wiring BGL. Note that the back gate and the gate of the transistor M**4** may be electrically connected to each other. Alternatively, the transistor M**4** does not necessarily include the back gate.
(537) Note that each of the transistor M**5** and the transistor M**6** may be an n-channel Si transistor or a p-channel Si transistor. Alternatively, the transistor M**4** to the transistor M**6** may be OS transistors, in which case the circuit of the memory cell array **1470** can be formed using only n-channel transistors.

(538) In the case where the semiconductor device described in the above embodiments is used in the memory cell **1478**, the transistor **200** can be used as the transistor M**4**, the transistor **300** can be used as the transistor M**5** and the transistor M**6**, and the capacitor **100** can be used as the capacitor CC. When an OS transistor is used as the transistor M**4**, the leakage current of the transistor M**4** can be extremely low.
(539) Note that the structures of the peripheral circuit **1411**, the memory cell array **1470**, and the like described in this embodiment are not limited to the above. Positions and functions of these circuits, wirings connected to the circuits, circuit elements, and the like can be changed, deleted, or added as needed.
(540) In general, a variety of storage devices (memory) are used as semiconductor devices such as a computer in accordance with the intended use. FIG. **28** is a hierarchy diagram showing various storage devices with different levels. The storage devices at the upper levels of the diagram require high access speeds, and the storage devices at the lower levels require large memory capacity and high record density. In FIG. **28**, sequentially from the top level, a memory combined as a register in an arithmetic processing device such as a CPU, an SRAM (Static Random Access Memory), a DRAM (Dynamic Random Access Memory), and a 3D NAND memory are shown.
(541) A memory combined as a register in an arithmetic processing device such as a CPU is used for temporary storage of arithmetic operation results, for example, and thus is very frequently accessed by the arithmetic processing device. Accordingly, high operation speed is required rather than memory capacity. In addition, the register also has a function of retaining setting information of the arithmetic processing device, for example.
(542) An SRAM is used for a cache, for example. A cache has a function of duplicating and retaining part of information retained in a main memory. When the frequently used data is duplicated and retained in the cache, the access speed to the data can be increased.
(543) A DRAM is used for a main memory, for example. A main memory has a function of retaining a program or data read from a storage. A DRAM has a recording density of approximately 0.1 Gbit/mm.sup.2 to 0.3 Gbit/mm.sup.2.
(544) A 3D NAND memory is used for a storage, for example. A storage has a function of retaining data that needs to be retained for a long time or a variety of programs used in an arithmetic processing device, for example. Therefore, a storage needs to have high memory capacity and a high recording density rather than operation speed. A storage device used for a storage has a recording density of approximately 0.6 Gbit/mm.sup.2 to 6.0 Gbit/mm.sup.2.
(545) The storage device of one embodiment of the present invention has a high operation speed and can retain data for a long time. The storage device of one embodiment of the present invention can be suitably used as a storage device in a boundary region **901** that includes both the level to which a cache belongs and the level to which a main memory belongs. The storage device of one embodiment of the present invention can be suitably used as a storage device in a boundary region **902** that includes both the level to which the main memory belongs and the level to which a storage belongs.
(546) The storage device of one embodiment of the present invention can be suitably used as a storage device used for a server, a notebook PC, a smartphone, a game machine, an image sensor, IoT (Internet of Things), healthcare, or the like.
(547) The structure described in this embodiment can be used in an appropriate combination with the structures described in the other embodiments, example, and the like.
Embodiment 5
(548) In this embodiment, an example of a chip **1200** on which the semiconductor device of the present invention is mounted is described with reference to FIG. **29**A and FIG. **29**B. A plurality of circuits (systems) are mounted on the chip **1200**. The technique for integrating a plurality of circuits (systems) on one chip as described above is referred to as system on chip (SoC) in some cases.

(549) As illustrated in FIG. **29**A, the chip **1200** includes a CPU **1211**, a GPU **1212**, one or more of analog arithmetic units **1213**, one or more of memory controllers **1214**, one or more of interfaces **1215**, one or more of network circuits **1216**, and the like.
(550) A bump (not illustrated) is provided on the chip **1200** and is connected to a first surface of a printed circuit board (PCB) **1201** as shown in FIG. **29**B. A plurality of bumps **1202** are provided on the rear side of the first surface of the PCB **1201**, and the PCB **1201** is connected to a motherboard **1203**.
(551) A storage device such as a DRAM **1221** or a flash memory **1222** may be provided over the motherboard **1203**. For example, the DOSRAM described in the above embodiment can be used as the DRAM **1221**. For example, the NOSRAM described in the above embodiment can be used as the flash memory **1222**.
(552) The CPU **1211** preferably includes a plurality of CPU cores. Furthermore, the GPU **1212** preferably includes a plurality of GPU cores. The CPU **1211** and the GPU **1212** may each include a memory for storing data temporarily. Alternatively, a common memory for the CPU **1211** and the GPU **1212** may be provided in the chip **1200**. The NOSRAM or the DOSRAM described above can be used as the memory. The GPU **1212** is suitable for parallel computation of a number of data and thus can be used for image processing or product-sum operation. When an image processing circuit or a product-sum operation circuit including an oxide semiconductor of the present invention is provided in the GPU **1212**, image processing and product-sum operation can be performed with low power consumption.
(553) Since the CPU **1211** and the GPU **1212** are provided in the same chip, a wiring between the CPU **1211** and the GPU **1212** can be shortened; accordingly, the data transfer from the CPU **1211** to the GPU **1212**, the data transfer between the memories included in the CPU **1211** and the GPU **1212**, and the transfer of arithmetic operation results from the GPU **1212** to the CPU **1211** after the arithmetic operation in the GPU **1212** can be performed at high speed.
(554) The analog arithmetic unit **1213** includes one or both of an A/D (analog/digital) converter circuit and a D/A (digital/analog) converter circuit. The analog arithmetic unit **1213** may include the above-described product-sum operation circuit.
(555) The memory controller **1214** includes a circuit functioning as a controller of the DRAM **1221** and a circuit functioning as the interface of the flash memory **1222**.
(556) The interface **1215** includes an interface circuit for an external connection device such as a display device, a speaker, a microphone, a camera, or a controller. Examples of the controller include a mouse, a keyboard, and a game controller. As such an interface, USB (Universal Serial Bus), HDMI (registered trademark) (High-Definition Multimedia Interface), or the like can be used.
(557) The network circuit **1216** includes a circuit for a network such as a LAN (Local Area Network). The network circuit **1216** may include a circuit for network security.
(558) The circuits (systems) can be formed in the chip **1200** in the same manufacturing process. Thus, even when the number of circuits needed for the chip **1200** is increased, there is no need to increase the number of steps in the manufacturing process; thus, the chip **1200** can be manufactured at low cost.
(559) The motherboard **1203** provided with the PCB **1201** on which the chip **1200** including the GPU **1212** is mounted, the DRAM **1221**, and the flash memory **1222** can be referred to as a GPU module **1204**.
(560) The GPU module **1204** includes the chip **1200** formed using the SoC technology, and thus can have a small size. Furthermore, the GPU module **1204** is excellent in image processing, and thus is suitably used in a portable electronic device such as a smartphone, a tablet terminal, a laptop PC, or a portable (mobile) game console. Furthermore, the product-sum operation circuit using the GPU **1212** can perform a method such as a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), an autoencoder, a deep Boltzmann machine

(DBM), or a deep belief network (DBN); hence, the chip **1200** can be used as an AI chip or the GPU module **1204** can be used as an AI system module.
(561) The structure described in this embodiment can be used in an appropriate combination with the structures described in the other embodiments, example, and the like.
Embodiment 6
(562) In this embodiment, examples of electronic components and electronic devices in which the storage device or the like described in the above embodiment is incorporated will be described.
(563) <Electronic Components>
(564) First, examples of electronic components in which the storage device **720** is incorporated will be described with reference to FIGS. **30**A and **30**B.
(565) FIG. **30**A is a perspective view of an electronic component **700** and a substrate on which the electronic component **700** is mounted (a mounting board **704**). The memory device **700** illustrated in FIG. **30**A includes the storage device **720** in a mold **711**. FIG. **30**A omits part of the electronic component to show the inside of the electronic component **700**. The electronic component **700** includes a land **712** outside the mold **711**. The land **712** is electrically connected to an electrode pad **713**, and the electrode pad **713** is electrically connected to the storage device **720** via a wire **714**. The electronic component **700** is mounted on a printed circuit board **702**, for example. A plurality of such an electronic component are combined and electrically connected to each other on the printed circuit board **702**, whereby the mounting board **704** is completed.
(566) The storage device **720** includes a driver circuit layer **721** and a memory circuit layer **722**.
(567) FIG. **30**B is a perspective view of an electronic component **730**. The electronic component **730** is an example of a SiP (System in package) or an MCM (Multi Chip Module). In the electronic component **730**, an interposer **731** is provided on a package substrate **732** (a printed circuit board), and a semiconductor device **735** and a plurality of storage devices **720** are provided on the interposer **731**.
(568) The electronic component **730** using the storage devices **720** as high bandwidth memory (HBM) is shown as an example. An integrated circuit (a semiconductor device) such as a CPU, a GPU, or an FPGA can be used as the semiconductor device **735**.
(569) As the package substrate **732**, a ceramic substrate, a plastic substrate, a glass epoxy substrate, or the like can be used. As the interposer **731**, a silicon interposer, a resin interposer, or the like can be used.
(570) The interposer **731** includes a plurality of wirings and has a function of electrically connecting a plurality of integrated circuits with different terminal pitches. The plurality of wirings are provided in a single layer or multiple layers. Moreover, the interposer **731** has a function of electrically connecting an integrated circuit provided on the interposer **731** to an electrode provided on the package substrate **732**. Accordingly, the interposer is referred to as a “redistribution substrate” or an “intermediate substrate” in some cases. A through electrode may be provided in the interposer **731** and used for electrically connecting an integrated circuit and the package substrate **732**. For a silicon interposer, a TSV (Through Silicon Via) can also be used as the through electrode.
(571) A silicon interposer is preferably used as the interposer **731**. A silicon interposer can be manufactured at lower cost than an integrated circuit because it is not necessary to provide an active element. Meanwhile, since wirings of a silicon interposer can be formed through a semiconductor process, formation of minute wirings, which is difficult for a resin interposer, is easy.
(572) In order to achieve a wide memory bandwidth, many wirings need to be connected to HBM. Therefore, formation of minute and high-density wirings is required for an interposer on which HBM is mounted. For this reason, a silicon interposer is preferably used as the interposer on which HBM is mounted.
(573) In a SiP, an MCM, or the like using a silicon interposer, the decrease in reliability due to a

difference in expansion coefficient between an integrated circuit and the interposer does not easily occur. Furthermore, the surface of a silicon interposer has high planarity, so that a poor connection between the silicon interposer and an integrated circuit provided on the silicon interposer does not easily occur. It is particularly preferable to use a silicon interposer for a 2.5D package (2.5-dimensional mounting) in which a plurality of integrated circuits are arranged side by side on an interposer.

(574) A heat sink (a radiator plate) may be provided to overlap the electronic component **730**. In the case of providing a heat sink, the heights of integrated circuits provided on the interposer **731** are preferably equal to each other. For example, in the electronic component **730** described in this embodiment, the heights of the storage devices **720** and the semiconductor device **735** are preferably equal to each other.

(575) To mount the electronic component **730** on another substrate, an electrode **733** may be provided on the bottom portion of the package substrate **732**. FIG. **30**B illustrates an example in which the electrode **733** is formed of a solder ball. Solder balls are provided in a matrix on the bottom portion of the package substrate **732**, whereby BGA (Ball Grid Array) mounting can be achieved. Alternatively, the electrode **733** may be formed of a conductive pin. When conductive pins are provided in a matrix on the bottom portion of the package substrate **732**, PGA (Pin Grid Array) mounting can be achieved.

(576) The electronic component **730** can be mounted on another substrate by various mounting methods not limited to BGA and PGA. For example, a mounting method such as SPGA (Staggered Pin Grid Array), LGA (Land Grid Array), QFP (Quad Flat Package), QFJ (Quad Flat J-leaded package), or QFN (Quad Flat Non-leaded package) can be employed.

(577) This embodiment can be implemented in an appropriate combination with the structures described in the other embodiments, example, and the like.

Embodiment 7

(578) In this embodiment, application examples of the storage device using the semiconductor device described in the above embodiment are described. The semiconductor device described in the above embodiment can be applied to, for example, storage devices of a variety of electronic devices (e.g., information terminals, computers, smartphones, e-book readers, digital cameras (including video cameras), video recording/reproducing devices, and navigation systems). Here, the computers refer not only to tablet computers, notebook computers, and desktop computers, but also to large computers such as server systems. Alternatively, the semiconductor device described in the above embodiment is applied to removable storage devices such as memory cards (e.g., SD cards), USB memories, and SSDs (solid state drives). FIG. **31**A to FIG. **31**E schematically show some structure examples of removable storage devices. The semiconductor device described in the above embodiment is processed into a packaged memory chip and used in a variety of storage devices and removable memories, for example.

(579) FIG. **31**A is a schematic view of a USB memory. A USB memory **1100** includes a housing **1101**, a cap **1102**, a USB connector **1103**, and a substrate **1104**. The substrate **1104** is held in the housing **1101**. For example, a memory chip **1105** and a controller chip **1106** are attached to the substrate **1104**. The semiconductor device described in the above embodiment can be incorporated in the memory chip **1105** or the like.

(580) FIG. **31**B is a schematic external view of an SD card, and FIG. **31**C is a schematic view of the internal structure of the SD card. An SD card **1110** includes a housing **1111**, a connector **1112**, and a substrate **1113**. The substrate **1113** is held in the housing **1111**. For example, a memory chip **1114** and a controller chip **1115** are attached to the substrate **1113**. When the memory chip **1114** is also provided on the rear surface side of the substrate **1113**, the capacity of the SD card **1110** can be increased. In addition, a wireless chip with a radio communication function may be provided on the substrate **1113**. With this, data can be read from and written in the memory chip **1114** by radio communication between a host device and the SD card **1110**. The semiconductor device described

in the above embodiment can be incorporated in the memory chip **1114** or the like.
(581) FIG. **31**D is a schematic external view of an SSD, and FIG. **31**E is a schematic view of the internal structure of the SSD. An SSD **1150** includes a housing **1151**, a connector **1152**, and a substrate **1153**. The substrate **1153** is held in the housing **1151**. For example, a memory chip **1154**, a memory chip **1155**, and a controller chip **1156** are attached to the substrate **1153**. The memory chip **1155** is a work memory for the controller chip **1156**, and a DOSRAM chip may be used, for example. When the memory chip **1154** is also provided on the rear surface side of the substrate **1153**, the capacity of the SSD **1150** can be increased. The semiconductor device described in the above embodiment can be incorporated in the memory chip **1154** or the like.
(582) This embodiment can be implemented in an appropriate combination with the structures described in the other embodiments, example, and the like.
Embodiment 8
(583) The semiconductor device of one embodiment of the present invention can be used as a processor such as a CPU and a GPU or a chip. FIG. **32**A to FIG. **32**H show specific examples of electronic devices including a chip or a processor such as a CPU or a GPU of one embodiment of the present invention.
(584) <Electronic Devices and Systems>
(585) The GPU or the chip of one embodiment of the present invention can be mounted on a variety of electronic devices. Examples of electronic devices include a digital camera, a digital video camera, a digital photo frame, an e-book reader, a mobile phone, a portable game machine, a portable information terminal, and an audio reproducing device in addition to electronic devices provided with a relatively large screen, such as a television device, a monitor for a desktop or notebook information terminal or the like, digital signage, and a large game machine like a pachinko machine. In addition, when the GPU or the chip of one embodiment of the present invention is provided in the electronic device, the electronic device can include artificial intelligence.
(586) The electronic device of one embodiment of the present invention may include an antenna. When a signal is received by the antenna, a video, data, or the like can be displayed on the display portion. When the electronic device includes the antenna and a secondary battery, the antenna may be used for contactless power transmission.
(587) The electronic device of one embodiment of the present invention may include a sensor (a sensor having a function of measuring force, displacement, position, speed, acceleration, angular velocity, rotational frequency, distance, light, liquid, magnetism, temperature, a chemical substance, sound, time, hardness, an electric field, current, voltage, power, radioactive rays, flow rate, humidity, a gradient, oscillation, odor, or infrared rays).
(588) The electronic device of one embodiment of the present invention can have a variety of functions. For example, the electronic device can have a function of displaying a variety of data (a still image, a moving image, a text image, and the like) on the display portion, a touch panel function, a function of displaying a calendar, date, time, and the like, a function of executing a variety of software (programs), a wireless communication function, and a function of reading out a program or data stored in a recording medium. FIG. **32**A to FIG. **32**H show examples of electronic devices.
(589) [Information Terminal]
(590) FIG. **32**A illustrates a mobile phone (smartphone), which is a type of information terminal. An information terminal **5100** includes a housing **5101** and a display portion **5102**. As input interfaces, a touch panel is provided in the display portion **5102** and a button is provided in the housing **5101**.
(591) When the chip of one embodiment of the present invention is applied to the information terminal **5100**, the information terminal **5100** can execute an application utilizing artificial intelligence. Examples of the application utilizing artificial intelligence include an application for

recognizing a conversation and displaying the content of the conversation on the display portion **5102**; an application for recognizing letters, figures, and the like input to the touch panel of the display portion **5102** by a user and displaying them on the display portion **5102**; and an application for performing biometric authentication using fingerprints, voice prints, or the like.
(592) FIG. **32**B illustrates a notebook information terminal **5200**. The notebook information terminal **5200** includes a main body **5201** of the information terminal, a display portion **5202**, and a keyboard **5203**.
(593) Like the information terminal **5100** described above, when the chip of one embodiment of the present invention is applied to the notebook information terminal **5200**, the notebook information terminal **5200** can execute an application utilizing artificial intelligence. Examples of the application utilizing artificial intelligence include design-support software, text correction software, and software for automatic menu generation. Furthermore, with use of the notebook information terminal **5200**, novel artificial intelligence can be developed.
(594) Note that although FIG. **32**A and FIG. **32**B illustrate a smartphone and a notebook information terminal, respectively, as examples of the electronic device in the above description, an information terminal other than a smartphone and a notebook information terminal can be used. Examples of information terminals other than a smartphone and a notebook information terminal include a PDA (Personal Digital Assistant), a desktop information terminal, and a workstation.
(595) [Game Machines]
(596) FIG. **32**C illustrates a portable game machine **5300** as an example of a game machine. The portable game machine **5300** includes a housing **5301**, a housing **5302**, a housing **5303**, a display portion **5304**, a connection portion **5305**, an operation key **5306**, and the like. The housing **5302** and the housing **5303** can be detached from the housing **5301**. When the connection portion **5305** provided in the housing **5301** is attached to another housing (not shown), an image to be output to the display portion **5304** can be output to another video device (not shown). In that case, the housing **5302** and the housing **5303** can each function as an operating unit. Thus, a plurality of players can perform a game at the same time. The chip described in the above embodiment can be incorporated into the chip provided on a substrate in the housing **5301**, the housing **5302** and the housing **5303**.
(597) FIG. **32**D illustrates a stationary game machine **5400** as an example of a game machine. A controller **5402** is wired or connected wirelessly to the stationary game machine **5400**.
(598) Using the GPU or the chip of one embodiment of the present invention in a game machine such as the portable game machine **5300** and the stationary game machine **5400** achieves a low-power-consumption game machine. Moreover, heat generation from a circuit can be reduced owing to low power consumption; thus, the influence of heat generation on the circuit, a peripheral circuit, and a module can be reduced.
(599) Furthermore, when the GPU or the chip of one embodiment of the present invention is applied to the portable game machine **5300**, the portable game machine **5300** including artificial intelligence can be achieved.
(600) In general, the progress of a game, the actions and words of game characters, and expressions of an event and the like occurring in the game are determined by the program in the game; however, the use of artificial intelligence in the portable game machine **5300** enables expressions not limited by the game program. For example, questions posed by the player, the progress of the game, time, and actions and words of game characters can be changed for various expressions.
(601) In addition, when a game requiring a plurality of players is played on the portable game machine **5300**, the artificial intelligence can create a virtual game player; thus, the game can be played alone with the game player created by the artificial intelligence as an opponent.
(602) Although the portable game machine and the stationary game machine are shown as examples of game machines in FIG. **32**C and FIG. **32**D, the game machine using the GPU or the chip of one embodiment of the present invention is not limited thereto. Examples of the game

machine to which the GPU or the chip of one embodiment of the present invention is applied include an arcade game machine installed in entertainment facilities (a game center, an amusement park, and the like), and a throwing machine for batting practice installed in sports facilities.
(603) [Large Computer]
(604) The GPU or the chip of one embodiment of the present invention can be used in a large computer.
(605) FIG. **32**E illustrates a supercomputer **5500** as an example of a large computer. FIG. **32**F illustrates a rack-mount computer **5502** included in the supercomputer **5500**.
(606) The supercomputer **5500** includes a rack **5501** and a plurality of rack-mount computers **5502**. The plurality of computers **5502** are stored in the rack **5501**. The computer **5502** includes a plurality of substrates **5504** on which the GPU or the chip shown in the above embodiment can be mounted.
(607) The supercomputer **5500** is a large computer mainly used for scientific computation. In scientific computation, an enormous amount of arithmetic operation needs to be processed at a high speed; hence, power consumption is large and chips generate a large amount of heat. Using the GPU or the chip of one embodiment of the present invention in the supercomputer **5500** achieves a low-power-consumption supercomputer. Moreover, heat generation from a circuit can be reduced owing to low power consumption; thus, the influence of heat generation on the circuit, a peripheral circuit, and a module can be reduced.
(608) Although a supercomputer is shown as an example of a large computer in FIG. **32**E and FIG. **32**F, a large computer using the GPU or the chip of one embodiment of the present invention is not limited thereto. Other examples of large computers in which the GPU or the chip of one embodiment of the present invention is usable include a computer that provides service (a server) and a large general-purpose computer (a mainframe).
(609) [Moving Vehicle]
(610) The GPU or the chip of one embodiment of the present invention can be applied to an automobile, which is a moving vehicle, and the periphery of a driver's seat in the automobile.
(611) FIG. **32**G illustrates an area around a windshield inside an automobile, which is an example of a moving vehicle. FIG. **32**G illustrates a display panel **5701**, a display panel **5702**, and a display panel **5703** that are attached to a dashboard and a display panel **5704** that is attached to a pillar.
(612) The display panel **5701** to the display panel **5703** can provide a variety of kinds of information by displaying a speedometer, a tachometer, mileage, a fuel gauge, a gear state, air-condition setting, and the like. The content, layout, or the like of the display on the display panels can be changed appropriately to suit the user's preferences, so that the design can be improved. The display panel **5701** to the display panel **5703** can also be used as lighting devices.
(613) The display panel **5704** can compensate for view obstructed by the pillar (a blind spot) by showing an image taken by an imaging device (not shown) provided for the automobile. That is, displaying an image taken by the imaging device provided outside the automobile leads to compensation for the blind spot and an increase in safety. In addition, display of an image that complements the area that cannot be seen makes it possible to confirm safety more naturally and comfortably. The display panel **5704** can also be used as a lighting device.
(614) Since the GPU or the chip of one embodiment of the present invention can be applied to a component of artificial intelligence, the chip can be used for an automatic driving system of the automobile, for example. The chip can also be used for a system for navigation, risk prediction, or the like. The display panel **5701** to the display panel **5704** may display information regarding navigation, risk prediction, and the like.
(615) Although an automobile is described above as an example of a moving vehicle, moving vehicles are not limited to an automobile. Examples of a moving vehicle include a train, a monorail train, a ship, and a flying object (a helicopter, an unmanned aircraft (a drone), an airplane, and a rocket), and these moving vehicles can include a system utilizing artificial intelligence when

equipped with the chip of one embodiment of the present invention.

(616) [Household Appliance]

(617) FIG. **32**H illustrates an electric refrigerator-freezer **5800** as an example of a household appliance. The electric refrigerator-freezer **5800** includes a housing **5801**, a refrigerator door **5802**, a freezer door **5803**, and the like.

(618) When the chip of one embodiment of the present invention is applied to the electric refrigerator-freezer **5800**, the electric refrigerator-freezer **5800** including artificial intelligence can be obtained. Utilizing the artificial intelligence enables the electric refrigerator-freezer **5800** to have a function of automatically making a menu based on foods stored in the electric refrigerator-freezer **5800** and the food expiration dates, for example, a function of automatically adjusting the temperature to be appropriate for the foods stored in the electric refrigerator-freezer **5800**, and the like.

(619) Although the electric refrigerator-freezer is described in this example as a household appliance, examples of other household appliances include a vacuum cleaner, a microwave oven, an electric oven, a rice cooker, a water heater, an IH cooker, a water server, a heating-cooling combination appliance such as an air conditioner, a washing machine, a drying machine, and an audio visual appliance.

(620) The electronic device and the functions of the electronic device, the application example of the artificial intelligence and its effects, and the like described in this embodiment can be combined as appropriate with the description of another electronic device.

(621) This embodiment can be implemented in an appropriate combination with the structures described in the other embodiments, example, and the like.

EXAMPLE

(622) In this example, an aggregation state of a metal oxide film was analyzed. Specifically, selected area electron diffraction (SAED) was performed on a sample including a metal oxide film. The sample was observed by a dark field observation method and a bright field observation method.

(623) First, the samples used in this example are described.

(624) First, the samples are fabricated. As illustrated in FIG. **33**A, the sample includes a substrate **800**, an oxide film **801** over the substrate **800**, and a metal oxide film **802** over the oxide film **801**. The substrate **800** is a substrate containing silicon. The oxide film **801** is a 100-nm-thick silicon oxide film formed by performing heat treatment on a surface of the substrate **800** in a hydrogen chloride (HCl) atmosphere. The metal oxide film **802** is a 3-μm-thick IGZO film formed by a sputtering method. In the deposition of the metal oxide film **802**, an oxide target with In:Ga:Zn=4:2:4.1 [atomic ratio] was used; 30 sccm of an argon gas and 15 sccm of an oxygen gas were used as a deposition gas; the deposition pressure was 0.4 Pa; the deposition power was 500 W; the substrate temperature was 200° C.; and the distance between the target and the substrate was 60 nm.

(625) Next, the samples were processed using focused ion beam (FIB), whereby a sample for cross-sectional observation and a sample for plan-view observation were fabricated.

(626) The above is the description of the samples used in this example.

(627) In order to observe the crystal structure of the metal oxide film **802**, SAED was performed on the sample for cross-sectional observation. Note that a transmission electron microscope “H-9500” manufactured by Hitachi High-Technologies Corporation was used for the SAED. The diameter of a region measured by SAED (referred to as a selected area in some cases) is approximately 3 μm.

(628) Note that since the metal oxide film **802** has a large thickness, in a selected area diffraction pattern obtained by SAED, the intensity of an electron beam spot can be high.

(629) FIG. **33**B and FIG. **33**C show selected area diffraction patterns of the sample for cross-sectional observation, which were obtained by SAED. Note that selected areas in the selected area diffraction patterns shown in FIG. **33**B and FIG. **33**C correspond to a region **810** and a region **811**,

respectively, denoted by dotted circles in FIG. **33**A. That is, FIG. **33**B shows a selected area diffraction pattern in the case where the selected area is positioned in the metal oxide film **802**. FIG. **33**C shows a selected area diffraction pattern in the case where the selected area is positioned across the metal oxide film **802**, the oxide film **801**, and the substrate **800**.

(630) In each of the selected area diffraction patterns shown in FIG. **33**B and FIG. **33**C, a spot observed at the center is a spot of a transmission wave (000). An arched spot which is a region surrounded by the dotted line and observed above the center in FIG. **33**B is a spot of a diffraction wave (009). FIG. **33**B shows that the metal oxide film **802** is a CAAC-IGZO film.

(631) Next, a bright field electron microscope image (also referred to as a bright field image) of the sample for cross-sectional observation was observed by the bright field observation method, which is a technique of forming an image by extracting a transmitted wave. Note that the transmitted wave used for forming an image is extracted by disposing an objective diaphragm so that the transmitted wave is allowed to transmit and the diffraction wave is blocked. The transmission electron microscope "H-9500" manufactured by Hitachi High-Technologies Corporation was used for the bright field image observation.

(632) FIG. **34** is a bright field image of the sample for cross-sectional observation. FIG. **34** is a bright field image of the metal oxide film **802** and the oxide film **801**.

(633) In the bright field image of the sample for cross-sectional observation, differential contrast was observed in the metal oxide film **802**, as shown in FIG. **34**. This contrast probably contains information derived from different crystal orientation. Accordingly, it is suggested that regions each having a size of several tens of nanometers and a different orientation order exist in the metal oxide film **802**.

(634) Next, in order to capture partial orientation state of the metal oxide film **802**, a dark field electron microscope image (also referred to as a dark field image) of the sample for cross-sectional observation was observed by the dark field observation method, which is a technique of forming an image by extracting a particular diffraction wave. Note that the diffraction wave used for forming an image is extracted by disposing an objective diaphragm so that the diffraction wave is allowed to transmit and the transmitted wave is blocked. Since the diffraction wave is observed as a spot in the selected area diffraction pattern, the dark field image is sometimes referred to as a dark field image in the case where the spot of the diffraction wave observed in the selected area diffraction pattern is extracted using the objective diaphragm. The transmission electron microscope "H-9500" manufactured by Hitachi High-Technologies Corporation was used for the dark field image observation.

(635) FIG. **35**A shows a selected-area electron diffraction pattern of the sample for cross-sectional observation. FIG. **35**A shows a selected area diffraction pattern of the sample for cross-sectional observation near a surface of the metal oxide film **802**.

(636) FIG. **35**B to FIG. **35**D show dark field images of the sample for cross-sectional observation. FIG. **35**B to FIG. **35**D are dark field images near the surface of the metal oxide film **802** utilizing the spot of the diffraction wave (009) observed in the selected area diffraction pattern in FIG. **35**A. FIG. **35**B is a dark field image in the case where a region **820** (the left side of the spot of the diffraction wave (009)) denoted by the dotted line in FIG. **35**A is extracted with the objective diaphragm. FIG. **35**C is a dark field image in the case where a region **821** (the center of the spot of the diffraction wave (009) and the vicinity thereof) denoted by the dotted line in FIG. **35**A is extracted with the objective diaphragm. FIG. **35**D is a dark field image in the case where a region **822** (the right side of the spot of the diffraction wave (009)) denoted by the dotted line in FIG. **35**A is extracted with the objective diaphragm. Note that the dark field images shown in FIG. **35**B to FIG. **35**D are formed near the surface of the metal oxide film **802**, which is the uppermost portion, and are all the same field of view.

(637) As seen from FIG. **35**C, in the dark field image in the case where the region **821** is extracted, a region with a differential orientation state is not clearly observed. On the other hand, as seen from

FIG. **35**B and FIG. **35**D, in the dark field images in the case where the region **820** or the region **822** is extracted, band-like contrast which is slightly tilted to the normal direction of the substrate surface was observed. Note that in this cross-sectional observation, the thickness information of the sample fabricated by FIB processing is included as information of the depth direction, and thus there is a possibility that a crystal layer is formed while orientation order of the CAAC structure is tilted in the normal direction of the substrate surface.

(638) Next, in order to reduce information about the depth direction as much as possible, a sample for cross-sectional observation processed to be thinner (thinned) only in the depth direction was fabricated, and the sample for cross-sectional observation was subjected to selected area diffraction pattern observation and dark field image observation.

(639) FIG. **36**A shows a selected area diffraction pattern of the thinned sample for cross-sectional observation. FIG. **36**A shows a selected area diffraction pattern of the thinned sample for cross-sectional observation near the oxide film **801**.

(640) FIG. **36**B to FIG. **36**D show dark field images of the thinned sample for cross-sectional observation. FIG. **36**B to FIG. **36**D are dark field images near the oxide film **801** utilizing the spot of the diffraction wave (009) observed in the selected area diffraction pattern in FIG. **36**A. FIG. **36**B is a dark field image in the case where a region **830** (the left side of the spot of the diffraction wave (009)) denoted by the dotted line in FIG. **36**A is extracted with the objective diaphragm. FIG. **36**C is a dark field image in the case where a region **831** (the center of the spot of the diffraction wave (009) and the vicinity thereof) denoted by the dotted line in FIG. **36**A is extracted with the objective diaphragm. FIG. **36**D is a dark field image in the case where a region **832** (the right side of the spot of the diffraction wave (009)) denoted by the dotted line in FIG. **36**A is extracted with the objective diaphragm. Note that the dark field images shown in FIG. **36**B to FIG. **36**D are formed near the surface of the oxide film **801** and are all the same field of view.

(641) As seen from FIG. **36**B to FIG. **36**D, band-like contrast having a width of approximately 10 nm and showing an oriented region that extends in the normal direction of the substrate was obtained. Furthermore, it was also found that the growth of a crystal layer of the metal oxide film **802** begins at a point less than 1 nm from the interface with the oxide film **801**, and the crystal layer is oriented while tilted in the normal direction of the substrate by approximately 2° to 3°.

(642) In order to evaluate the distribution of the crystal layer having a long-range order observed in the thinned sample for cross-sectional observation as band-like contrast, a dark field image of the sample for plan-view observation in the case where the spot of the diffraction wave (100) was extracted with the objective diaphragm was observed. Note that an atomic resolution analytical electron microscope “JEM-ARM200F” manufactured by JEOL Ltd. was used for the sample for the selected-area electron diffraction pattern observation and the dark field image observation performed on the sample for plan-view observation.

(643) FIG. **37**A shows a selected-area electron diffraction pattern of the sample for plan-view observation. FIG. **37**A shows a selected area diffraction pattern of the sample for plan-view observation of the metal oxide film **802**.

(644) FIG. **37**B to FIG. **37**D show dark field images of the sample for plan-view observation. FIG. **37**B to FIG. **37**D are dark field images of the metal oxide film **802** utilizing the spot of the diffraction wave (100) observed in the selected area diffraction pattern in FIG. **37**A. FIG. **37**B is a dark field image in the case where a region **840** (the left side of one spot of the diffraction wave (100)) denoted by the dotted line in FIG. **37**A is extracted with the objective diaphragm. FIG. **37**C is a dark field image in the case where a region **841** (the center of one spot of the diffraction wave (100) and the vicinity thereof) denoted by the dotted line in FIG. **37**A is extracted with the objective diaphragm. FIG. **37**D is a dark field image in the case where a region **842** (the right side of one spot of the diffraction wave (100)) denoted by the dotted line in FIG. **37**A is extracted with the objective diaphragm. Note that the dark field images shown in FIG. **37**B to FIG. **37**D are all the same field of view.

(645) As seen from FIG. **37**B to FIG. **37**D, in the metal oxide film **802**, orientation orders shown as the spots of the diffraction wave (100) each having a size of approximately 100 nm are distributed, and contrast in the dark field image formed by connecting this vertical oriented region and oriented regions tilted from the vertical oriented region by several degrees was observed.
(646) As described above, it was found that the CAAC-IGZO film has continuous crystal orientation even in a region between the level of nanometers, which is larger than the atomic level, and a bulk (what is called a mesoscopic region).
(647) At least part of the structure, the method, and the like described in this example can be implemented in appropriate combination with other embodiments described in this specification.
REFERENCE NUMERALS
(648) **100**: capacitor, **110**: conductor, **112**: conductor, **115**: conductor, **120**: conductor, **125**: conductor, **130**: insulator, **140**: conductor, **142**: insulator, **145**: insulator, **150**: insulator, **152**: insulator, **153**: conductor, **154**: insulator, **156**: insulator, **200**: transistor, **200**_*n*: transistor, **200_1**: transistor, **200***a*: transistor, **200***b*: transistor, **200**T: transistor, **205**: conductor, **205***a*: conductor, **205***b*: conductor, **210**: insulator, **211**: insulator, **212**: insulator, **214**: insulator, **216**: insulator, **217**: insulator, **218**: conductor, **222**: insulator, **224**: insulator, **230**: oxide, **230***a*: oxide, **230**A: oxide film, **230***b*: oxide, **230**B: oxide film, **230***c*: oxide, **230***c***1**: oxide, **230***c***2**: oxide, **230**C: oxide film, **240**: conductor, **240***a*: conductor, **240***b*: conductor, **241**: insulator, **241***a*: insulator, **241***b*: insulator, **242**: conductor, **242***a*: conductor, **242**A: conductive film, **242***b*: conductor, **242**B: conductive layer, **242***c*: conductor, **243**: oxide, **243***a*: oxide, **243**A: oxide film, **243***b*: oxide, **243**B: oxide layer, **246**: conductor, **246***a*: conductor, **246***b*: conductor, **250**: insulator, **250**A: insulating film, **260**: conductor, **260***a*: conductor, **260**A: conductive film, **260***b*: conductor, **260**B: conductive film, **265**: sealing portion, **265***a*: sealing portion, **265***b*: sealing portion, **272**: insulator, **273**: insulator, **274**: insulator, **280**: insulator, **282**: insulator, **283**: insulator, **284**: insulator, **286**: insulator, **287**: insulator, **290**: memory device, **292**: capacitor device, **292***a*: capacitor device, **292***b*: capacitor device, **294**: conductor, **294***a*: conductor, **294***b*: conductor, **300**: transistor, **311**: substrate, **313**: semiconductor region, **314***a*: low-resistance region, **314***b*: low-resistance region, **315**: insulator, **316**: conductor, **320**: insulator, **322**: insulator, **324**: insulator, **326**: insulator, **328**: conductor, **330**: conductor, **350**: insulator, **352**: insulator, **354**: insulator, **356**: conductor, **400**: transistor, **405**: conductor, **411**: element layer, **413**: transistor layer, **415**: memory device layer, **415_1**: memory device layer, **415_3**: memory device layer, **415_4**: memory device layer, **420**: memory device, **424**: conductor, **430***c*: oxide, **431***a*: oxide, **431***b*: oxide, **432***a*: oxide, **432***b*: oxide, **440**: conductor, **442***a*: conductor, **442***b*: conductor, **443***a*: oxide, **443***b*: oxide, **450**: insulator, **460**: conductor, **460***a*: conductor, **460***b*: conductor, **470**: memory unit, **600**: semiconductor device, **601**: semiconductor device, **610**: cell array, **610_1**: cell array, **610**_*n*: cell array, **700**: electronic component, **702**: printed circuit board, **704**: mounting board, **711**: mold, **712**: land, **713**: electrode pad, **714**: wire, **720**: storage device, **721**: driver circuit layer, **722**: memory circuit layer, **730**: electronic component, **731**: interposer, **732**: package substrate, **733**: electrode, **735**: semiconductor device, **800**: substrate, **801**: oxide film, **802**: metal oxide film, **810**: region, **811**: region, **820**: region, **821**: region, **822**: region, **830**: region, **831**: region, **832**: region, **840**: region, **841**: region, **842**: region, **901**: boundary region, **902**: boundary region, **1001**: wiring, **1002**: wiring, **1003**: wiring, **1004**: wiring, **1005**: wiring, **1006**: wiring, **1007**: wiring, **1008**: wiring, **1009**: wiring, **1010**: wiring, **1100**: USB memory, **1101**: housing, **1102**: cap, **1103**: USB connector, **1104**: substrate, **1105**: memory chip, **1106**: controller chip, **1110**: SD card, **1111**: housing, **1112**: connector, **1113**: substrate, **1114**: memory chip, **1115**: controller chip, **1150**: SSD, **1151**: housing, **1152**: connector, **1153**: substrate, **1154**: memory chip, **1155**: memory chip, **1156**: controller chip, **1200**: chip, **1201**: PCB, **1202**: bump, **1203**: motherboard, **1204**: GPU module, **1211**: CPU, **1212**: GPU, **1213**: analog arithmetic unit, **1214**: memory controller, **1215**: interface, **1216**: network circuit, **1221**: DRAM, **1222**: flash memory, **1400**: storage device, **1411**: peripheral circuit, **1420**: row circuit, **1430**: column circuit, **1440**: output circuit, **1460**: control logic circuit, **1470**: memory cell array, **1471**: memory cell, **1472**: memory cell, **1473**: memory cell, **1474**:

memory cell, **1475**: memory cell, **1476**: memory cell, **1477**: memory cell, **1478**: memory cell, **5100**: information terminal, **5101**: housing, **5102**: display portion, **5200**: notebook information terminal, **5201**: main body, **5202**: display portion, **5203**: keyboard, **5300**: portable game machine, **5301**: housing, **5302**: housing, **5303**: housing, **5304**: display portion, **5305**: connection portion, **5306**: operation key, **5400**: stationary game machine, **5402**: controller, **5500**: supercomputer, **5501**: rack, **5502**: computer, **5504**: substrate, **5701**: display panel, **5702**: display panel, **5703**: display panel, **5704**: display panel, **5800**: electric refrigerator-freezer, **5801**: housing, **5802**: refrigerator door, **5803**: freezer door

Claims

1. A semiconductor device comprising a transistor, wherein the transistor comprises: a first insulator; an oxide provided with a groove portion over the first insulator; a first conductor and a second conductor disposed in a region that does not overlap with the groove portion of the oxide; a second insulator disposed between the first conductor and the second conductor and in the groove portion of the oxide; and a third conductor over the second insulator, wherein a first bottom surface of the third conductor is lower than a top surface of the oxide outside the groove portion, wherein a bottom surface of the groove portion does not reach the first insulator, wherein in a cross-sectional view of the transistor in a channel length direction, an end portion of the bottom surface of the groove portion comprises a curvature, and wherein in a cross-sectional view of the transistor in a channel width direction, a second bottom surface of the third conductor is lower than a bottom surface of the oxide.
2. The semiconductor device according to claim 1, wherein a depth of the groove portion is greater than or equal to 5 nm and less than or equal to 30 nm.
3. The semiconductor device according to claim 1, wherein a distance between the top surface of the oxide and the first bottom surface of the third conductor in a depth direction is longer than a distance between the bottom surface of the groove portion and a bottom surface of the oxide in a depth direction.
4. The semiconductor device according to claim 1, further comprising a fourth conductor under the first insulator.
5. A semiconductor device comprising a transistor, wherein the transistor comprises: a first insulator; a first oxide over the first insulator; a second oxide over the first oxide, the second oxide comprising a groove portion; a first conductor and a second conductor over the second oxide; a second insulator over the second oxide; a third conductor over the second insulator; and a third insulator over the first conductor and the second conductor; the third insulator comprising an opening reaching the groove portion of the second oxide, wherein the second insulator and the third conductor are provided inside the opening of the third insulator and the groove portion of the second oxide, wherein a top surface of the third conductor is substantially aligned with a top surface of the second insulator, wherein a sidewall of the groove portion is substantially aligned with a sidewall of the opening, wherein a first bottom surface of the third conductor is lower than a top surface of the second oxide outside the groove portion, wherein a bottom surface of the groove portion does not reach the first oxide and the first insulator, wherein in a cross-sectional view of the transistor in a channel length direction, an end portion of the bottom surface of the groove portion comprises a curvature, and wherein in a cross-sectional view of the transistor in a channel width direction, a second bottom surface of the third conductor is lower than a bottom surface of the second oxide.
6. The semiconductor device according to claim 5, wherein the second oxide comprises indium, wherein the first oxide comprises indium, an element M, and zinc, and wherein the element M is gallium, aluminum, yttrium, or tin.
7. The semiconductor device according to claim 5, wherein an atomic ratio of the indium to a metal

element which is a main component in the second oxide is larger than an atomic ratio of the indium to a metal element which is a main component in the first oxide.
8. The semiconductor device according to claim 5, wherein a depth of the groove portion is greater than or equal to 5 nm and less than or equal to 30 nm.
9. The semiconductor device according to claim 8, wherein the second oxide comprises indium, wherein the first oxide comprises indium, an element M, and zinc, and wherein the element M is gallium, aluminum, yttrium, or tin.
10. The semiconductor device according to claim 9, wherein an atomic ratio of the indium to a metal element which is a main component in the second oxide is larger than an atomic ratio of the indium to a metal element which is a main component in the first oxide.
11. The semiconductor device according to claim 5, wherein a distance between the top surface of the second oxide and the first bottom surface of the third conductor in a depth direction is longer than a distance between the bottom surface of the groove portion and a bottom surface of the second oxide in a depth direction.
12. The semiconductor device according to claim 5, further comprising a fourth conductor under the first insulator.